

RIFT VALLEY UNIVERSITY



HEALTH SCIENCES FACULTY

DEPARTMENT OF NURSING

Pediatrics & Child Health Nursing

For 3rd Year BSc Nursing Students'

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UNIT ONE

INTRODUCTION TO PEDIATRIC NURSING

GOOD MORNING?



Introduction to pediatric Nursing

❖ **Definition of terms:**

➤ The word “pediatrics” means “healer of children.

Pediatrics is a Greek word,

Pedia – **Child**

iatrike– **Treatment**

ics– **Branch of medicine**

Introduction....

- ❖ **Pediatrics** can be defined as the branch of medical science that deals with the care of children from **conception** to **adolescence** in health & illness.
- ❖ It is concerned with preventive, promotive, curative and rehabilitative care of children.
- ✓ Their growth and development; and
- ✓ Their opportunity to achieve full potential as adults.

Cont...

- **Child health** is an area of medical practice with a focus on **providing holistic care to** infants, children and adolescents.
- It differs from **pediatrics**, which emphasis **illness and the alleviation** of symptoms or disease.

Introduction ...

- **Pediatric nursing is the art and science of giving nursing care to children** from birth through adolescent with emphasis on the physical growth, mental, emotional and psycho-social development.

WHO IS A CHILD?

- ❖ According to the United Nations Convention on the Rights of the Child (UNCRC) , a child is any person aged below 18 years.
- ❖ The word child refers to all the various stages of growth and development that a person goes through before attaining adulthood.
- ❖ The stages are divided into two groups; those before birth and those after birth.

Stages before birth

- ❖ Before birth we have the **Embryo and Foetus**.
- ❖ The Embryo is the product of conception **up to 8 weeks of pregnancy**
- ❖ While the Foetus is the product of **conception up to birth**.

Stages after birth

- The stages of the child after birth are:
 - ✓ **Neonatal period;** from birth up to 28 daies
 - ✓ **Infancy;** from birth to end of 1st year.
 - ✓ **Toddlerhood;** period from end of 1st year to end of 2nd year.
 - ✓ **Preschool;** period between 3 & 5 years.
 - ✓ **Middle childhood;** period between 6 & 11 years
 - ✓ **Adolescence;** period between 10 & 20 years

Introduction

❖ Common Pediatric terminologies

1. *Prenatal* → (conception to birth)

- ✓ **Embryo** → from conception up to 8 wks of gestations
- ✓ **Fetus** → from 8 wks of gestation to birth.

2. *Neonatal* → (birth to 28 days)

3. *Infant* → (month to 1 year)

4. *Toddler* → (1 – 3 years)

Cont...

- 5. *Preschool* → (3 -6 years)
- 6. *School age* → (6 – 12 years)
- 7. *Adolescent* → (12 – 18 years)
- 8. *Young Adolescent* → 13 - 15yrs
- 9. *Older Adolescent* → 16 - 19yrs
- 10. *Puberty* → 14 yrs and above (up to 17 yrs of age).

Differences between adult and pediatric

- ❖ Pediatrics differs from adult medicine in many respects.
- ❖ The obvious **body size** differences are paralleled by maturational changes.
- ❖ The smaller body of an **infant or neonate** is substantially different **physiologically** from that of an adult.
- ❖ Children are **not small** adults.
- ❖ Treating a child is **not like treating** a miniature adult.

Cont...

- ❖ A major difference between pediatrics and adult medicine is that children **cannot make decisions** for themselves.
- ❖ The issues of guardianship, privacy, legal responsibility and informed consent must always be considered in every pediatric procedure.

Cont...

- ❖ Treating the parents and sometimes, the family, rather than just the child is often necessary
- ❖ Adolescents are in their own legal class, having rights to their own health care decisions in certain circumstances.

I AM NOT LITTLE ADULT



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History of Neonatal and Pediatric Nursing

Pediatrics became a medical specialty in the mid -19th century.

Before that time the care and treatment of childhood diseases was included with in general medicine and obstetrics (midwifery).

Virtually all nations have practicing departments of pediatrics or child health.

Why pediatrics car, Count...

- ❖ With some **reasons**, pediatrics become an independent medical specialty:
 - Large numbers:
 - High mortality & High Morbidity:
- The health problems of children differ from adults.
- Children response to illness is affected by age.
- Vulnerable due to developmental drawbacks.
- Preventable disease are common in children.

1.2 An overview Child Health in Ethiopia

⇒ < 5 years & < 1 years are 16-18% & 3-4 % of the total population respectively (**MOH, 2007**)

⇒ Infants and < 5 years mortality rate 48 and 67 Per 1000 live birth respectively (**EDHS, 2016**)

⇒ These counts 50% of total recorded death

Causes of Mortality

- Infection and malnutrition are major cause

- ✓ **Infection:** Diarrhea disease, ARI, Measles, Malaria and HIV/AIDS

- ✓ **Malnutrition:** (PEM & Micronutrient Deficiency)

The major causes of under five mortality in Ethiopia...

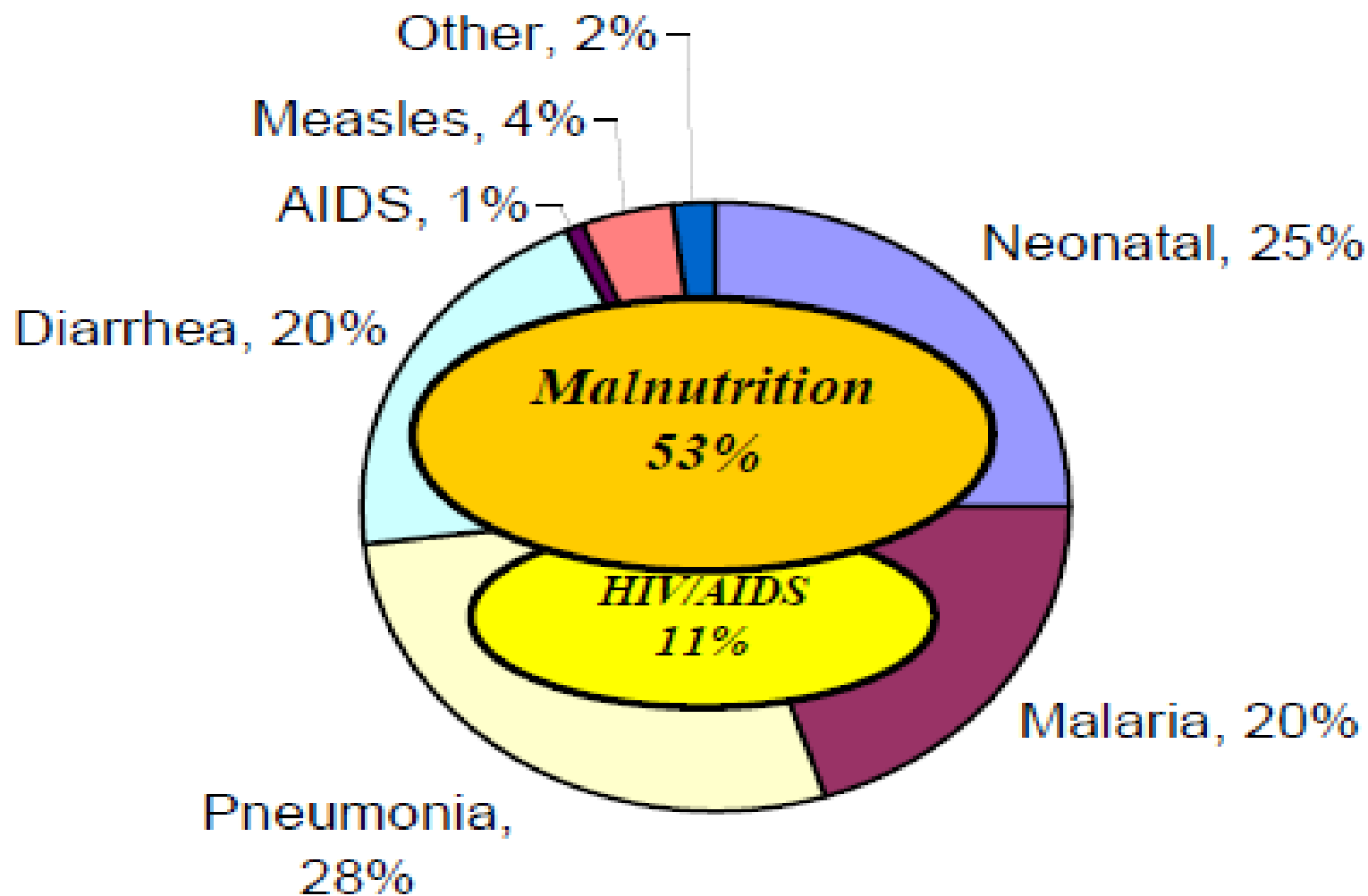
❑ The Child Survival Strategy identifies the direct causes that are responsible for under-five mortality :

- ❖ Pneumonia (28%),
- ❖ Neonatal causes (25%)
- ❖ Malaria (20%)
- ❖ Diarrhea (20%)
- ❖ Measles (4%) and AIDS (1%).
- ❖ **Underlying** conditions are also identified, mainly malnutrition and HIV/AIDS .

Cont...

- ❖ Infections and tetanus account for 47% of neonatal deaths in Ethiopia
- ❖ The rest are due to Asphyxia (25%), Preterm birth (17%), diarrhea, congenital malformation and others.
- ❖ Fortunately, for each target condition there are **both preventive and curative interventions** that can **prevent around 72% of deaths** in under five children.

Major causes of under-five mortality in Ethiopia.



The trend analysis of under five, Infant and Neonatal Mortality in Ethiopia

Mortality	Years				2019 Mini rep.	Data source
	2000	2005	2011	2016		
U5MR	166	123	88	67	55	EDHS
IMR	97	77	59	48	43	
NMR	49	39	37	29	30	

U5MR = Under five mortality rate

IMR = Infant mortality rate

NMR = Neonatal mortality rate

- ❖ Child care have a scope of care before, during and after delivery.
- ❖ Factors affecting the health development of children related to the mother are:
 - *Maternal health before pregnancy*
 - *Maternal health during pregnancy, labor and post natal care*

Modern concept of child care

❖ Safe mother hood in relation to child health

➤ Up to 70% of neonatal deaths could be prevented by mother baby package intervention suggested by WHO

➤ These are:-

✓ ANC

✓ Labor & delivery

✓ Postnatal care (ENC)

A) Maternal health before pregnancy

- Genetic and environment (smoking, alcohol)
- *pre-pregnancy nutrition*
 - Strong influences on the: growth, Birth Weight of new born, and production sufficient breast milk.
- Infection prior to pregnancy
 - Maternal antibodies passed to the fetus and provide protection.

B) Maternal health during pregnancy and labor

■ *Nutrition & infection-*

- Pregnant mother should gain weight 10-12 kg .
- Acute infection can cause congenital deformities and abortion, e.g. syphilis, rubella.

C. Post natal care

- ❑ Good opportunity for health education and counseling: about hygiene, feeding & rest
- Prevent ophthalmic neonatorum & cord infection
- It enhanced neonatal exclusive breast feeding
- Delay washing of the new-born for 24 hours to avoid unnecessary chilling
- For immunization, FP, growth monitoring etc.

Changing Concepts in child health care:

- The role of the pediatric nurses is not only to see the child for acute illnesses but also to provide well comprehensive child care.
- Child health care can be done in different situation.

Institutional care:

- Hospital regulations have been changed and began to reflect the human needs.
- Nurses who works with children understand, value and use play as a therapeutic tool.

Family-centered care:

- Even if nursing taken place entirely in hospital wall, family centered care is given special attention.
- Children's need met only in a superficial manner among their family.

Regionalized care:

- To provide high quality care requires moving child-parent to medical center with the best resource for **diagnosis and treatment.**

Other innovative Child health-care Programs:

- Home-care programs (under hospital)
- E.g. leukemia, hemophilia, and cystic fibrosis - (hereditary disease- affects-thick mucus blocks & infections of respiration. Pancreas-deficiency of digestive enzymes-impaired nutrition).
- Some children's condition is monitored at home

The Nurses' Changing Role:

- In addition to help children to achieve optimal health, the nurse is also a teacher, advisor, and resource person.

Promotive, Preventive and Curative aspects of Child Health:

- **Promotive:** refers to increase healthiness through health education.
- **Preventive:** refers to prevention of diseases through: Health education, Immunization, and Environmental sanitation.
- **Curative:** refers to treatment of diseases through medication (drug use).

- Nurses' role here is supportive and educative.
- Prevention is the best measure for maintaining health.
- There are three levels of prevention:
 - ❖ Primarily
 - ❖ Secondary and
 - ❖ Tertiary types of prevention

1. Primary prevention:

- The aim is to avoid disease before its onset.
- *Growth monitoring*: nutritional status of the children.
- *Oral re-hydration*: prevent dehydration
- *Promotion of breast feeding*: prevent malnutrition.
- *Immunization*: developing immunity
- *Family planning*: pregnancy prevention
- *Health education on sanitation*: prevent infection

2. Secondary prevention

- Aim: - early detection and treatment of the precursors of the disease.
- Screening methods such as tuberculin surveys
- Vitamin A deficiency surveys etc
- Treating the respective health problem

3. Tertiary prevention

- Aim: - rehabilitation (e.g. polio mellitus), to prevent deformities.

Hospital environment for sick child

Assisting Hospitalized Child:

- Hospitalization begins with admitting a child to the health care unit (ward).

Purpose of admission:

- To help the new patient
- To adjust the newborn
- To alleviate fear and worry

Cont...

Indication for hospitalization:

- *For therapeutic purpose*
- *For clinical evaluation, assessment*
- *To break regular home care*
- *Investigation that cannot be done outside the hospital*
- When admission has been decided & accepted, rapport should be established b/n health care personnel & his family to gain confidence & trust.

General Principle of Hospitalization:

- The nurse should take full nursing, social history & record.
- Children's ward should be with "Mother -Child" accommodation, (family-centered care), to continue care about: Breast feeding & emotional support
- Assisting in nursing procedure
- Learning family the therapeutic & nursing procedures to facilitate home care.
- Therapeutic play (recreational facilities) should advised.
- Since the vulnerability, it should have all the curative preventive and promotive activities.

The roles of pediatric nurse in caring for hospitalized child:

- Role of the PN ranges from providing first aid to prepare for teaching, direct care, and referral as needed.
- Identify the pt: record the child's full name, record bed and room No etc
- Check the reason for hospitalization:
 - does the child need urgent medical or Nursing attention?

Cont...

- Check Consent form for respected procedure
- Introduce the hospital Environment to the child & family
- Provide Prevention by - Nutrition, Health education, and Immunization
- Restoration of health: includes the physical and emotional needs of children like feeding, bathing, toileting, dressing, security & socialization.

• Cont...

- Coordination/collaboration: with other professionals to provide high quality of health services.
- Prioritizing most critical aspects
 - Airway, breathing, circulation(ABC aproch)
- Using problem-solving skills
 - Inductive and deductive reasoning
 - Using critical thinking (intellectual criticism and balanced judgment)
- Apply nursing processes:

Principles of peri operative care of infant and children

Pre-operative Care:

❖ Assist on psychological preparation:

- The care includes provision of emotional support, psychological preparation, pre-operative teaching.
- Such care will minimize stress and help the child to cope with existed fears.
- *The nurse should be alert about:*
 - Potential threats for the hospitalized child from surgery.
(physical harm, separation loss of autonomy, confusion about his/her limit).

The nurse should be alert about....

- Familiarization of the patient and the family with ward.
- Encouraging questions and giving honest answers.
- Providing opportunity for clients to talk about their concern and feelings.
- Prepare the child to be aware his/her post operative expectations
- Obtaining consent from parents

❖ Assisting in physical preparation:

- Assist with necessary lab tests
- Keep the patient NPO (nothing per Os).
 - Small children may be given a dextrose drink 2 – 3 hours before anesthetic to minimize risk of hypoglycemia.
- Assist with fever reduction as fever increase risk of anesthesia and need for fluid calories.
- Administer appropriate preoperative medication as ordered

Cont...

- ❖ Establish good hydration.

Parental Support at time of crisis:

- Preparation and support should be integrated for parents and child as *parents may have feeling of fear and experiencing emotionally distressing.*
- Help parents how they can care for their child

During operation: OR nurse will be more responsible for child health care.

Post-operative care:

❖ Immediate post-operative care:

- Maintain patent air way by:
 - Positioning on his/her side or abdomen
 - prevent the tongue from back falling
 - Suction any secretions present
- watch for signs of shock:
 - pallor, coldness, rapid pulse and irregular respiration, and in older children decreased blood pressure (BP)

Cont...

- See all drainage are connected and function properly
- Monitor intravenous fluid as ordered.

❖ After Recovery:

- Continue frequent observations with *regard to behavior, vital signs, dressing on operative site, and special apparatus* (chest tube, oxygen).
- Note signs of dehydration:
 - dry skin and membrane, sunken eye, poor skin turgor, and sunken fontanel (infants).

- Record intake and output accurately
- Advance diet as tolerated, according to the child age, and surgeon's direction.
 - But before starting fluid diet check bowel sound.
- Prevent infection by:
 - Isolating from severe infections
 - Change the position frequent (every 2 – 3 hours), and
 - Encourage the patient to cough or deep breath or

Cont...

- Let infant cry for short period of time. Keep operative site clean: change dressing as needed and keep nappy away from wound.
- Providing general hygiene, skin care, proper rest and sleep,
- allow child exercise and movement out of bed when feels better.
- Offer the child measure of comfort:
 - keep warm, provide mouth care, frequent position change and holding infants.

- 
- Provide emotional support and psychological security to the child (let with parents)
 - Continue support and teaching for parents to maintain family relationship and to prepare for discharge:
 - any special procedures,
 - determine limits of activity for the child,
 - make follow up appointment.

CHAPTER TWO

Pediatric Health Assessment

❖ Presentation out line

- New born assessment
 - Essential New born Care
 - Pediatric Assessment
 - Pediatrics emergency
 - Emergency Triage Assessment and Treatment

New born assessment

➤ Objectives

At the end of this lesson students be able to:-

- ✓ Define the essential newborn care
- ✓ Describe APGAR score and its components
- ✓ Explain methods of newborn baby classification
- ✓ Classify newborn

What is involved in the immediate care of a newborn?

- Immediate care involves:
 - ✓ Drying the baby with warm towels or cloths, while being placed on the mother's abdomen or in her arms. ...
 - ✓ The baby's overall condition is recorded at 1 minute and at 5 minutes after birth using the Apgar Scale .
 - ✓ Vitamin K is also administered to prevent hemorrhagic disease of the newborn .

Essential Newborn Care(ENC)

- ENC is care that every newborn baby needs regardless of where it is born or its size.
- ENC should be applied immediately after the baby is born and continued for at least the first 7 days after birth.
- The main nursing goal for newborn baby care is the **promotion** and maintenance of **homeostasis**.

Steps ENC

1. Dry and stimulate
2. Evaluate breathing
3. Cord care
4. Keep the newborn warm(prevent hypothermia)
5. Initiate breast feeding with 1 hr
6. Eye care
7. Administer vitamin K
8. Apply Chlorohexidine on the cord daily for 7 days
9. Place identification band
10. Weigh the newborn when stable and warm
11. Recording all observations and treatments

Steps of Essential Newborn Care (ENC)

Step 1. Dry and stimulate

- Immediately dry the whole body including the head and limbs.
- Keep the newborn warm by placing on the abdomen of the mother
- Stimulate by rubbing the back or Slapping or flicking the soles of the feet

Cont...

- Remove the wet towel
- Let the baby stay in skin-to-skin contact on the abdomen and cover the baby quickly, including the head with a clean dry cloth.
- Don't let the baby remain wet, as this will cool the body and make it hypothermic

Step 2. Evaluate Breathing

- Check if the baby is crying while drying it
- If the baby does not cry, see if the baby is breathing properly
- If the baby is not breathing and/or is gasping: **Call for help**
- The assistant can provide basic care for the mother while you provide the more specialized care for the baby who is not breathing

Cont...

- Cut the cord rapidly and start resuscitation.
- If the baby breathes well, continue routine essential newborn care.
- Do not do suction of the mouth and nose *as a routine*.
- Do it only if there is:-
Meconium, thick mucus, or blood.

Normal breathing

- Normal breathing rate in a newborn baby is 30 to 60 breaths per minute.
- The baby should not have any chest in-drawing or grunting.
- Small babies (less than 2.5 kg at birth or born before 37 weeks gestation) may have some mild chest in-drawing and may periodically stop breathing for a few seconds.

Step 3. Cord care

- **Clamping /tying the cord:** If the baby does not need resuscitation, wait for cord pulsations to cease or approximately 1-3 minutes after birth, whichever comes first.
- Then place one metal clamp /cord tie 2 centimeters from the baby's abdomen and the second clamp / tie another 2 centimeters from the first clamp/tie
- Cutting the cord soon after birth can decrease the amount of blood that is transfused to the baby from the placenta and, in preterm

Cont...

- It is likely to result in subsequent anemia and increased chances of needing a blood transfusion
- **Cutting the cord:** Cut the cord with sterile scissors or surgical blade, under a piece of gauze in order to avoid splashing of blood
- At every delivery, a clean separate pair of scissors or blade should be designated for this purpose
- Counseling on cord care and apply 4% of chlorhexidine daily for 7 days.

Cont...

- Check for bleeding/oozing and retie if necessary
- The cord may be tied by using sterile cotton ties, elastic bands, or pre-sterilized disposable cord tie.
- Advise the mother not to cover the cord with the diaper.
- Don't use bandages as it may delay healing and introduce infection.

Cont...

- Don't use alcohol for cleansing as it may delay healing
- Don't apply traditional remedies to the cord as it may cause tetanus and other infections
- **Watch out for:**
 - Pus discharge from the cord stump
 - Redness around the cord especially if there is swelling
 - Fever (temperature more than 38°C) or other signs of infection

Step 4. Keep the newborn warm (Prevent Hypothermia)

- Keep the baby warm by placing it in skin-to-skin contact on the mother's chest
- Cover the baby's body and head with clean cloth
- If the room is cool ($<25\text{ }^{\circ}\text{C}$), use a blanket to cover the baby over the mother

Step 5. Initiate breastfeeding in the first one hour

- Early breastfeeding reduces the risk of postpartum hemorrhage for the mother.
- **Colostrum** (the "**first milk**") has many benefits for the baby, especially anti-infective properties.
- Skin to skin contact while feeding helps the baby to stay warm.
- Breastfeeding delays the mother's return to fertility because of lactation.
- Breastfeeding provides the best possible nutrition for the baby.

Cont...

- Feed day and night, at least 8 times in 24 hours, allowing on-demand sucking by the baby.
- If the baby is small (less than 2,500 grams), wake the baby to feed every 3 hours.
- If the baby is not feeding well, seek help.
- Successful breastfeeding requires support for the mother from the family and health institutions.
- There is no need for extra bottle feeds or water for normal babies, even in hot climates
- Avoid the use of the bottles and pacifiers

Step 6. Administer eye drops/eye ointment

- Wash your hands with soap and water
- Clean eyes immediately after birth with swab soaked in sterile water, using separate swab for each eye.
 - ▶ Clean from medial to lateral side.
- Give tetracycline eye ointment/drops within 1 hour of birth usually after initiating breast feeding.
- Don't put anything else in baby's eyes as it can cause infection.
- **Watch out for discharge** from the eyes, especially with redness and swelling around the eyes

Step 7. Administer vitamin K Intramuscularly (IM).

- 1 mg for babies with gestational age of 34 weeks or above
- 0.5 mg for premature babies less than 34 weeks gestation

Step 8. Place the newborn's identification bands on the wrist and ankle

- Putting the identification bands on the hands and ankle will save you from misshaping babies in busy delivery rooms.

Step 9. Weight the newborn when it is stable and warm

- Place a clean linen or paper on the pan of the weighing scale.
- Adjust the pointer to zero on the scale with the linen/paper on the pan.
- Place the naked baby on the paper/linen. If the linen is large, cover the baby with the cloth.
- Note the weight of the baby when the scale stops moving.
- Never leave the baby unattended on the scale.
- Record the baby 's weight in partographs/maternal/ newborn charts and delivery room
- Register and inform the mother

Step 10. Record all observations and treatment provided in the registers/appropriate chart/cards

- **Note** Defer the bath for at least 24 hours.
- Clean the newborn of an HIV-infected mother as recommended
- Organize transport if necessary
- Inform the mother of the newborn's weight

Summery

● These are steps of ENC:

- ▶ Dry and stimulate
- ▶ Evaluate Breathing
- ▶ Cord care
- ▶ Keep the newborn warm (Prevent Hypothermia)
- ▶ Initiate breastfeeding in the first one hour
- ▶ Administer eye drops/eye ointment
- ▶ Administer vitamin K Intramuscularly (IM) .
- ▶ Place the newborn's identification bands on the wrist and ankle.
- ▶ Anthropometry measurement (Weight, height measurement)

Assessment of the newborn

Assessment of newborn can be divided into four phases:

1. *The initial assessment /APGAR scoring system/.*
2. *Transitional assessment (6-8 hours after birth)*
3. *Assessment of gestational age (pre-term assessment)*
4. *Systematic physical assessment (general assessment)*

A. The initial assessment:

This occur along activities that included in- immediate care of the newborn:

- ✓ **Maintain patent air way** (*Positioning, Suctioning*)
- ✓ **Care of the cord** (*tie, GV application*)
- ✓ **Maintain stable body temperature** (*Drying, cloth*)
- ✓ **Eye Care** (*eye ointment*)

Initial assessment...

- ✓ **Evaluate the Baby** (APGAR score, Breathing, Skin Color :
 - ❖ (**Acrocyanosis** is normal for a newborn during the first few hours, but not Central cyanosis) and Heart rate (Normal HR is over 100 BPM.
- Use Pediatric stethoscopes))

Cont..

- ❖ **APGAR scoring:** is a commonly-used method to quantitatively evaluate the newborn's adjustment for extra uterine environment.

The score is based on observation of the chx:

- ❖ ***A-Appearance/color/- peripheral tissue oxygenation***
- ❖ ***P-Pulse/heart rate/- status of asphyxia***
- ❖ ***G-Grimace/Response to stimuli/- response***
- ❖ ***A-Activity/Muscle tone/- Degree of muscle strength***
- ❖ ***R-Respiratory effort- adequacy of ventilation***

- ❖ Usually evaluated at **1 and 5-minutes** after birth.
- ❖ N.B. A **1min** APGAR score might signal the need for *resuscitation* and the 5 min indicate the probability of *successful resuscitation*.
- ❖ Done by assigning points from 0 to 2 for each category.

APGAR Score

Sign	0	1	2
Heart rate	Absent	Slow <100	Over 100
Respiratory	Absent	Weakly	Strong cry
Muscle tone	Flaccid	Some flexion of extremities	Well flexed
Reflex response	No response	Grimace Grimace	Cough or sneeze, Cry
Color	Blue, pale	Body pink extremity blue	Completely pink

Key for scoring

- ❖ $\geq 7/10 \rightarrow$ Absence of difficulty /***no asphyxia***/
- ❖ $6/10 \rightarrow$ mild asphyxiated = needs follow up
- ❖ $4-5/10 \rightarrow$ Moderately asphyxiated = Needs clearing and oxygen.
- ❖ $0-3 \rightarrow$ ***severely asphyxiated/in distress***/ = needs resuscitation.

Neonatal resuscitation

- **Birth asphyxia** is defined simply as the **failure to initiate and sustain breathing at birth**, causes lack of oxygen and unconsciousness.

The common worry of health professionals and parents is the permanent brain damage that birth asphyxia can cause.

Management of baby with birth asphyxia includes ABC's of Resuscitation

ABC...

A - Establish open airway- position, suction

B - Initiate breathing- tactile stimulation, oxygen

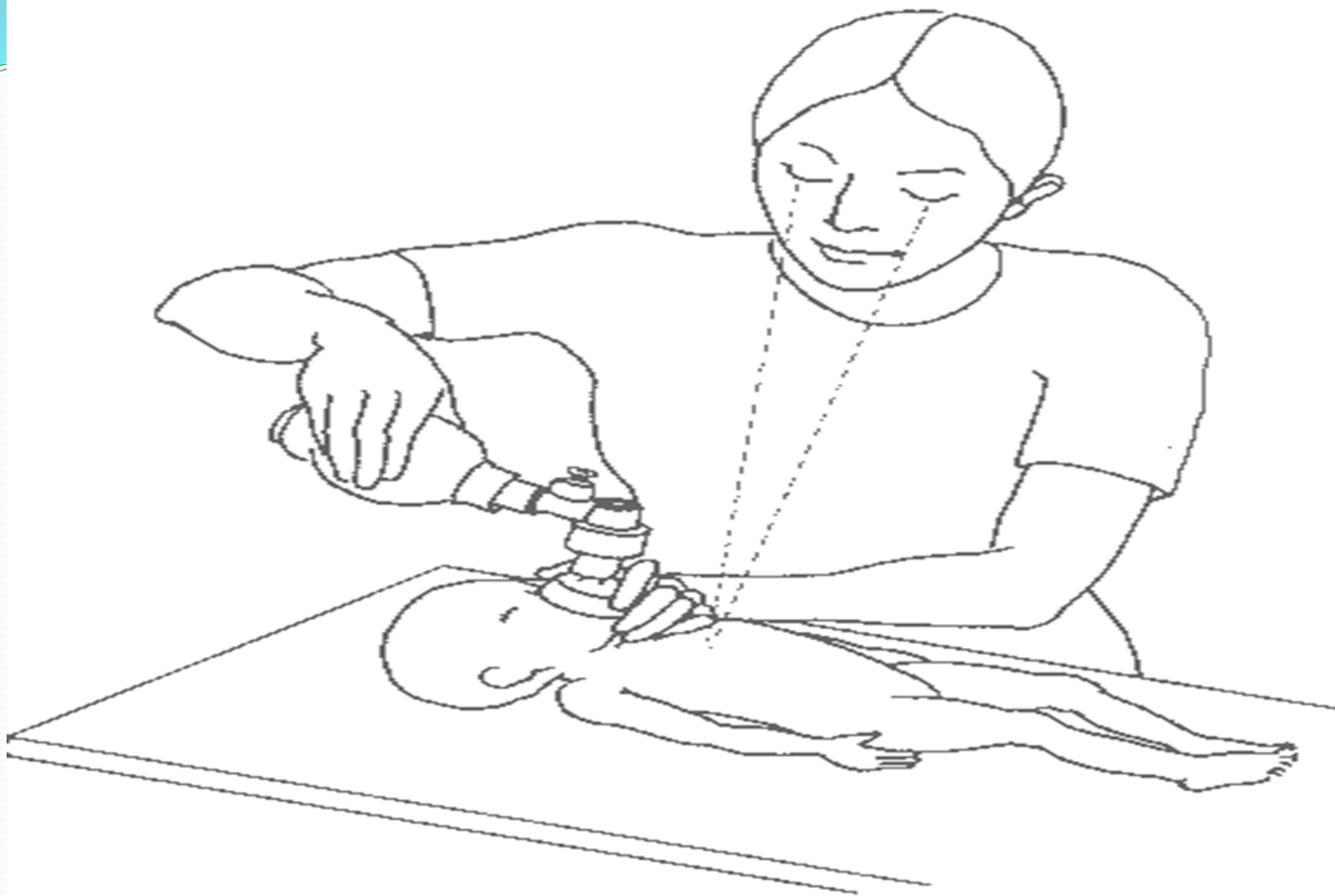
C -Maintain circulation- chest compressions

Critical for the success of resuscitation are:

❖ *Anticipation*

❖ *Adequate preparation*

❖ *Timely recognition*



- ❖ Quick and correct action If the newborn is not breathing or gasping: immediately start basic resuscitation.

Open the airway:

- ❖ Put the baby on its back: on warm, flat, firm and clean surface .
- ❖ Position the head so that it is slightly extended
- ❖ Suction- if any fluid

Positive pressure ventilation: Most important resuscitation for ensuring ventilation and initiation of breathing.

- Can be initiated using ***bag - mask with room air.***
- When **no** equipment is available: ***mouth to mouth***

Adequacy of ventilation is assessed by observing the chest movements.

A small proportion of infants fail to respond, when it does, advanced resuscitation must be taken.

Advanced resuscitation includes:

- ❖ **Endo-tracheal intubation,**
- ✓ Oxygen administration,
- ✓ Chest compression, and Drug use.
- ✓ Endo-tracheal intubation:
- ❖ **Provide effective ventilation .**
- ❖ **Convenient for prolonged resuscitation**
- ❖ **Since complicated process,** it requires good training.
- ❖ **Oxygen:** Additional oxygen is not necessary for basic resuscitation.
- ❖ However, when the newborn's color does not improve despite effective ventilation, oxygen should be given if available.

An increased concentration of oxygen is needed for:

- Meconium aspiration
- Immature lung,
- When the baby does not become pink despite adequate ventilation.
- Chest compressions
- Not recommended in basic newborn resuscitation.

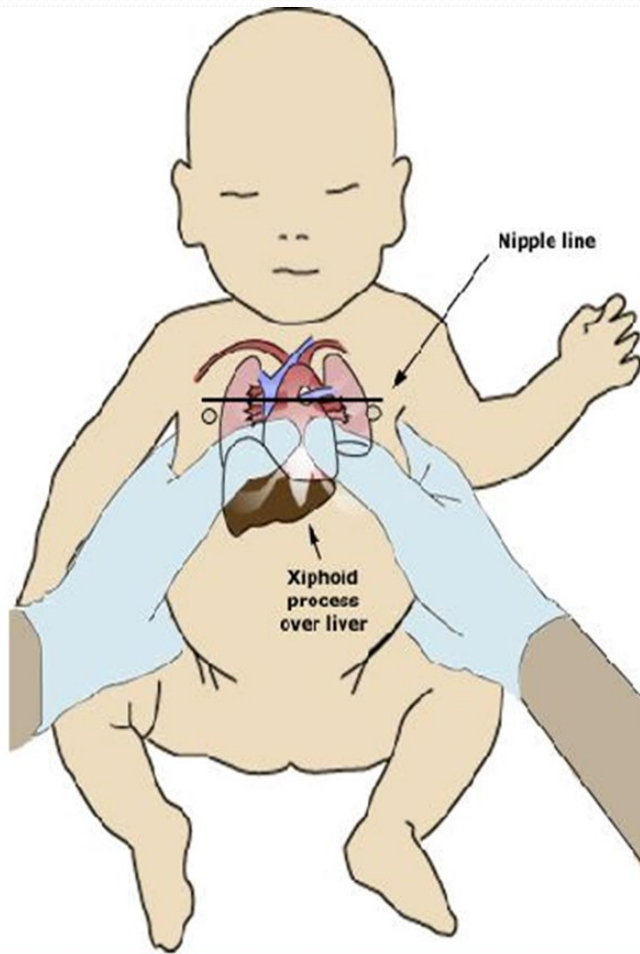
Oxygen can be administered if the heart rate is absent or <60 bpm despite adequate ventilation for 30 seconds

The 2-thumb-encircling-hands method is preferred, with a depth of compression $\frac{1}{3}$ the AP diameter of the chest, compared to the two-finger method of compressing the sternum.

Two people are needed for effective chest.

Compression techniques of chest

Thumb technique



Two-finger technique



Drugs

Drugs are seldom needed to:

- Stimulate the heart.
- Increase tissue perfusion
- Restore acid-base balance.
- Required in newborns who do not respond to adequate ventilation with 100% oxygen and chest compressions.

Cont...

Epinephrine:

Epinephrine (0.1-0.3 mL/kg of 1:10,000 solution) should be administered if the heart rate remains <60 bpm after a minimum of 30 seconds of adequate ventilation and chest compressions.

relax the airways, constrict blood vessels- increase pumping power of the heart.

B. Transitional assessment

A Period takes place 6-8 hours after birth.

It has three stages:

❖ **First stage:** is during the first 30 minutes; which is initial stage of alertness and activity.

Changes- like cry, eyes open, vital signs = RR- 80 breath/min, HR-180 beat/min, temp may decrease, active bowel sound, and mucus secretions are increased.

❖ **Second stage:** Lasts to 2-4 hours and infant is in state of sleep and relative calm.

Changes: Heart and respiration rate decrease, temperature continues to fall.

❖ **Third stage:** Lasts about 2-5 hours; which is period of reactivity when the infant awakes from deep sleep,

Changes: Stabilization of physiologic systems continued

C. Gestational assessment

Two methods can be used:

- ❖ **Physical maturity:** (Skin, Lanugo, Plantar Surface, Breast, Eye/Ear, Genitals)
- ❖ **Neuromuscular maturity:** (Posture, Square Window, Arm Recoil, Popliteal Angle, Scarf Sign, and Heel to Ear).
- The score is from -1 up to 5.

Cont...

- The relation is $-5 (-10) \text{ score} = 20\text{wk}(22)$,
increase by 2wk for every increase in 5 score
- The cumulative score is -5 up to 55 relates with
a gestational age from 20 to 44 weeks of
gestation.
- Physical assessment
- Every system will be assessed

Neonatal Classification

- A neonate is a newborn of 28 days old or less.
- The neonatal period is the time between birth and the 28th day.
- There are several classifications of neonates based on different criteria

A. Based on the neonatal period

- Very Early (birth – 24 hrs)
- Early (1 day – 7 days)
- Late (7 days – 28 days)

B. Based on Gestation age

- Preterm (<37 weeks)
- Term (37 – 42 weeks)
- Post-term (>42 weeks)

C. Based on Birth Weight

- Normal birth weight (2500 – 4000 gm)
- Low birth weight (1500 – 2500 gm)
- Very low birth weight (<1500 gm)
- Extreme Low Birth Weight (ELBW) - born at less than 1000g/1Kg.

D. Based on weight to Gestational age

- AGA- if the birth weight lies between 10 to 90 % for the specified GA.
- LGA – if the weight lies above 90% for the specified GA
- SGA –if the weight lies below 10% for the specified GA.

Physiological Response of Newborn to Birth

Neonatal Transition

- ✓ Surfactant production
- ✓ Breathing initiated by mechanical, sensory, chemical, thermal changes.
- ✓ Mechanical- pressure from chest compression, recoil
- ✓ Remaining fluid pulled into interstitial fluid
- ✓ Problems r/t inadequate chest compression

Chemical Stimuli

- Fetal acidosis, decrease in PO_2 , increase PCO_2
- Cord clamped, stress of labor
- Stimulation of medulla

Thermal and Sensory

- ✓ Shock of cold environment
- ✓ Nerve ending stimulated
- ✓ Cold stress
- ✓ Tactile, auditory and visual stimulation
- ✓ Encourage skin to skin contact

Cardiopulmonary Adaptation

- ✓ Blood flow to lungs established
- ✓ Oxygen saturation
- ✓ Newborn anatomy restricts lung expansion
- ✓ Periodic breathing
- ✓ Obligate nose breathers

Cardiopulmonary Adaptation

- ✓ Clamping of cord, shut off the hose, increase in blood pressure
- ✓ Perfusion increased by pulmonary blood flow and dilation pulmonary vessels.
- ✓ Decrease in vascular resistance
- ✓ Pressure change closes foramen ovale
- ✓ Ductus arteriosus closes due to PO_2
- ✓ Ductus venosus closure leads to liver profuse

Cardiac Function

- ✓ Normal rate
- ✓ BP
- ✓ Most murmurs transient and benign
- ✓ Right ventricle versus left ventricle
- ✓ Pressure gradient changes from left to right

Hemodynamics

- ✓ RBC shorter life span
- ✓ Rise in HCT
- ✓ Stress response
- ✓ Cord clamping
- ✓ Gestational age
- ✓ Presence of hemorrhage
- ✓ Site of blood sample

Temperature

- ✓ Metabolism and O₂ consumption increase with heat loss
- ✓ Thin skin, decreased fat
- ✓ Blood vessels close to surface
- ✓ Flexed posture
- ✓ Premature infants

Heat Loss

- ✓ Large body surface Vessels close to skin
- ✓ **Convection**- cool air currents, door open
- ✓ **Radiation**- heat transfers to cooler surface, place cool object on warmer
- ✓ **Evaporation**-water converted to vapor, wet baby
- ✓ **Conduction**- loss of heat due to direct contact with object

Thermogenesis

- ✓ Increase in BMR, and activity generate heat
- ✓ Non shivering thermogenesis (NST)
- ✓ Infant uses stores of brown fat
- ✓ Brown fat metabolized quickly to produce heat
- ✓ Do not chill newborns
- ✓ Cold stress can delay drug metabolism

Hepatic Adaptation

- ✓ Iron stored in fetal liver, last for 6 months
- ✓ Energy crunch caused by labor and loss of maternal glucose
- ✓ Newborn converts from use of carbohydrate metabolism to fat metabolism

Conjugation of Bilirubin

- ✓ Bilirubin is by product of break down of RBCs
- ✓ In utero bilirubin excreted by placenta
- ✓ Bilirubin needs to be conjugated to be excreted.
- ✓ Enzymes in liver conjugate bilirubin- bacteria transforms into urobilirubin
- ✓ Low levels of glucuronyl transferase and immature liver function

Physiologic Jaundice

- ✓ Normal response 2-3 days after birth
- ✓ Caused by increase volume and RBC destruction
- ✓ Bilirubin not flushed from plasma
- ✓ Bilirubin not conjugated
- ✓ Decreased bacterial flora and motility

Nursing

- ✓ Room environment-avoid pink
- ✓ Head to toe, blanch and assess for yellow
- ✓ Maintain temp
- ✓ Monitor for excretion
- ✓ Feed early
- ✓ Phototherapy for newborns over 13mg/dl

Breastfeeding Jaundice

- ✓ Peaks in 2-3 weeks
- ✓ Composition of breast milk may interfere with conjugation
- ✓ If above 20mg.dl may be asked to cease
- ✓ Continue to pump
- ✓ Reassure moms

Coagulation

- ✓ Some coagulation factors are Vit. K dependent
- ✓ At birth bacteria in colon not present for Vit. K synthesis
- ✓ Dilantin and Coumadin associated with bleeding issues

GI

- ✓ Lactose(carb.),proteins easily digested
- ✓ Starches not easily digested
- ✓ Lack of pancreatic enzyme limits fat digest.
- ✓ Some regurge nl, burp do not overfeed
- ✓ Need 120 cal/kg/day, 5-10 weight loss.
- ✓ Meconium- debris, dark, tarry
- ✓ Differentiate breast feed from bottle fed

Kidney Fx

- ✓ Inability to concentrate urine
- ✓ Decrease in GFR, unable to diuresis quickly
- ✓ Most void within 24 hrs., 6/day
- ✓ Increase 5-25/day after 2 days
- ✓ Blood in female diaper due to pseudo menstruation

Immunity

- ✓ Immunoglobulin IgG transferred to fetus, passive acquired immunity
- ✓ Length of immunity to bacteria and virus vary
- ✓ Lack of IgM, fetus susceptible to gram -
- ✓ IgA protects, gi, int, eyes, high concentration in colostrum

Neurological Fx

- ✓ Able to habituate
- ✓ Able to fixate on faces or objects with contrast
- ✓ Blinking reflex
- ✓ Growth is cephalocaudal
- ✓ Reflexes present
- ✓ Defensive motor ability

Triage

Objective:

At the end of this session, learners will be able to:

- Define what triage means
- Demonstrate triage of pediatric problems
- Identify emergency signs and priority signs
- Manage emergency problems of children via ABCD

Triage

❖ What is triage?

➤ Triage, from the French word “trier”, literally means:

“to sort”

- The aim depends on the situation of the event:

1. Severity

2. number of victims/medical or trauma

3. and capacity of the facility or the state

Case scenario

- ❖ A four-year old girl is carried in her mother's arms wrapped in a blanket, in the queue. Her airway and breathing are normal. She has cold hands. Her capillary refill is 1.5 seconds. She is alert. Asked if the child has had diarrhoea, the mother answered "YES. Four loose stools per day". The skin pinch takes 3 seconds. How do you triage this child?

Cont...Triage

- Triage is putting the patient in the right place at the right time to receive the right level of care, the allocation of appropriate resources to meet the patient's medical needs.
- Triage is the process of rapid assessment of a patient with a view to define urgency of care & priorities in treatment.
- It also allows for the allocation of the patient to the most appropriate assessment and treatment area .
- The triage system is based on available medical services, community need and load of emergency departments.

Literatures in Ethiopia

- A study in Woliata Ethiopia (*Gargamo DB et al, 2019*) about "Quality of Pediatrics Emergency Triage" indicated that 41.7% of HCWs did not correctly define triage, 81% did not know triage duration, 85.7% did not identify all triage categories and 64% did not categorize child with urgent signs. Conclusion: The overall quality of pediatric emergency triage service was poor.
- Poor quality of pediatric emergency triage: delay in child triage, no adherence to national guidelines, and low level of triage knowledge, low confidence and lack of basic triage infrastructure.

Cont...

- A study in Addis Ababa Ethiopia (Kerie et al, 2018) revealed Above half of the participants {52.9%) had a moderate level of triage skill.
- ✓ A strong positive relationship was found between nurses' level of triage knowledge and skill.
- ✓ Knowledge about triage, educational level and training experience had a significant relationship with triage skill.
- ✓ This study revealed that most triage nurses had a moderate level of skills. Therefore, the ministry of health and hospitals should provide training and education to improve triage skill

➤ When should Triage take place? ^{Cont...}

For a child who does not have emergency signs, it takes on average 20 seconds.

- As soon as the sick child arrives at hospital.
- Before any administrative procedure such as registration.

➤ Where should Triaging be done?

- Can be carried out anywhere.
- At Emergency Room, at OPD, at ICU or at ward.

Cont..

➤ Who should triage?

- All the staff working in a health care facility should be trained to carry out rapid assessment of sick child and triage

Triage is needed to place sick children in one of the following categories:

1. **EMERGENCY SIGNS** : If you find any emergency signs, do the following immediately
 - Start to give appropriate emergency treatment.
 - Call a senior health worker and other health workers to help.

Cont...

- Carry out emergency laboratory investigations.

2. PRIORITY SIGNS: they should be given priority in the queue, so that they can rapidly be assessed and treated without delay.

3. NONURGENT (queue) cases: These children can wait their turn in the queue for assessment and treatment.

The ABCD concept

❖ Triage of patients involves looking for signs of serious illness or injury. These emergency signs relate to the Airway-Breathing Circulation/Consciousness, and Dehydration ("ABCD")

A, B

- Airway
- Breathing

C

- Circulation
- Coma, Convulsion

D

- Dehydration (severe)

Priority signs

- There are priority signs, which should alert you to a child who needs prompt, but not emergency assessment.
- **Tiny baby:** any sick child aged **under two months**
- **Temperature:** child is very hot
- **Trauma** or other urgent surgical condition
- **Pallor (severe)**
- **Poisoning**
- **Pain (severe)**

Priority signs

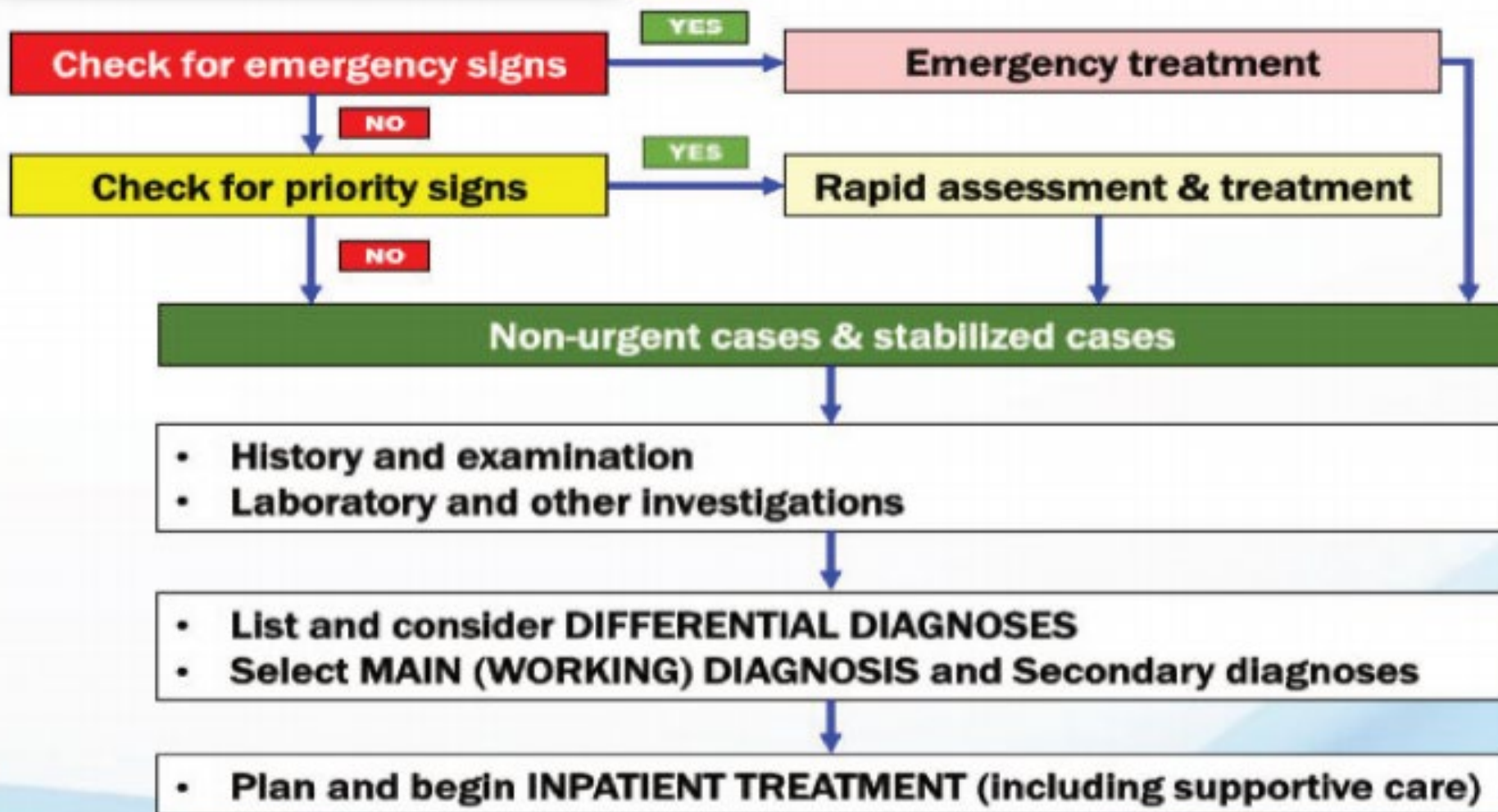
- **Respiratory distress**
- **Restless,** continuously irritable, or lethargic
- **Referral (urgent)**
- **Malnutrition:** Visible severe wasting, Oedema of both feet
- **Burns**

The triage process

STEPS IN THE MANAGEMENT OF THE SICK CHILD ADMITTED TO HOSPITAL

TRIAGE

Triage is the first step in the management of a sick child admitted to a hospital.



Cont...

INPATIENT TREATMENT

```
graph TD; A[INPATIENT TREATMENT] --> B[Monitor for<br/>• Response to treatment<br/>• Complications]; B --> C[IMPROVING]; B --> D[NOT IMPROVING OR<br/>NEW PROBLEM]; C --> E[• Continue treatment<br/>• Plan for discharge]; D --> F[• Revise treatment<br/>• Treat complications<br/>• Refer if not possible];
```

Monitor for

- Response to treatment
- Complications

IMPROVING

- Continue treatment
- Plan for discharge

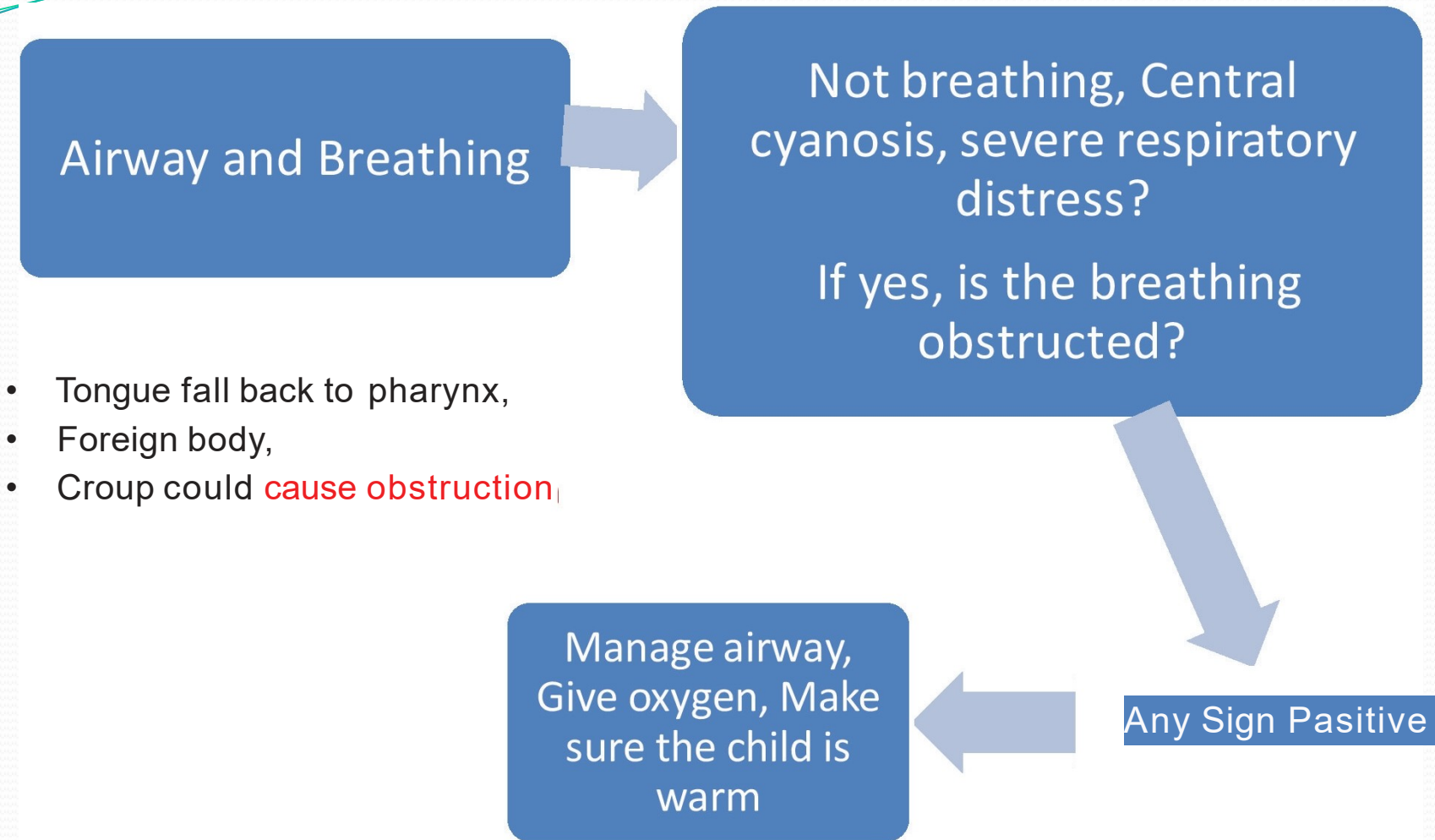
**NOT IMPROVING OR
NEW PROBLEM**

- Revise treatment
- Treat complications
- Refer if not possible

SUMMARY

- Triage is sorting of patients into priority groups according to their need.
- With practice, a complete triage (if no emergency treatment is needed) takes less than a minute.
- All tiny babies of under two months should be seen as a priority.

Emergency management



NB. Check for **head/neck trauma** before treating the child; do not move neck if cervical spine injury is possible.

Management of the airway

MANAGEMENT OF THE CHOKING CHILD

- A child with a history of aspiration of a foreign body who shows increasing respiratory distress is in immediate danger of choking.
- Attempts to remove the foreign body should be made instantly.
- Apply back slaps or Heimlich manoeuvre (a first-aid procedure for dislodging an obstruction from a person's windpipe in which a sudden strong pressure is applied on their abdomen, b/n the navel & the ribcage).
- The treatment differs depending on whether there is a foreign body causing respiratory obstruction or some other cause for the obstruction or respiratory distress.

Cont...

- If a **foreign body** is causing the obstruction, the treatment depends on the **age** of the child.

➤ **Management of young infant**

- Lay the infant on your arm or thigh in a head down position
- Give 5 blows to the infant's back with heel of hand
- If obstruction persists, turn infant over and give 5 chest thrusts with 2 fingers, one finger breadth below nipple level in midline

- 
- If obstruction persists, check infant's mouth for any obstruction which can be removed
 - If necessary, repeat sequence with back slaps again

Cont...

back slaps



Chest thrusts



Management of young infant

6/10/2022

Management of the child

- Give 5 blows to the child's back with heel of hand with child sitting, kneeling or lying
- If the obstruction persists, go behind the child and pass your arms around the child's body; form a fist with one hand immediately below the child's sternum; place the other hand over the fist and pull upwards into the abdomen; repeat this Heimlich manoeuvre 5 times

Cont...

- If the obstruction persists, check the child's mouth for any obstruction which can be removed
- If necessary, repeat this sequence with backslaps again

Cont...

**Slapping the back to clear
airway obstruction in a
choking child**



**Heimlich
manoeuvre in a
choking older
child**



Management of child

Cont....

POSITIONING TO IMPROVE THE AIRWAY

- To treat an airway or breathing problem you should **first open the airway** and then begin giving the child **oxygen**.
- **Neutral position**: place your hand on the child's forehead and apply a little pressure to achieve the tilt. The fingers of the other hand are used to gently **lift the chin**

Cont...

- Know the child's age because you will position an infant (under 12 months of age) differently from a child.



Neutral position to open airway in an infant
("nose up")



Sniffing position to open up airway in an older
child ("chin up")

Cont...

- **NB - Always** ask and check for head or neck trauma for **any child with an emergency sign** who needs emergency treatment before treating them.

To check for **head or neck** trauma:

- Ask if the child has had trauma to the head or neck, or a fall which could have damaged the spine
- Look for bruises or other signs of head or neck trauma & Stabilize the neck if trauma is suspect

Cont...

- To open and manage the airway when **head trauma** is suspected a **jaw thrust** is used. This is a way of opening the airway **without moving the head.** It is safe to use in cases of trauma for children of **all ages.** The jaw thrust is achieved by placing two or three fingers under the angle of the jaw on both sides, and lifting the-



Jaw thrust without head tilt

Assessment of breathing

- Look, Listen, Feel



Look, Listen, Feel for breathing

Cont...

- If no breathing, start **resuscitation.**

DOES THE CHILD SHOW CENTRAL CYANOSIS?

- A bluish or purplish discoloration of the tongue and the inside of the mouth indicates central cyanosis.



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DOES THE CHILD HAVE SEVERE RESPIRATORY DISTRESS?

Signs of severe respiratory distress

- Very fast breathing
- Severe lower chest wall indrawing
- Use of auxiliary muscles
- Head nodding or bob with every inspiration
- Inability to feed because of respiratory problems

Summary

- In all cases of airway or breathing problems:
Give oxygen: 0.5 to 1 litre/min (<1-year-olds)
and
1 to 2 litres/minute for (older children).

Circulation

To assess the circulation, you need to know:

1. Does the child have warm hands?
2. If not, is the capillary refill more than 3 seconds?
3. And is the pulse weak and fast?

In other words, is the child shocked?

NB. It is not recommended blood pressure to assess for shock because low blood pressure is a late sign in children and may not help identify treatable cases and the BP cuff necessary in children of different age groups is mostly unavailable in many district hospitals.

Assessment and management of shock

- If the child's hands are warm, there is no problem with the circulation and you can move to the next assessment.
- If Cold hand with: []Capillary refill longer than 3 seconds, and Weak and fast pulse, Check for severe malnutrition

Stop any bleeding, Give oxygen, Make sure the child is warm.

Treatment of shock

❖ If the child with shock has **NO severe malnutrition**,

1. Infuse 20 ml/kg as rapidly as possible.

➤ Improvement: warmer hands, pulse slows and capillary refill faster.

2. If there is NO improvement: Give another 20 ml/kg of Ringer's lactate or normal saline as quickly as possible.

Reassess the circulation again, and

Cont..

3. If there is still no improvement. Give another 20 ml/kg of Ringer's lactate or normal saline, as quickly as possible. The circulation should be assessed again.

4. If there is still NO improvement: give **20 ml/kg of blood over 30 minutes unless there is profuse watery diarrhoea.**

In this case, repeat Ringer's lactate. The circulation should be assessed again.

5. If there is still NO improvement: treat underlying condition

- ❖ If the child with shock has **severe malnutrition**,
- **Avoid IV**, find out if the child can drink or use a nasogastric tube (NGT).

Weigh the child.

- **Give ReSoMal** rehydration fluid orally or by NGT: -
5 ml/kg every 30 min for 2 hours, then - 5-10
ml/kg/hour for 4-10 hours, or - give half strength
normal saline

Cont.

- (or half strength Darrows with 5% glucose) with 5% glucose at 15 ml/kg give over 1 hour.
- Stay with the child and check the pulse and breathing rate **every 5-10 minutes.**
- Discontinue the intravenous infusion if either of these increase (pulse by 15, respiratory rate by 5/ min).

➤ **If there is improvement:**

- pulse and breathing rate fall. Repeat 15ml/kg over 1 hour.
- Switch to oral or NGT rehydration with ReSoMal 10ml/kg/hour.

➤ **If there is NO improvement:**

- Call for help other health worker. give maintenance IV fluid 4ml/kg/hour while waiting for blood.
- **Transfuse fresh whole blood at 10ml/kg/hour slowly over 3 hours** (use packed cells if in cardiac failure).

Cont...

Maintenance fluids

if the child has NO severe malnutrition
See Chart 11

Give 70 ml/kg of Ringer's lactate solution (or, if not available, normal saline)

- over 5 hours in infants (aged <12 months);
- over 2½ hours in children (aged 12 months to 5 years).

Reassess the child every 1-2 hours. If the condition is not improving, give the IV fluids more rapidly.

Also give ORS solution (about 5 ml/kg/hour) as soon as the child can drink; this is usually:

- after 3-4 hours (in infants);
- after 1-2 hours (in children).

Encourage breastfeeding:

Reassess after 6 hours (infants) and after 3 hours (children)

If the child HAS severe malnutrition

Continue ReSoMal 5-10ml/kg/hour for the next 4-10 hours.

See guidelines in the "Pocket book of hospital care for children" or the manual "Management of the child with a serious infection or severe malnutrition."

Summary

- If the child HAS severe malnutrition,
Assess if _____ child can
drink oral or NGT fluids or IV fluids if child
unable to tolerate oral or fluids by
nasogastric tube

Coma and convulsion

- A child who is awake and alert, or is playing and talking, obviously does not have a dangerous or disturbed **level of consciousness**
- AVPU=
 - A=Is the child Alert? If not,
 - V=Is the child responding to Voice? If not,
 - P=Is the child responding to Pain?
 - U=The child who is Unresponsive to voice (or being shaken) AND to pain is Unconscious.

Cont...

- If the child is unconscious (coma), Manage the airway,

Position the child (if there is a history of trauma, stabilize neck first), check the blood sugar, and give IV glucose.

- If the child is convulsing now, Manage the airway, position the child, Check blood sugar, Give IV glucose, and Give anticonvulsant



Positioning the unconscious child (recovery position)

Cont....

- Hypoglycaemia is present if the measured blood glucose level is low <2.5 mmol/l (**45 mg/dl**) in a well nourished or <3 mmol/litre (**55 mg/dl**) in a severely malnourished child).
- When the blood glucose cannot be measured, hypoglycaemia should be assumed to be present in all children in coma or having a convulsion AND should be treated.

Cont...

Coma and convulsion: GIVE IV GLUCOSE

- Give 5 ml/kg of 10% glucose solution rapidly by IV injection
- Recheck the blood glucose in 30 minutes.
- If it is still low, repeat 5 ml/kg of 10% glucose solution.
- Feed the child as soon as conscious.

If the child is not able to feed without danger of aspiration, give:

- IV fluid containing 5-10% glucose (dextrose), or
- Milk or sugar solution via nasogastric tube.
- To make sugar solution, dissolve four level teaspoons of sugar (20 grams) in a 200-ml cup of clean water.

Cont...

Amount of glucose to give by age			
Age/weight	Volume of 10% glucose solution to be given as bolus (5 ml/kg)	To make up 10% glucose using 50% glucose solution	
		Volume of 50% glucose	With added volume of water
Less than 2 months (<4kg)	15 ml	3ml	12ml
2 - <4 months (4 - <6 kg)	25 ml	5ml	20ml
4 - <12 months (6 - <10 kg)	40 ml	8ml	32ml
1 - <3 years (10 - <14 kg)	60 ml	12ml	48ml
3 - <5 years (14 - <19 kg)	80 ml	16ml	64ml

Convulsion

- Diazepam is used to stop convulsion (anticonvulsant) for ≥ 2 weeks (0.1 to 0.3 mg/kg)
- Phenobarbital is the drug of first choice in infants < 2 weeks of age to control convulsions (15-20 mg/kg loading then 5 mg/kg/day 1-2 divided dose)
- Reassess the child after 10 minutes: If still convulsing, give a second dose of diazepam, rectally. If the convulsion continues in spite of this second dose, a third dose can be given.
- Diazepam can affect the child's breathing, so it is important to reassess the airway and breathing regularly.

❖ SEVERE DEHYDRATION (in child with diarrhoea only)

- Diarrhoea plus any two of these:

Lethargy, sunken eyes, very slow skin pinch

IF NO SEVERE MALNUTRITION: insert IV line and begin giving fluids rapidly following Plan C (For infants: - 30 ml/kg in the first hour - 70 ml/kg in the next 5 hours & for children > 1 year of age: - 30 ml/kg in the first 30 minutes - 70 ml/kg in the next 2.5 hours).

Cont..

- Assess every 1 to 2 hours.
- If the signs of dehydration are not improving: give fluid more rapidly and inform senior staff
- As soon as the child can drink: give oral fluids in addition to the drip, give ORS 5 ml/kg every hour

Maintenance fluids: severe DHN without severe malnutrition

- Give 70 ml/kg of Ringer's lactate solution (or, if not available, normal saline) over 5 hours in infants (aged <12 months); 1>vear 2½ hours in children (aged 12 months to 5 years).
- Reassess the child every 1-2 hours.
- If the condition is not improving, give the IV fluids more rapidly.
- Also give ORS solution (about 5 ml/kg/hour) as soon as the child can drink; this is usually after 3-4 hours (in infants); after 1-2 hours (in children). Encourage breastfeeding:
- Reassess after 6 hours (infants) and after 3 hours (children)

Cont...

- IF SEVERE MALNUTRITION: Do not insert IV, but proceed immediately to assessment and treatment.
- For all children: Give ReSoMal 5ml/kg every 30 minutes for the first 2hours. Then, 5-10ml/kg/hour for the next 4-10hours.
- Give more ReSoMal if child wants more or large stool loss or vomiting.
- Check blood glucose: treat if $<3\text{mmol/l}$
- Maintenance fluids: Severe DHN with severe malnutrition

ReSoMal – modified ORS

- **ReSoMal: Rehydration Solution for Malnutrition**
- **Preparation of ReSoMal from ORS**

- 1. Add two pkts of ORS in 2 litres of water (instead of 1 litre)*
- 2. Add 50gm (10 rounded 5mls teaspoon of sugar)*
- 3. Add 3 vials of the 10 ml vial of 15% KCL (20mmol/10ml) in the 2 liters*

Oral rehydration salt (ORS)

❖ Prepare at home from

1. One liter clean drinking or boiled and cooled water 5 capfuls (each cup about 200ml)
2. Six (6) level teaspoon of sugar
3. Half ($1/2$) level teaspoon of salt

Chapter Three

Growth and Development

➤ Objective

- At the end of this lesson you will be able to:
 - Define growth and development of child
 - Describe developmental stages of children
 - Describe growth monitoring methods

Definition

- **Growth:** Increase in physical size (process).
- **Development:**
 - It is change in shape, form or structure so as to improve
 - in working or functioning.
 - It is the process that
 - Turns babies into adults
 - Has stages
 - Begins at conception
 - Ends at death

Growth and Development

- Growth and development follows a predictable pattern
- Always is **cephal to caudal** and **proximal to distal**
- Affected by **biological, emotional, social** and **environmental** factors
- Varies based on the age of the child and is sequential.

Principles of Growth and Development

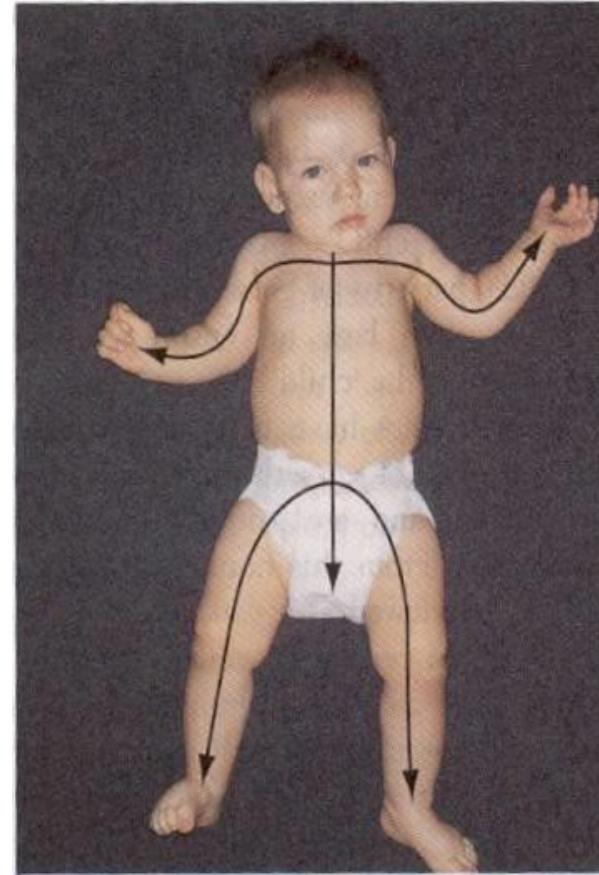
- Growth is an orderly process, occurring in systematic fashion.
- Rates and patterns of growth are specific to certain parts of the body.
- Wide individual differences exist in growth rates.
- Growth and development are influenced by a multiple factors.

Cont...

- Development proceeds from the simple to the complex and from the general to the specific.
- Development occurs in a **cephalocaudal** and a **proximodistal** progression.
- There are critical periods for growth and development.
- Rates in development vary.
- Development continues throughout the individual's life span.

Growth Pattern

- The child's pattern of growth is in:-
 - **Head-to-toe direction, or**
 - **Cephalocaudal, and**
 - In an inward to outward pattern called **proximodistal**.



Growth monitoring

- ❑ Growth: increase in physical size (process).
- ❑ Measurable parameters:
 - ❖ **Weight**
 - ❖ **Height**
 - ❖ **Head circumference**
 - ❖ **MUAC**
- ❑ Analysis of growth pattern: continuous process.
- ❑ Single measurements: screening purpose.

Growth monitoring ...

Rationale:

- ❑ Growth monitoring can greatly strengthen **preventive health programs**.
- ❑ Growth is **the best general index** of the health of an individual child.
- ❑ **Regular measurements of growth** permit the **early detection of malnutrition**.

Stages of Growth and Development

- **Infancy(Birth to 1 year)**
 - Neonate
 - Birth to 1 month
 - Post neonatal
 - 1 month to 1 year
- **Early Childhood (1yr to 6 yr)**
 - Toddler
 - 1-3 years
 - Preschool
 - 3-6 years
- **Middle Childhood (6yr to 12 yr)**
 - School age
 - 6 to 12 years
- **Late Childhood (13yr to 18 yr)**
 - Adolescent
 - 13 years to approximately 18 years

Normal growth

- Trend helps to define if growth is within acceptable limits (serial /single measurements)
- Most powerful tool in growth assessment is **growth chart**
- When caloric intake is inadequate, the wt percentile falls first, then the height and last the head circumference.

FIVE HEALTH BEHAVIOR THEORIES

- Freud: psychosexual_ body centered
- Erikson: psychosocial
- Piaget: cognitive
- Kohlberg: moral
- **Biopsychosocial Models**

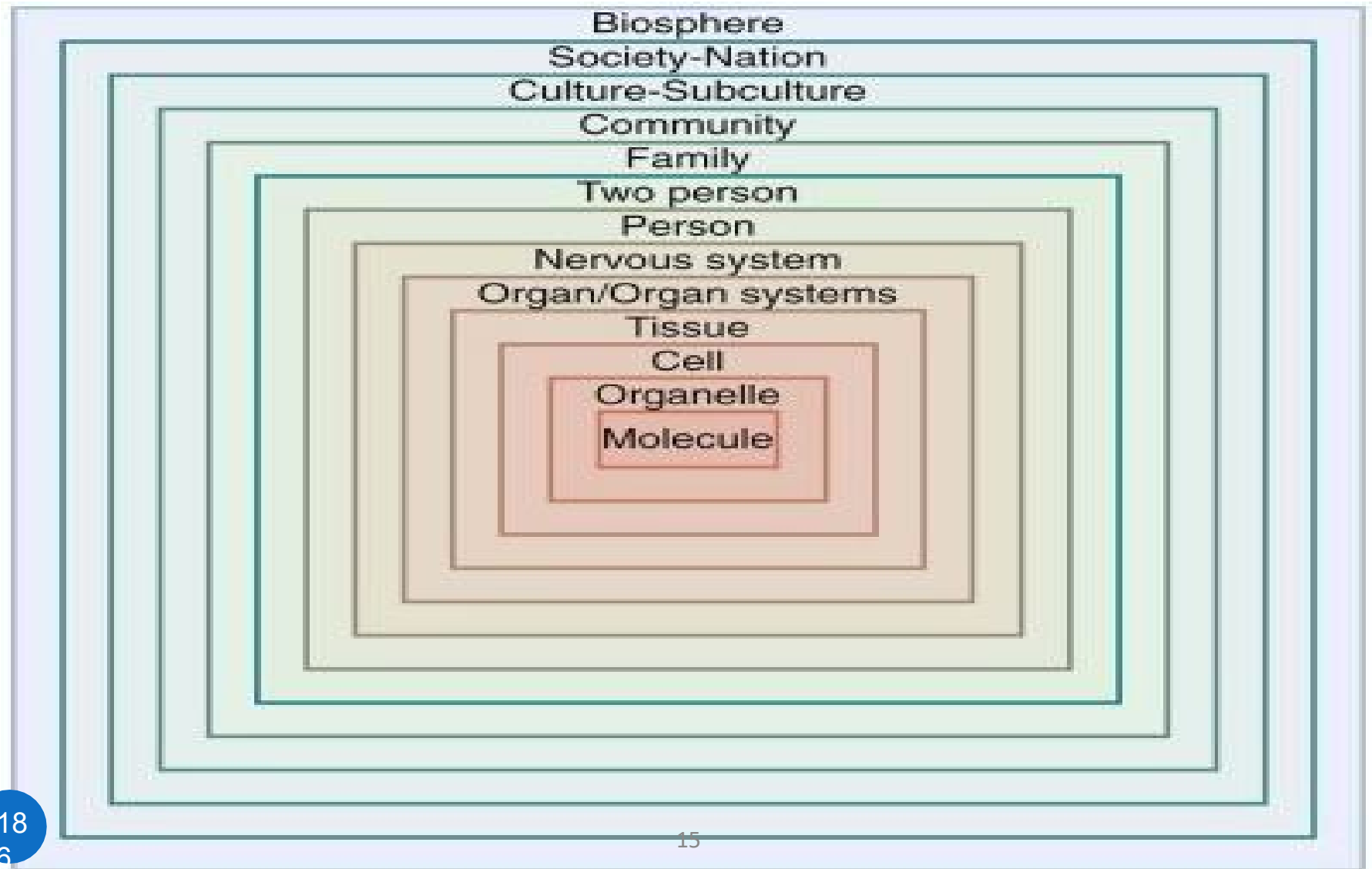
Classic Stage Theories

	INFANCY (0–1 YR)	TODDLERHOOD (2–3 YR)	PRESCHOOL (3–6 YR)	SCHOOL AGE (6–12 YR)	ADOLESCENCE (12–20 YR)	20-24 years	25-64 years	65-death
Freud: psycho sexual	Oral	Anal	Phallic/ oedipal	Latency	Genital			
Erikson: psycho social	Basic trust vs mistrust	Autonomy vs shame and doubt	Initiative vs guilt	Industry vs inferiority	Identity vs role diffusion	Intimacy vs. isolation	Generativity vs. stagnation	Ego integrity vs. despair
	Hope	Will	Purpose	Competence	Fidelity	Love	Care	wisdom

Piaget's four stages of intellectual (or cognitive) development are:

- **Sensorimotor.** Birth through ages 18-24 months
- **Preoperational.** Toddlerhood (18-24 months) through early childhood (age 7)
- **Concrete operational.** Ages 7 to 12
- **Formal operational.** Adolescence through adulthood

Biopsychosocial Models of Development



Why developmental assessment? And what will be assessed?

- **Early detection** of deviation in child's pattern of development
- **Simple and time efficient** mechanism to ensure adequate surveillance of developmental progress
- Domains assessed:
 - **Cognitive**
 - **Motor**
 - **Language**
 - **Social / behavioral and adaptive**

COGNITIVE

Stares momentarily at spot where object disappeared	2	Lack of object permanence (out of sight, out of mind) [e.g., yarn ball dropped]
Stares at own hand	4	Self-discovery, cause and effect
Bangs 2 cubes	8	Active comparison of objects
Uncovers toy (after seeing it hidden)	8	Object permanence
Egocentric symbolic play (e.g., pretends to drink from cup)	12	Beginning symbolic thought
Uses stick to reach toy	17	Able to link actions to solve problems
Pretend play with doll (e.g., gives doll bottle)	17	Symbolic thought

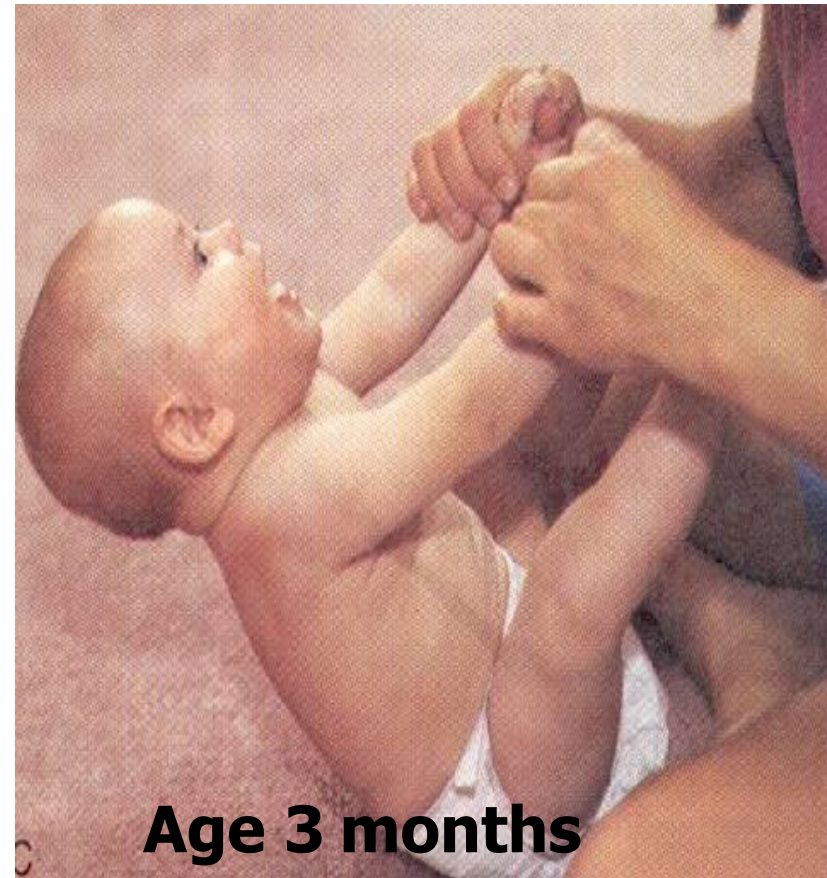
Gross Motor Skills

- The acquisition of **gross motor** skill precedes the development of **fine motor** skills.
- Both processes occur in a cephalocaudal fashion
 - Head control preceding arm and hand control
 - Followed by leg and foot control.

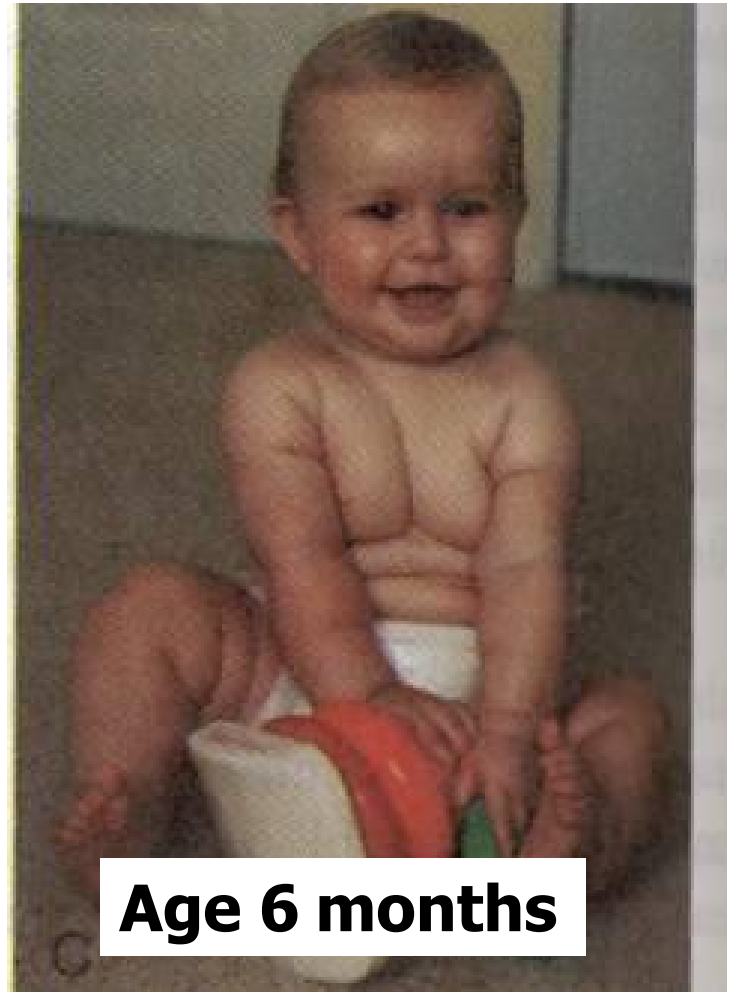
Gross motor

GROSS MOTOR	AGE (MO)	DEVELOPMENTAL IMPLICATIONS
Holds head steady while sitting	2	Allows more visual interaction
Pulls to sit, with no head lag	3	Muscle tone
Brings hands together in midline	3	Self-discovery of hands
Asymmetric tonic neck reflex gone	4	Can inspect hands in midline
Sits without support	6	Increasing exploration
Rolls back to stomach	6.5	Truncal flexion, risk of falls
Walks alone	12	Exploration, control of proximity to parents
Runs	16	Supervision more difficult

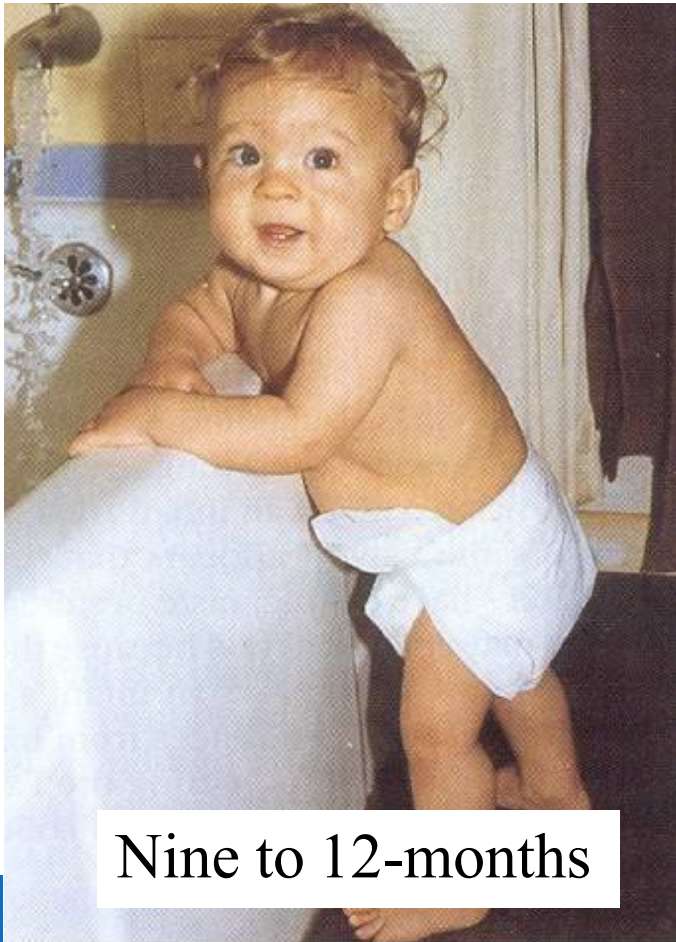
Head Control



Sitting Up



Ambulation



Nine to 12-months



12 month old

Fine motor

FINE MOTOR	AVERAGE AGE (MO)	DEVELOPMENTAL IMPLICATIONS
Grasps rattle	3.5	Object use
Reaches for objects	4	Visuomotor coordination
Palmar grasp gone	4	Voluntary release
Transfers object hand to hand	5.5	Comparison of objects
Thumb-finger grasp	8	Able to explore small objects
Turns pages of book	12	Increasing autonomy during book time
Scribbles	13	Visuomotor coordination

COMMUNICATION AND LANGUAGE		
Smiles in response to face, voice	1.5	More active social participant Monosyllabic
babble	6	Experimentation with sound, tactile sense
Inhibits to “no”	7	Response to tone (nonverbal)
Follows one-step command with gesture	7	Nonverbal communication
Follows one-step command without gesture	10	Verbal receptive language (e.g., “Give it to me”)
Says “mama” or “dada”	10	Expressive language
Points to objects	10	Interactive communication
Speaks first real word	12	Beginning of labeling
Speaks 4–6 words	15	Acquisition of object and personal names
Speaks 10–15 words	18	Acquisition of object and personal names
Speaks 2-word sentences (e.g., “Mommy shoe”)	19	Beginning grammaticization, corresponds with 50+ word vocabulary

Red Flags in infant development

- Unable to sit alone by age 9 months
- Unable to transfer objects from hand to hand by age 1 year
- Abnormal pincer grip or grasp by age 15 months
- Unable to walk alone by 18 months
- Failure to speak recognizable words by 2 years.

Fine Motor – Older Toddler and preschools

- **3 year old**: copy a circle and a cross – build using small blocks
- **4 year old**: use scissors, color within the borders
- **5 year old**: write some letters and draw a person with body parts

Toddler



Adele Pilitterri, Child Health Nursing, Lippincott

Fine motor and cognitive abilities

- Buttoning clothing
- Holding a crayon / pencil
- Building with small blocks
- Using scissors
- Playing a board game
- Have child draw picture of himself

Red flags: preschool

- **Inability to perform self-care tasks**, hand washing
simple dressing, daytime toileting
- **Lack of socialization**
- **Unable to play with other children**
- **Unable to follow directions during exam**

School Years: fine motor

- Writing skills improve
- Fine motor is refined
- Fine motor with more focus
 - Building: models – Legos
 - Sewing
 - Musical instrument
 - Painting
 - Typing skills
 - Technology: computers

School performance

- Ask about favorite subject
- How they are doing in school
- Do they like school
- By parent report: any learning difficulties, attention problems, homework
- Parental expectations

School Age: gross motor

- 8 to 10 years: team sports
- Age ten: match sport to the physical and emotional development

School Age: cognitive

- Greater ability to concentrate and participate in self-initiating quiet activities that challenge cognitive skills, such as reading, playing computer and board games.

Red flags: school age

- School failure
- Lack of friends
- Social isolation
- Aggressive behavior: fights, fire setting, animal abuse

Adolescent

- As teenagers gain independence they begin to challenge values
- Critical of adult authority
- Relies on peer relationship
- Mood swings especially in early adolescents

Adolescent behavioral problems

- Anorexia
- Attention deficit
- Anger issues
- Suicide

Adolescent Teaching

- Relationships
- Sexuality – STD's / AIDS
- Substance use and abuse
- Gang activity
- Driving
- Access to weapons

Growth Evaluation

- Weight
- Height
- Head circumference
- BMI
- Body proportions
- Height velocity
- Mid parental height and target height

Growth monitoring

- Measurement
- Plotting
- Interpretation

Weighing Infants and Toddlers



☒ Up to 36 months if unable to stand without assistance

Weighing scale



Measure Lying or Standing?

New born Length

48 cm-52 cm (18 inches to 22 inches)



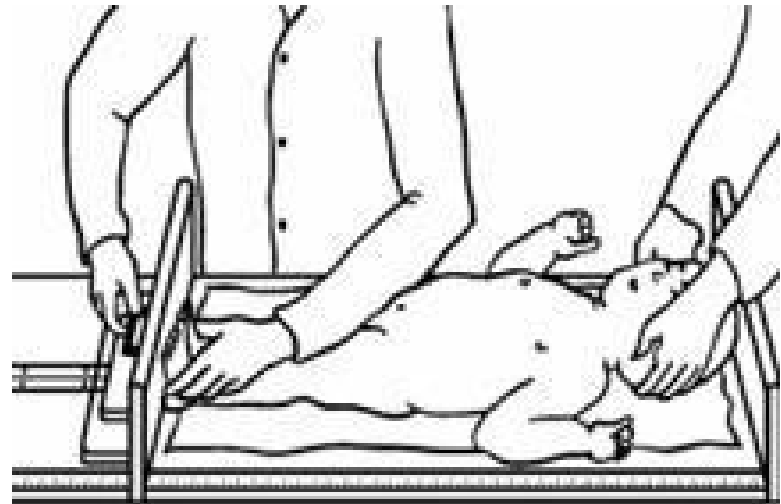
Length

- **Lying**
- Unable to stand without assistance
- Use **Birth -24** months growth chart

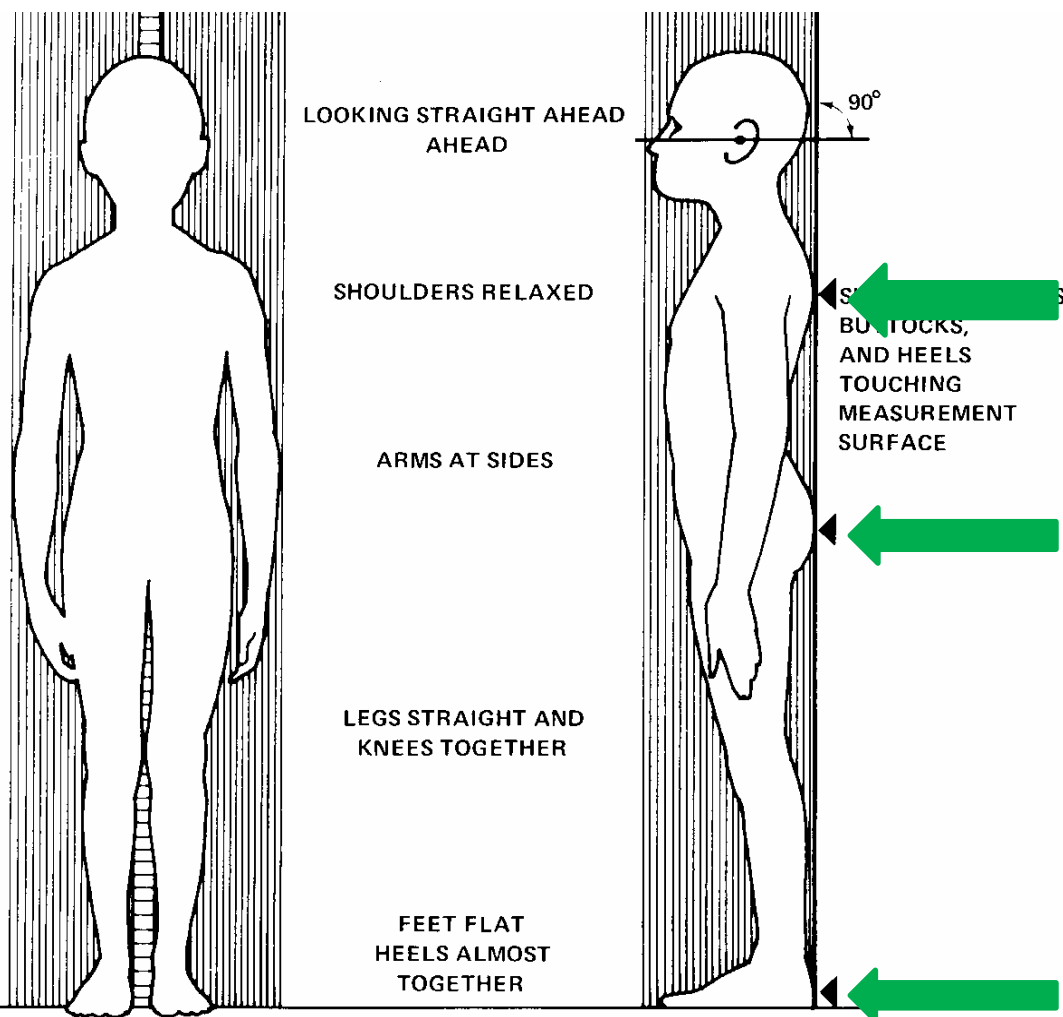
Stature

- Standing
- Able to stand without assistance
- Use **2-20** years growth chart

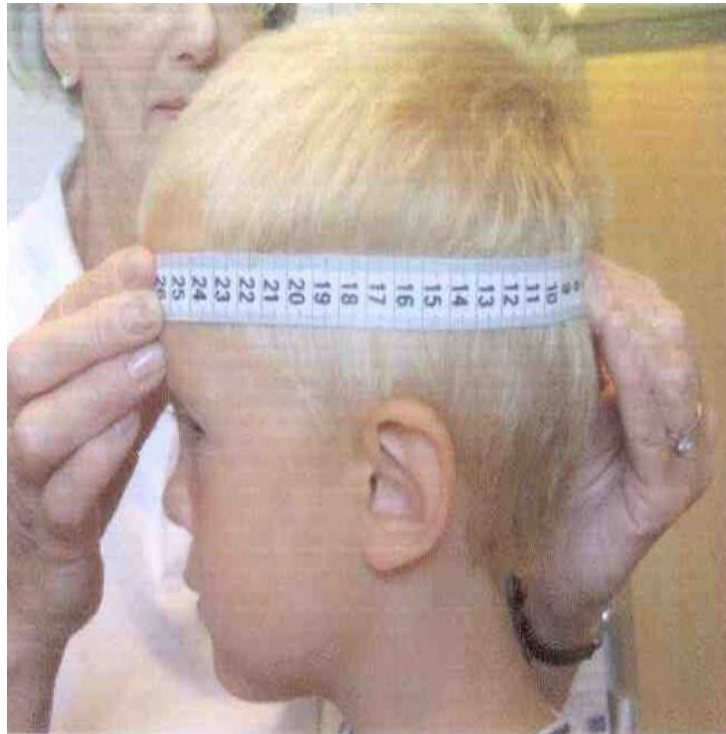
Measuring Length



Measuring Stature (Height)



Measuring head circumference(OFC)



What to measure?

1. **Weight – for – Age**
2. **Length/Height – for – Age**
3. **Weight – for – Height**
4. **Mid – Upper Arm Circumference**
5. **Head Circumference**

- Indices derived from growth measurements

$$\text{Weight-for-age} = \frac{\text{Weight of the child}}{\text{Weight of the reference child of the same age}} \times 100$$

$$\text{Weight-for-height} = \frac{\text{Weight of the child}}{\text{Weight of the reference child of the same height}} \times 100$$

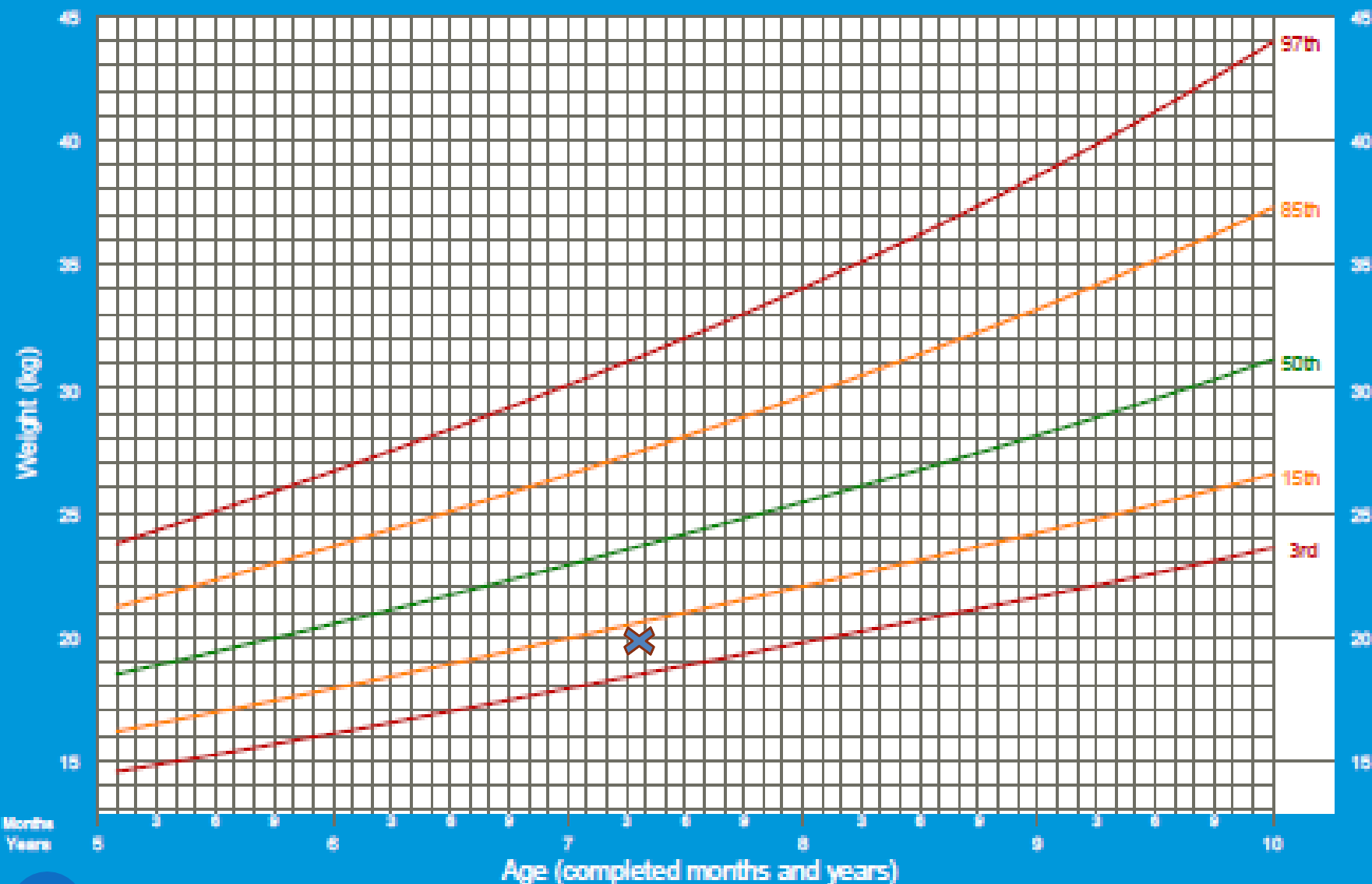
$$\text{Height-for-age} = \frac{\text{Height of the child}}{\text{Height of the reference child of the same age}} \times 100$$

What to measure...

1. **Weight – for – age**: weight of the child (in kg) is compared with that of a healthy child of the same age from a reference population.

Weight-for-age BOYS

5 to 10 years (percentiles)

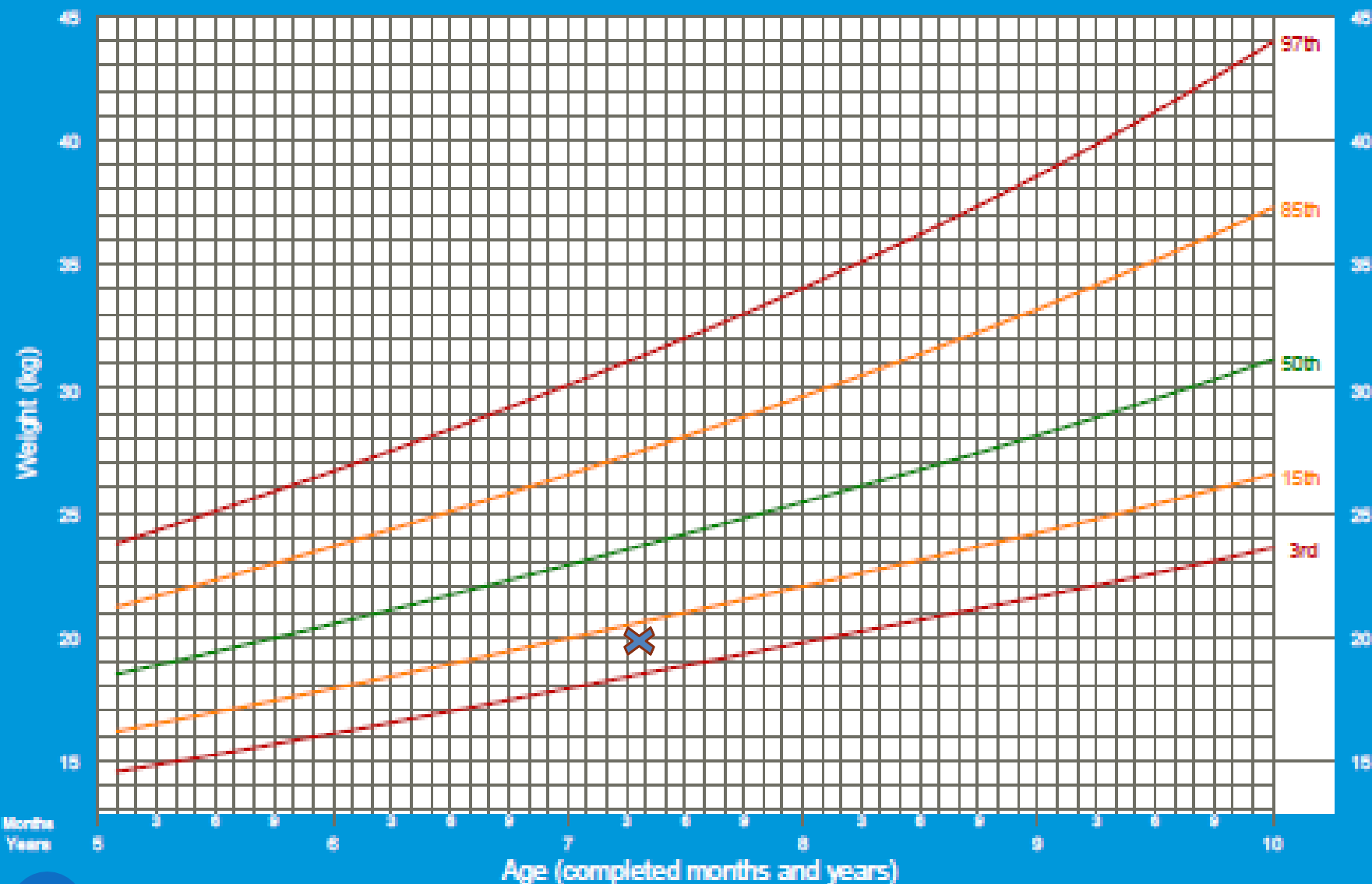


Analysis Growth Patterns

- Growth is a process rather than a static quality
- An infant at the 5th percentile of weight for age may be growing **normally**, **may be failing to grow**, or **may be recovering from growth failure**, depending on the pattern of the growth curve.

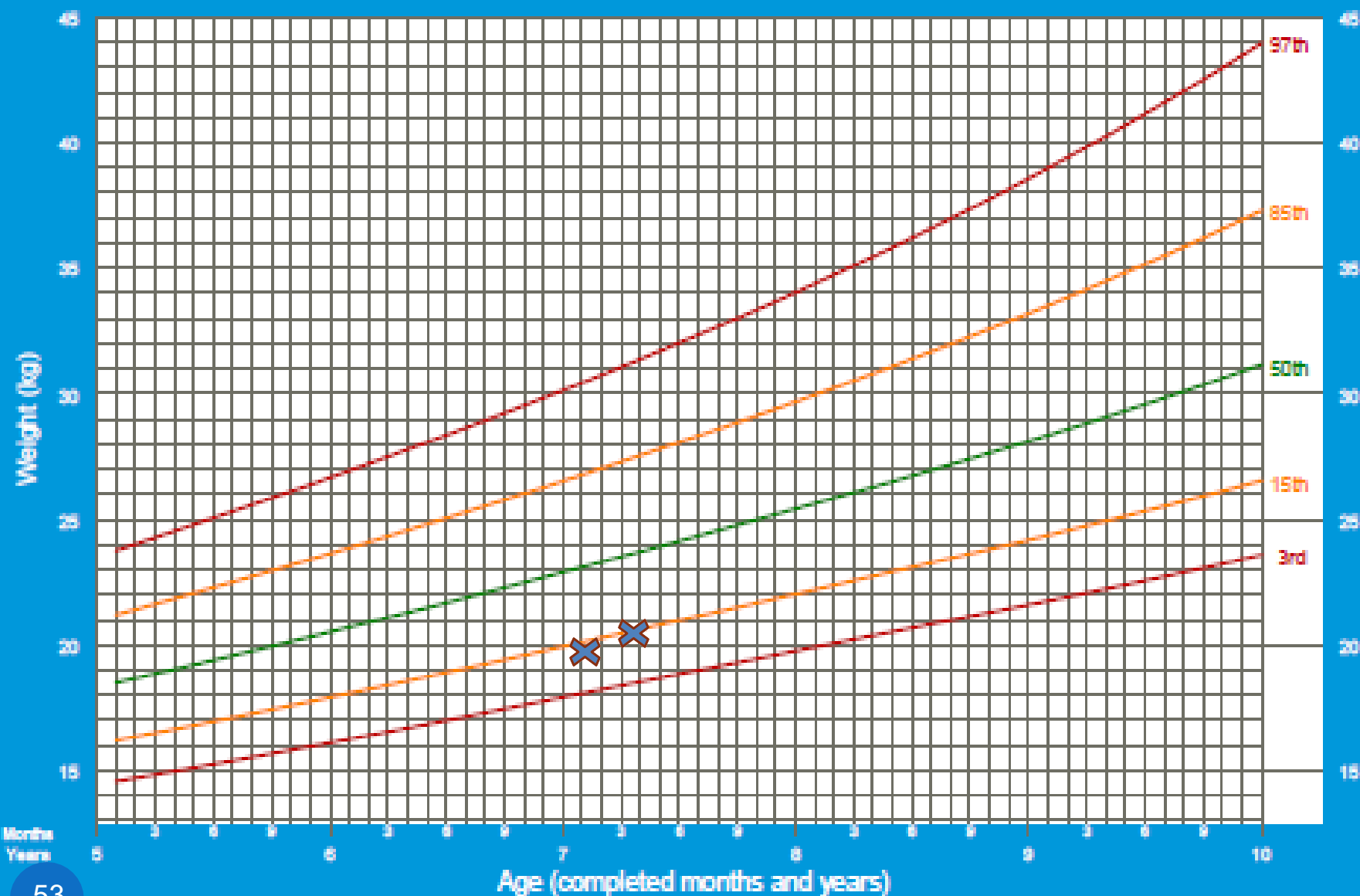
Weight-for-age BOYS

5 to 10 years (percentiles)



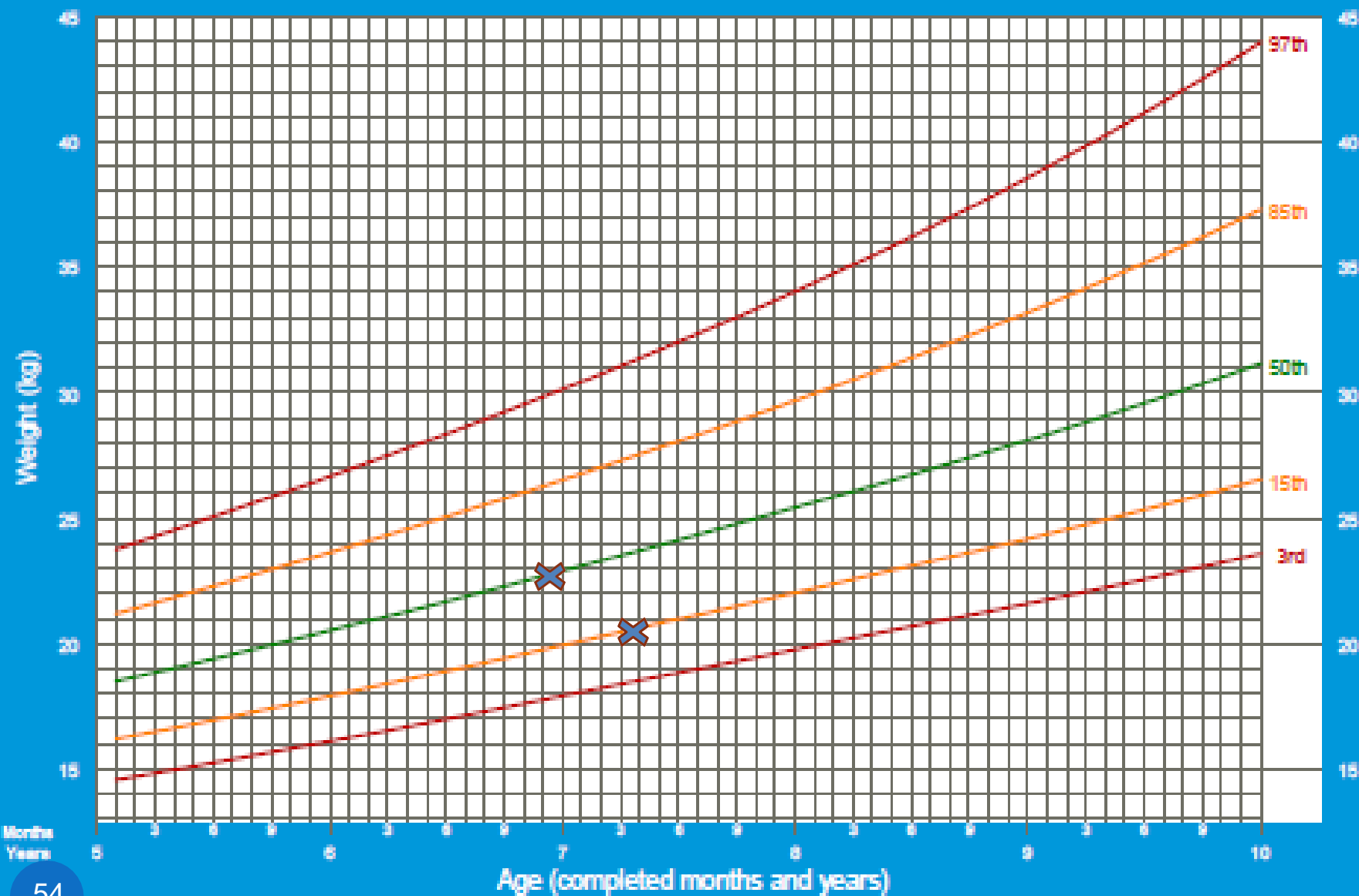
Weight-for-age BOYS

5 to 10 years (percentiles)



Weight-for-age BOYS

5 to 10 years (percentiles)



2. Length – for – age:

- Compares height/length with the expected height of a healthy reference child of the same age.
- Used to assess **stunting** which may be an index of **long-term** nutritional deprivation.

3. Weight – for – length/height:

- ❖ Degree of **wasting** assessed by comparing the child's weight with the weight that would be expected for a healthy child of the same height.
- ❖ A child who is $< 70\%$ of the expected weight for height is classified as **severely wasted**.

Waterlow Classification: percent weight for height =
((weight of patient) / (weight of a normal child of the
same height))

	Weight for Height (wasting)	Height for Age (stunting)
Normal	> 90	> 95
Mild	80 - 90	90 - 95
Moderate	70 - 80	85 - 90
Severe	< 70	< 85

Advantages:

- ▣ Identifies the very thin or wasted child with definite malnutrition who requires immediate attention.
- ▣ Good indicator to distinguish children who are well proportioned from those who are wasted for their height.
- ▣ Doesn't require age data.

Disadvantages:

- ❑ Stunted children with reasonable body proportion are classified as not malnourished.
- ❑ Two pieces of equipments required.
- ❑ Takes longer time.
- ❑ Measuring length in infants may require two persons.
- ❑ More difficult for unskilled health workers.

4. Head circumference

□ Increases in size during the 1st year of life, but little increment between 1 and 5 years of age.

- Average Head circumference (cm)
 - ❖ At birth = 35 (33 – 37) cm
 - ❖ At 6mo = 44cm
 - ❖ At 1 yr = 47cm
 - ❖ At 2 yr = 49 cm

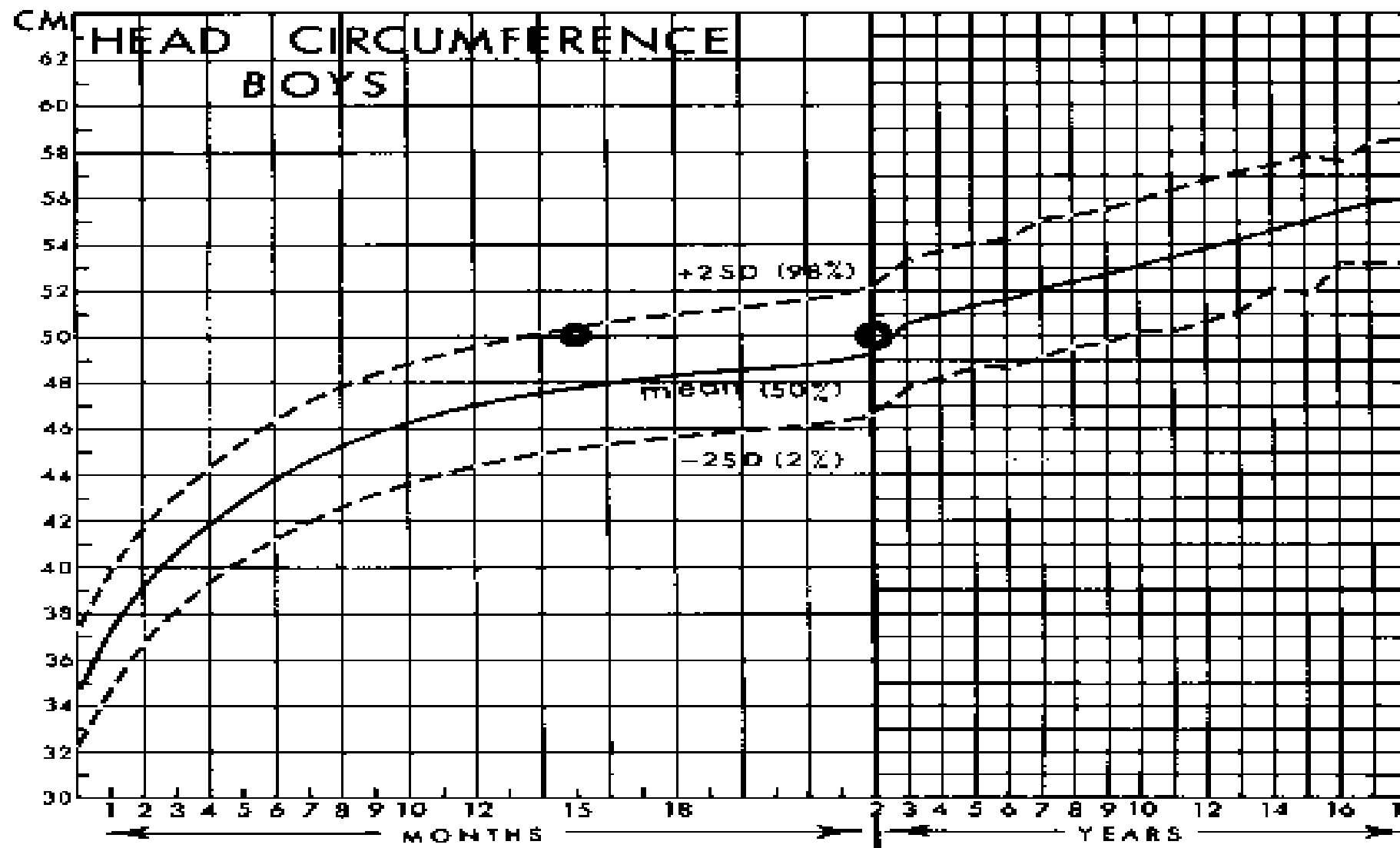


Fig. 4-2A. - Head circumference for boys from birth to 18 years. (From Nellhaus.⁴⁷)

5. Mid – upper – arm circumference

- ❑ At birth, average MUAC= 10.5cm
- ❑ By the age of 1 year, MUAC=16.5cm
- ❑ Between 1 and 5 years of life, average increment is only about 1cm, to reach 17.5cm at 5 years.
- ❑ Between 1 and 5 years of age, MUAC of <12.5cm, considered malnourished between 12.5 and 13.5 cm is probably malnourished.
- ❑ Low MUAC is more closely associated with acute malnutrition (wasting) than stunting.

MUAC...

- ▣ Useful as method of screening large number of children, during nutritional emergencies.
- ▣ less useful in long-term growth monitoring.

Advantages:

- ▣ Indicates severe current malnutrition.
- ▣ Quick to use.

Advantages...

- ☐ The measuring tape can be color – coded, for use by illiterate workers or community.
- ☐ Doesn't require exact age data.
- ☐ The equipment is portable and cheap.

Disadvantages:

- ☐ Only identifies children with severe malnutrition and is not a sensitive early predictor of malnutrition.
- ☐ Great variability in measurement.

SUMMARY

- Children are different as they are in the process of growth and development
- Be patient and opportunistic in examining a child.
- Pattern of growth should be followed
- Significant deviation from normal pattern should alert the clinician.

Chapter Four

Feeding of New born, Infant & Children

❖ Session objectives

- Identify the nutritional needs of infants.
- Describe the impact of poor diet on child's life
- Demonstrate breastfeeding technique
- Explain about complementary feeding
- Determine when artificial (formula) feeding is require

Infant feeding

- Infant feeding is of great importance.
- It is a “must” to meet nutritional as well as emotional and psychological needs of the infant.
- The basic food for infant feeding is milk.
- **Breast feeding** is the most natural method and, in fact, the supreme gift.
- **Artificial feeding** should not be the first choice, except for a compulsive situation.

- **Artificial feeding** exposes the infant to risks, especially infections, leading to considerable ***morbidity & mortality***, especially in the developing world.

ANATOMICAL ASPECTS OF LACTATION

➤ ***Milk*** is produced in the sac-like spaces, alveoli, of the glandular tissue of breast.

From alveoli, about 20 small ducts carry milk to their own dilated ends, lactiferous sinuses, which lie under the areola and store milk.

➤ From these sinuses, milk passes on to the nipple for supply to the infant.

➤ However, it is important for the infant to suckle the nipple before milk gets drawn out of the sinuses and the nipple.

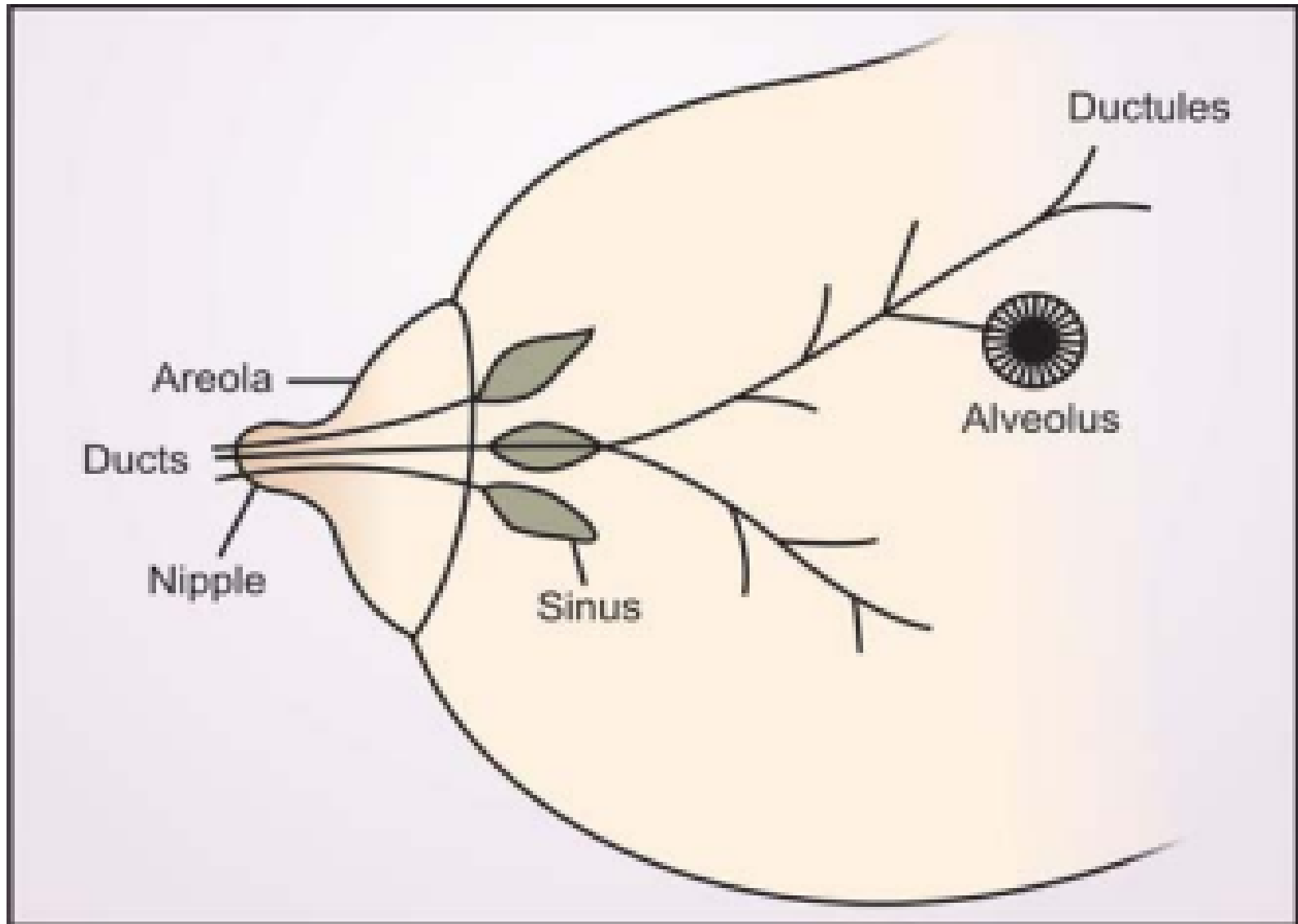


Fig. 12.1: Anatomy of breast

PHYSIOLOGY OF LACTATION

❖ *In Relation to Mother*

➤ ***Prolactin Milk-secreting Reflex:*** Suckling by baby at breast stimulates alveolar cells of the breast to secrete milk through secretion of the hormone, ***prolactin***, by the anterior pituitary.

- Prolactin level reaches the peak around 30 minutes of initiation of breast feeding, thereby getting ready milk for the next feed.
- Since pituitary gland secretes more prolactin during the night, breastfeeding at night specially helps to keep good supply of milk.

PROLACTIN

*Secreted AFTER feed
to produce NEXT feed*

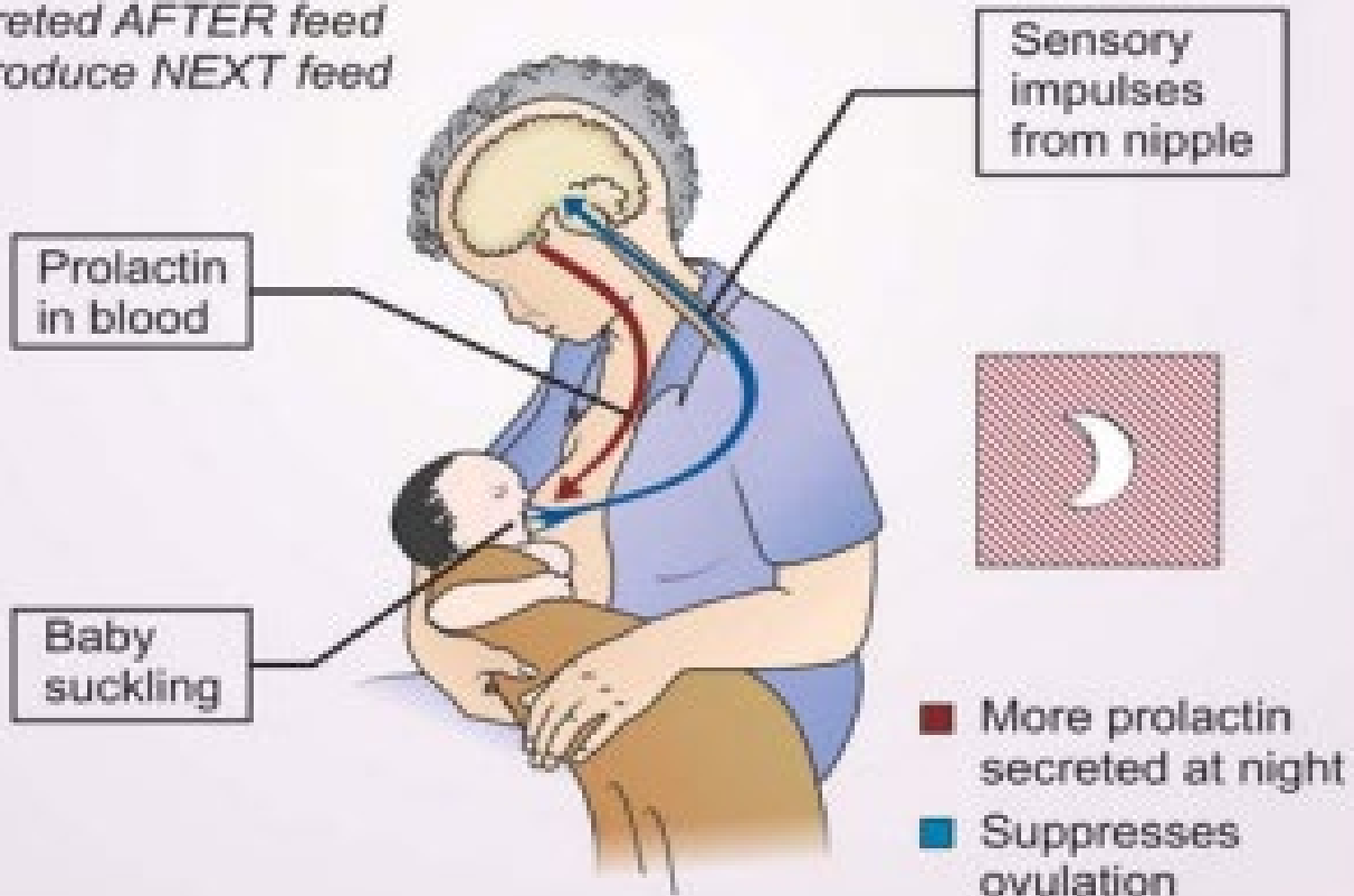


Fig. 12.2: Prolactin milk secreting reflex

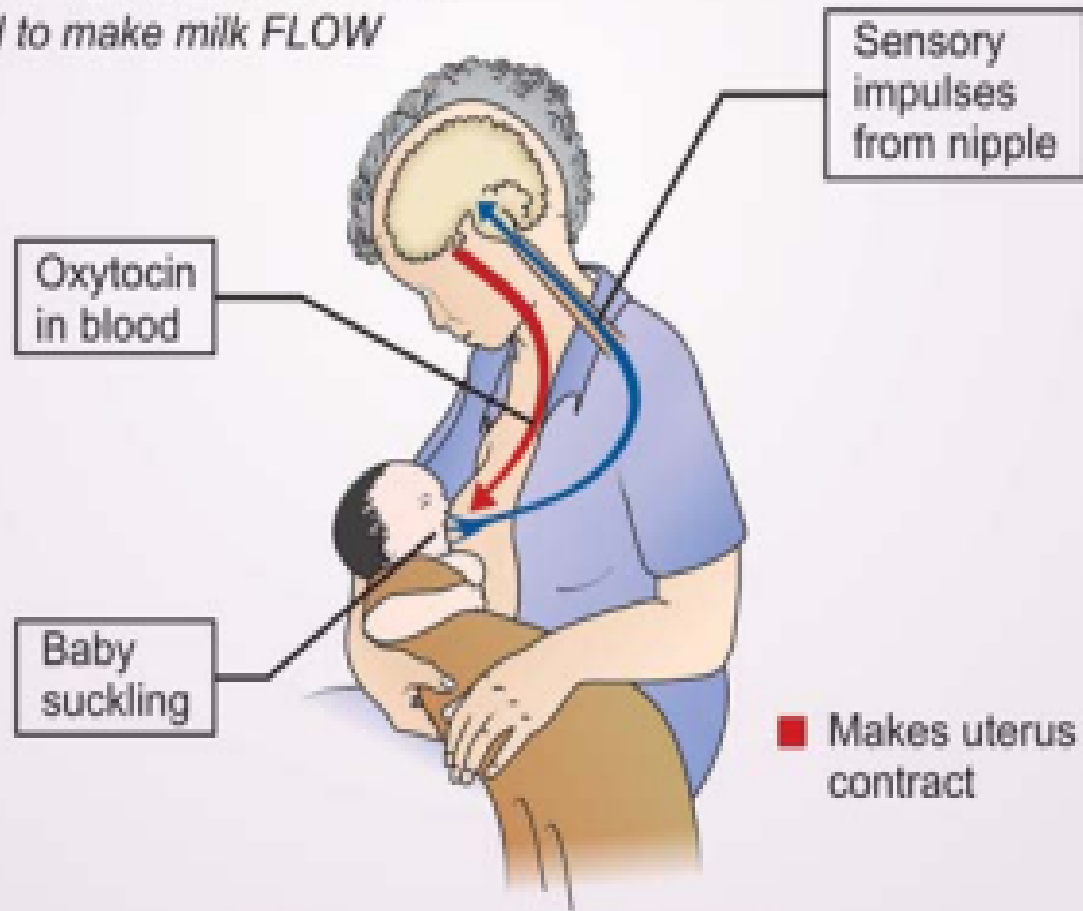
➤ **Oxytocin Milk Ejection Reflex:** Suckling by the baby sends sensory impulses from the nipple to the posterior pituitary gland.

oxytocin, secreted by the gland reaches through blood to the breast, making the muscle cells around the alveolar cells contract.

Thus, milk, which has collected in the alveoli, flows along the ducts to the lactiferous sinuses.

OXYTOCIN REFLEX

Works BEFORE or DURING feed to make milk FLOW



: Breastfeeding: It is not only the best but the "must" for the baby

Fig. 12.3: Oxytocin milk ejection reflex



In Relation to the Infant

❖ ***Rooting reflex*** guides the infant to reach the nipple and to have his mouth properly attached to the breast.

A good attachment (termed “**latching**”) with nipple and enough of areola into infant’s mouth is essential for effective suckling.

❖ ***Suckling reflex*** helps the infant to draw out milk from mother’s breast.

It consists of drawing the nipple and areola into the mouth, compressing it between jaw tongue and palate and then drawing out milk by peristaltic movements of the tongue.

❖ ***Swallowing reflex*** helps the baby to swallow milk when mouth is full of it (after one to three suckles).

He takes the breath after swallowing.

It takes about a second or so for the “suckle swallow-breathe” cycle.

Nutritional Needs of Infants.

- In the first few years, child has specific nutritional needs to ensure his/her growth.
- Human milk and infant formula have all the nutrients your baby needs for the first 6 months - protein, fats, carbohydrate, vitamins and minerals.

- Adequate nutrition during infancy and early childhood is essential to ensure the growth, health, and development of children to their full potential.
- Inappropriate nutrition can also lead to childhood obesity which is an increasing public health problem in many countries.

What impact can a poor diet have on a child's nutritional needs?

- Children need a wide range of nutritious foods, with high intake of important minerals and vitamins such as protein and calcium.
- If your child's intake of good food is poor, they can fail to gain or lose weight.
- May get different diseases Poor growth & Development

Breast feeding

Breast feeding, also known as **nursing**, is the feeding of babies and young children with milk from a woman's breast.

Human milk is decidedly superior to other milks.

It is remarkably adopted to the requirements of the infant and provides the best start in life.

Breast milk is the best food you can offer your new baby.

Preparation for breast feeding should start during pregnancy.

Advantages/Benefits of Human/Breast Milk

1. For the Infant

- Tailor-made composition Human milk has a composition that is ideally tailored to the requirements of a small infant.

Benefits of breast feeding for infants...

- ideal nutrients for the newborn
- Nutrients easily digestible, absorbable and metabolized
- Promote bonding
- Promote improved behavioural and neurodevelopment
- protection against various diseases
- It also promotes growth of Lactobacilli
- Promotes long term health

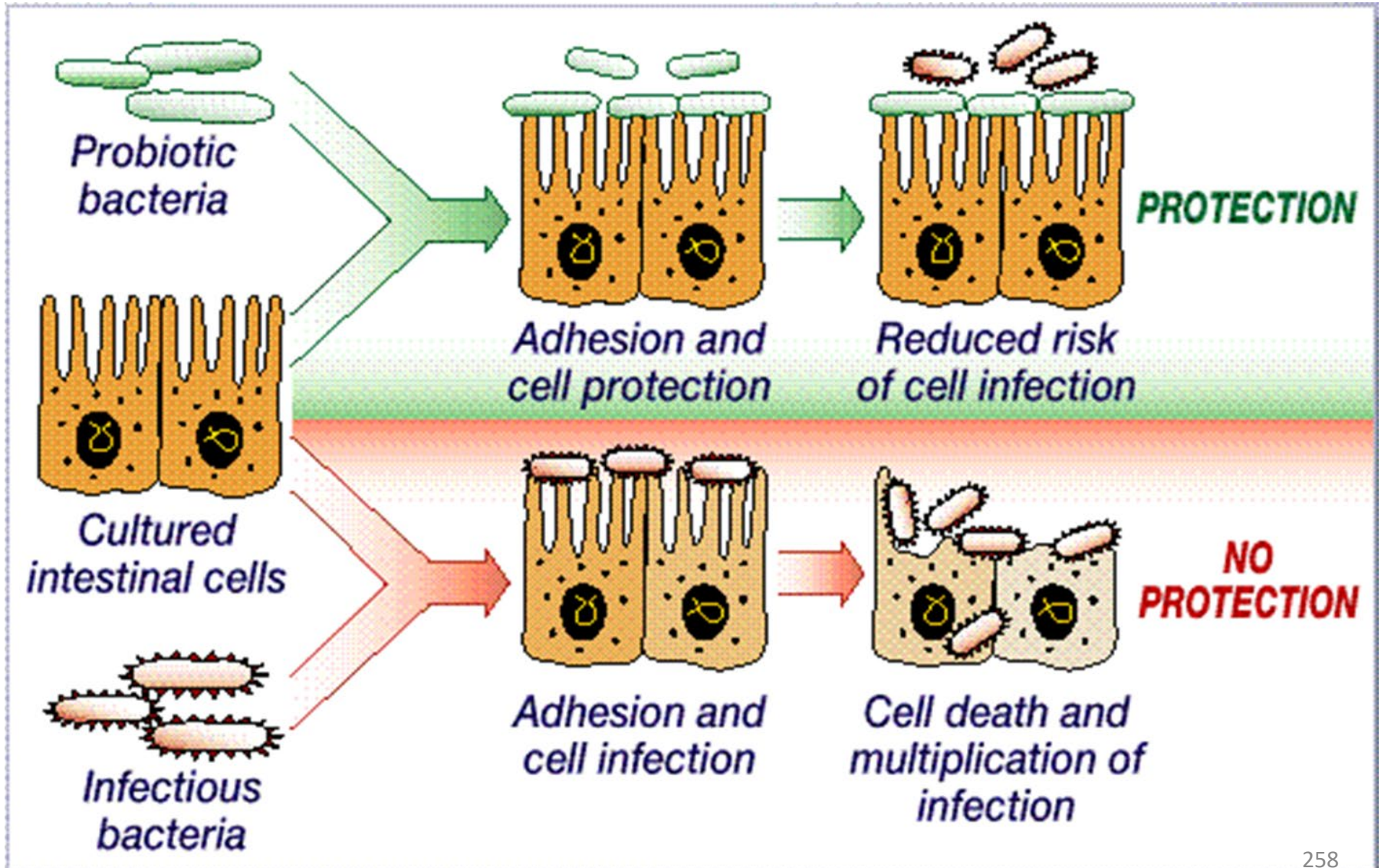
Compositions of Human Colostrum, Mature Breast Milk, and Cow's Milk (g/L)

	Human colostrum	Human breast milk	Cow's milk*
Total protein	23	11	31
Immunoglobulins	19	0.1	1
Fat	30	45	38
Lactose	57	71	47
Calcium	0.5	0.3	1.4
Phosphorus	0.16	0.14	0.90
Sodium	0.50	0.15	0.41

- It is always fresh, pure and ready made, requiring no preparations.
- It is at the right temperature.
- It is uncontaminated and aseptic.
- Protection against allergy.
- Breastfed babies have 7 times less chances of an allergy.

- It contains lacto ferrin, a substance that inhibits growth of E. coli, a common cause of infantile gastroenteritis.
- It assists in gradually establishing the organism, like Lactobacillus bifid us, in the baby's intestine. That help in digestion of sugar.
- it contains agents against Staphylococcus group of organisms which are responsible for septicemia of the newborn.

Protective barrier on gut cell wall



What is the role of Prebiotic's and Probiotics ?

Breast milk provides more than just nutrition; immune functions, digestive enzymes, and hormones are additional components of breast milk.

Additionally, research has found that breast milk contains both probiotics and prebiotics.

- ❖ **PREBIOTICS** are a special form of dietary fiber that acts as a fertilizer for the good bacteria in your gut.
- ❖ **PREBIOTICS** nourish the good bacteria that everyone already has in their gut.
- ❖ **PROBIOTICS** must compete with the over 1000 bacteria species already in the gut.

The good bacteria in your digestive tract help protect you from harmful bacteria and fungi.

They also send signals to your immune system and help regulate inflammation (1 , 2).

Additionally, some of your gut bacteria form vitamin K and short-chain fatty acids.

Protein

Whey : casein 60 ; 40
digestible, prevents
infection

0 -6 mo
Immunoglobulin
IgA, IgG IgM

Living cell
Enzyme,
Hormone

Fat, CH₂O

Energy, Immunity
brain, retina

Nuclotides
Immunity, growth, brain,
base for DNA & RNA

Minerals

Growth, bone and teeth

Lactoferrine
Iron uptake by intestine
bacteriostatic

Vitamins A, C,
B com.

Growth ,&immunity



Kills cancer cells
HAMLET

Water
80 %

Epidermal growth
factor
Proliferation, maturation
Mucosal repair

Prebiotes, Probiotics
Bifidobacterium & Lactobasillus

Benefits to mother

- ◆ Most Economical
- ◆ Gives her a sense of confidence and feeling of self sufficient to feed her baby
- ◆ **It enhances involution of the uterus & limit PPH**
- ◆ Family planning (child spacing)
- ◆ Protection against breast and cervical cancers

Breastfeeding Technique and Some Basic Principles

1. Breastfeeding should be done in as clean and safe a manner as possible.
2. The mother and the baby should be comfortable and relaxed at the feeding time.
3. She should be well conversant with “how to put the baby to breast and how to remove him off it.”
4. Correct positioning & attachment should maintained.

Sign of good positioning

- Positioning of baby and mother need to include
 1. Infants whole body supported
 2. Infants head, neck and body should be straight
 3. Infant facing mother
 4. Infants body close to mother



Sign of good attachment

Attachment of babies to the breast must include all the following :

- ❖ ***Mouth open widely***
- ❖ ***Lower lip turned outward***
- ❖ ***More areola visible on above than below***
- ❖ ***Chin touching the breast***



5. At least one breast should be completely emptied at every, sitting.

6. In the case of a working mother, her “expressed” milk can be spoon fed to the baby in her absence.

The mothers should be conversant with the technique of expressing milk. Using both hands, she should squeeze gently from the base of the breast towards the areola and nipple.

Then the breast and areola should be squeezed between fingers and thumbs and the milk collected in a clean container.

The container should be stored in a cool place.

If stored outside the refrigerator, it has got to be used within 12 hours.

Just before feeding, this milk should be warmed by placing the container in a bowl of hot water.

6. Practice exclusive breast feeding up to 6 months.

7. Practice **burping** after feeding baby, In order to “kick out” the swallowed air.

Failure to do so may cause regurgitation, vomiting, and even abdominal pain.

8. Mother should give adequate attention to her diet, personal hygienic and health and have sufficient rest.

avoid unnecessary use of drugs which may have adverse effects on the baby.

How to Burp a Baby



- Baby sits on lap
- Support chest with base of hand
- Cup chin with thumb and index fingers
- Bounce leg gently if needed



- Lay baby across lap
- Belly rests on your thigh
- Hold head higher than chest



- Baby's chin rests on your shoulder
- Support head and neck

Disadvantage of breast feeding

- **Transmission of maternal infection HIV, Hepatitis, rubella measles e.tc.**
- **Trans mission of material drug.**
- **Deficit of vit – K & vit – D.**
 - **1 mg I.m injection vit k to new born infant**
 - **sun shine exposure for vit – D.**

COMPLEMENTARY FEEDING

- ❖ ***complementary*** feeding means introduction of nutritious foods over and above breastfeeding.
- ❖ These energy-dense foods should be cost effective, affordable, easily available and well tolerated.
- ❖ The term **weaning** means “to be taken off the breasts” or “introduction of top feed”.

Recommended Age

- ❖ Just after **six** months of age is the most appropriate time for introducing complementary feeding.
- ❖ The whole process should be gradually completed by 9 months to 1 year of age when the child should be taking almost the adult diet.
- ❖ **delayed** introduction of complementary foods is known to cause malnutrition and growth retardation
- ❖ **premature** introduction of such foods exposes to the risk of infections and the resultant morbidity and mortality.²⁷⁵

Types of complementary foods used by the community

1. Appropriate (first line) weaning food

- ✓ Fresh home-made, locally available, culturally acceptable, hygienically prepared, energy dense and cost-effective foods like.
 - Combination of cereals and pulses (khichadi, dal-rice etc.), porridge, in any form fortified with sugar and oil or fried.
 - Mashed banana, sweet potato and potato.
 - Milk added to cereals preparations.
 - Sprouted legumes, nuts, dry fruits etc. depending upon the affordability.

2. Acceptable (second line) weaning foods

These includes home available foods like

- Breads, cakes, pastry, biscuits, cheese, ice cream, etc.
- Caloric dense fruits, etc.

3. Avoidable (third line) weaning foods

- Commercially available artificial foods or tinned foods.
- Fast food or junk food.
- Low caloric fruits juices, soups.
- Repeatedly fried foods containing trans fatty acids (which predispose to obesity, atherosclerosis, cardiac and neurological problems in future life).

Box 12.1: Types of complementary foods used by the community

Appropriate (first line) weaning food

Fresh home-made, locally available, culturally acceptable, hygienically prepared, energy dense and cost-effective foods like.

- Combination of cereals and pulses (khichadi, dal-rice etc.), porridge, suji, dalia, kheer, khaman dhokla, idli, dosa, ragi, rice preparations etc. in any form fortified with sugar and oil or fried.
- Mashed banana, sweet potato and potato.
- Milk added to cereals preparations.
- Sprouted legumes, nuts, dry fruits etc. depending upon the affordability.

Acceptable (second line) weaning foods

These includes home available foods like

- Breads, cakes, pastry, biscuits, cheese, icecream, etc.
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- Commercially available artificial foods or tinned foods.
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- Repeatedly fried foods containing trans fatty acids (which predispose to obesity, atherosclerosis, cardiac and neurological problems in future life).

ARTIFICIAL (FORMULA) FEEDING

The World Health Organization (WHO) recommends that infants are exclusively breastfed for the first six months of life to achieve optimal growth, development and health.

Breast milk is the optimal source of nutrition for the infant but, ***when breast milk is not available, infant formula feeding is an appropriate.***

Types of infant formula

- ❖ Commercial prepared formula such as NIDO, S-26 Nan lipto meal, etc.
- ❖ Cow's milk

Formula feeding/bottle feeding...

It is feeding an infant or children by preparing formula milk.

Purpose:

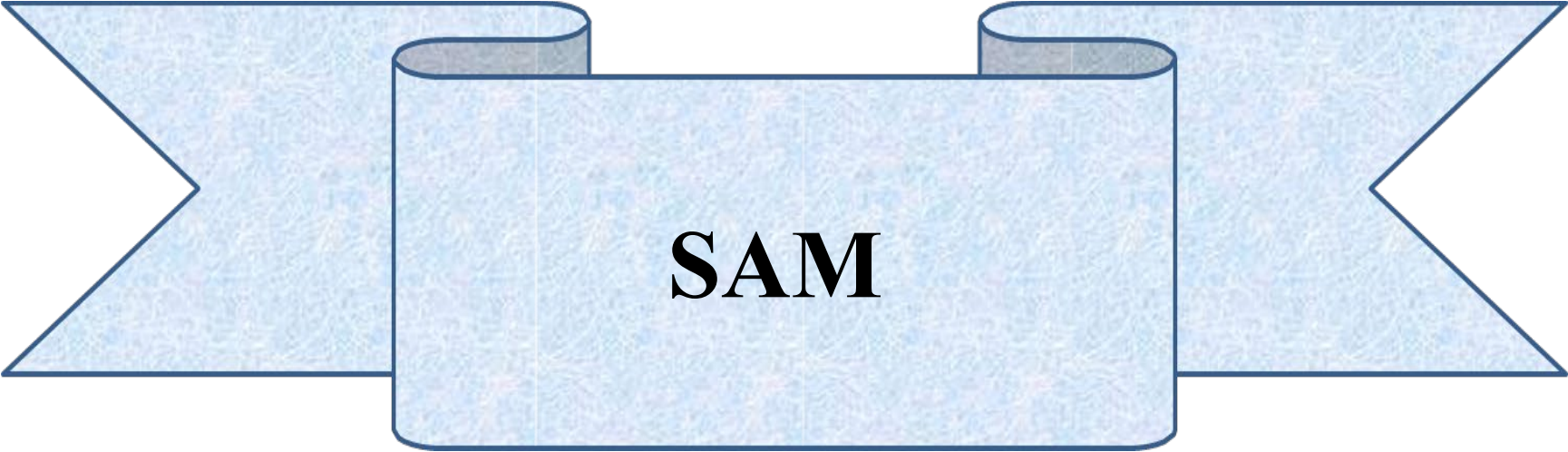
- ✓ To deliver dietary food for infant
- ✓ To increase thrive of infant
- ✓ To maintain health of the infant providing dietary foods
- ✓ To protect the infant from infection
- ✓ To enhance growth and development of the baby

Infants who required of formula feeding

- ❖ Child who lost his mother.
- ❖ Mother with severe illness especially breast infection or cancer.
- ❖ HIV positive mother.????
- ❖ Mother with low breast milk production.
- ❖ The mother is taking medication.
- ❖ Infants who are not breastfed.
- ❖ Very low-birth-weight babies.

CHAPTER FIVE

Child Nutrition



SAM

presentation Outline

- ☐ Introduction
- ☐ Types of malnutrition
- ☐ Definition of SAM
- ☐ Admission criteria
- ☐ management of SAM
- ☐ Discharge criteria
- ☐ Routine medicines

Introduction

- ❑ Malnutrition is a major public health problem to the World especially developing world.
 - ❑ **Stunting** over 149 million children < 5 years of age worldwide affected
 - ❑ **wasted** Over 49 million children
 - ❑ **overweight** 40 million children
- (Neufeld. L et al (2020))

Cont....

- ❑ Child malnutrition is the leading public health problem in developing countries
- ❑ In Ethiopia, malnutrition is a leading cause of child illness and death
- ❑ The prevalence of malnutrition among rural children in Ethiopia was 48.5% (Endris, N., H. Asefa and L. Dube (2017))

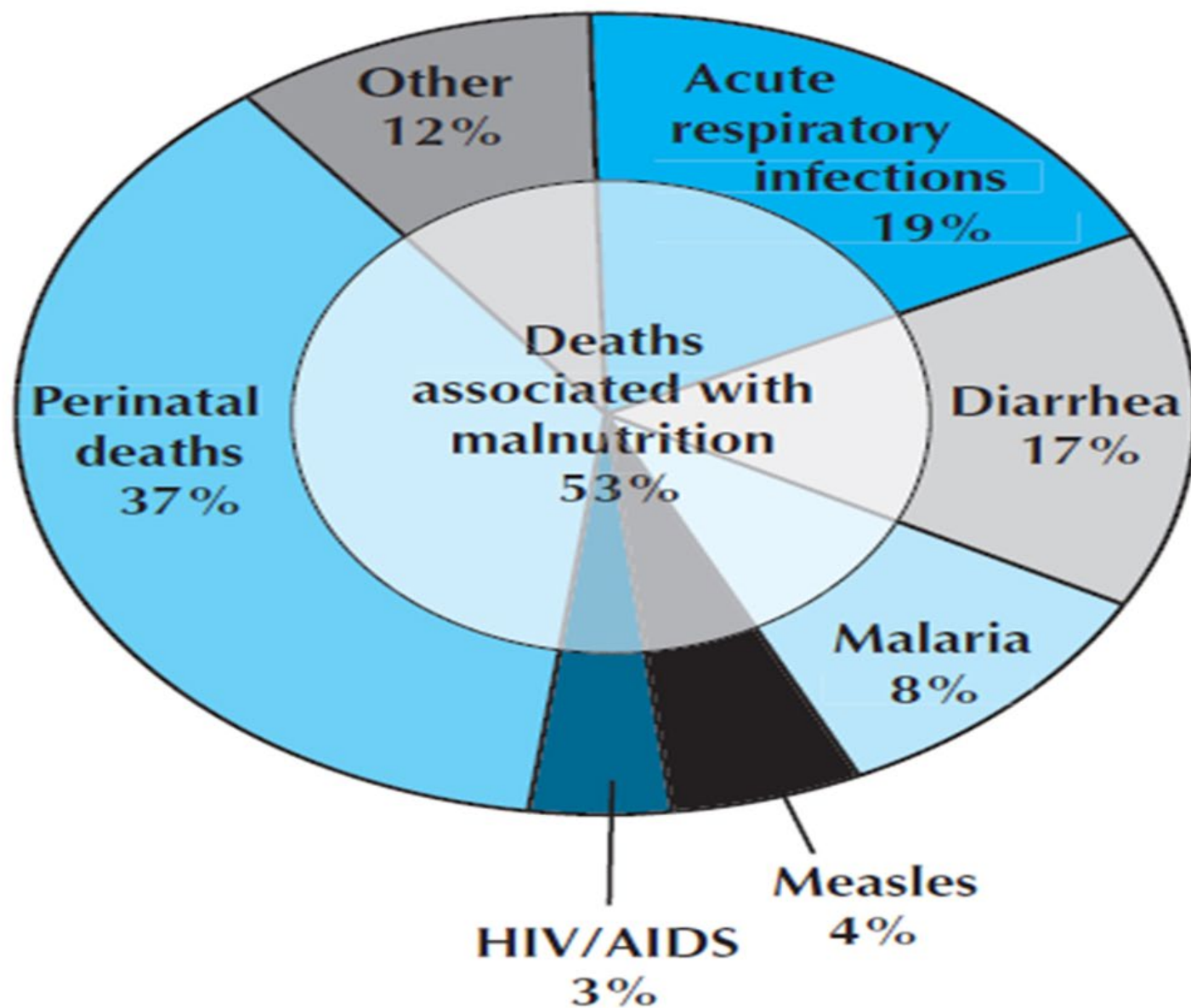


Fig. 1: Causes of death among children under 5 years of age,

Types of malnutrition

Acute malnutrition

- Wasting (MUAC or weight for height)

Chronic Malnutrition

- stunting (height for age)
- Underweight (weight for age)
- obesity
- micronutrient deficiency

Types of acute malnutrition

1. Severe acute malnutrition (SAM)

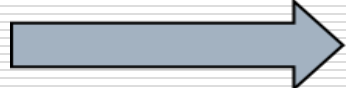
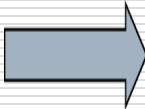
- All children with oedema
- Weight-for-height below - 3Z (below 70%)
- Severe wasting
- Obvious clinical problems

2. Moderate acute malnutrition(MAM)

- Weight-for-height between - 2Z and -3Z
(between 70-80%)
- Moderate wasting

Based on IMNCI 2016

INFANTS < 6 MONTHS

- ❑ WFL < -3Z score, and presence of complications OR Oedema of both feet  **COMPLICATED SAM**
- ❑ WFL < -3Z score AND no complications AND No oedema of both feet  **UNCOMPLICATED SAM**
- ❑ WFL $\geq$ -3Z to < -2Z score, AND No oedema of both feet  **MAM**
- ❑ WFL $\geq$ -2Z score AND No oedema of both feet  **NO ACUTE**
MALNUTRITION

SIGNS	CLASSIFY	TREATMENT
▶ MUAC less than 11.5 cm OR ▶ Pitting edema of both feet AND ▶ Any medical complication: Pneumonia OR Watery diarrhoea OR Dysentery OR Fever/low temperature OR Fail appetite test (RED)	COMPLICATED SEVERE ACUTE MALNUTRITION	▶ Give first dose of Amoxicillin ▶ maintain body temperature of the child ▶ Treat the child to prevent low blood sugar ▶ Advise the mother on the need of referral ▶ Refer URGENTLY to health facility (where there is a stabilization center)
➤ MUAC less than 11.5cm or ➤ Pitting edema of both feet AND ➤ No Medical complication: Pneumonia OR watery diarrhea OR Dysentery OR Fever/low temperature AND Pass appetite test (YELLOW)	UNCOMPLICATED SEVERE ACUTE MALNUTRITION	▶ Register in OTP and do the following ▶ Give RUTF and other medications accordingly and counsel the mother on how to feed a child with RUTF. ▶ Do Follow up.

▶ MUAC 11.5 cm-12.5 cm AND ▶ No edema of both feet (YELLOW)	MODERATE ACUTE MALNUTRITION	▶ give (TSFP)<u>therapeutic</u> supplementary feeding <u>programme</u>. ▶ Assess the child's feeding and counsel the mother on feeding according to the FOOD box on the COUNSEL THE MOTHER chart. ✓ If feeding problem, follow-up in 5 days ✓ If no feeding problem follow up in 15 days ▶ Advise mother when to return immediately.
▶ MUAC >12.5 cm or more , AND No edema of both feet (GREEN)	NO ACUTE MALNUTRITION	▶ If child is less than 2 years assess <u>child's</u> feeding and counsel the mother on feeding according to the FOOD box on the COUNSEL THE MOTHER chart ✓ if feeding problem, follow up in 5 days ✓ <u>if</u> no feeding problem praise the mother and appoint at 30th day for growth monitoring.

Definition of Severe Acute Malnutrition (SAM)

- ❑ Severe acute malnutrition (SAM) is a major cause of admission and in-patient mortality worldwide in children aged less than 5 years (Gachau, S etal (2018)
- ❑ It is both a medical and a social disorder
- ❑ Successful management of the severely malnourished patients requires that both medical and social problems be recognized and corrected(Pediatric Hospital Care ETHIOPIA)

Cont....

- ❑ Weight-for-height $<70\%$ or $< -3SD$ of the median ("wasted")
- ❑ Bilateral pitting oedema of nutritional origin ("oedematous malnutrition")
- ❑ MUAC $< 11.5\text{mm}$ in children between 6 – 59 months of age (WHO standards)

Protein energy malnutrition

- ❑ Protein Energy Malnutrition (PEM) is the deficiency of macronutrients or energy and protein in the diet.
- ❑ It is a nutritional disorder, which affects all the segments of population like children.

Different Types of PEM	
Clinical forms - o Kwashiorkor - o Marasmus - o Marasmic kwashiorkor	Sub-clinical forms - Underweight - Wasting - Stunting

Kwashiorkor

- ❑ Kwashiorkor is protein deficiency majorly
 - ❑ Mostly in children between the ages of **1** and **3** years
 - ❑ The three essential manifestations or signs of kwashiorkor are:
 1. Oedema (swelling of feet)
 2. Growth failure, and
 3. Mental changes- child has no interest in the surrounding
-

Other signs which may be present are

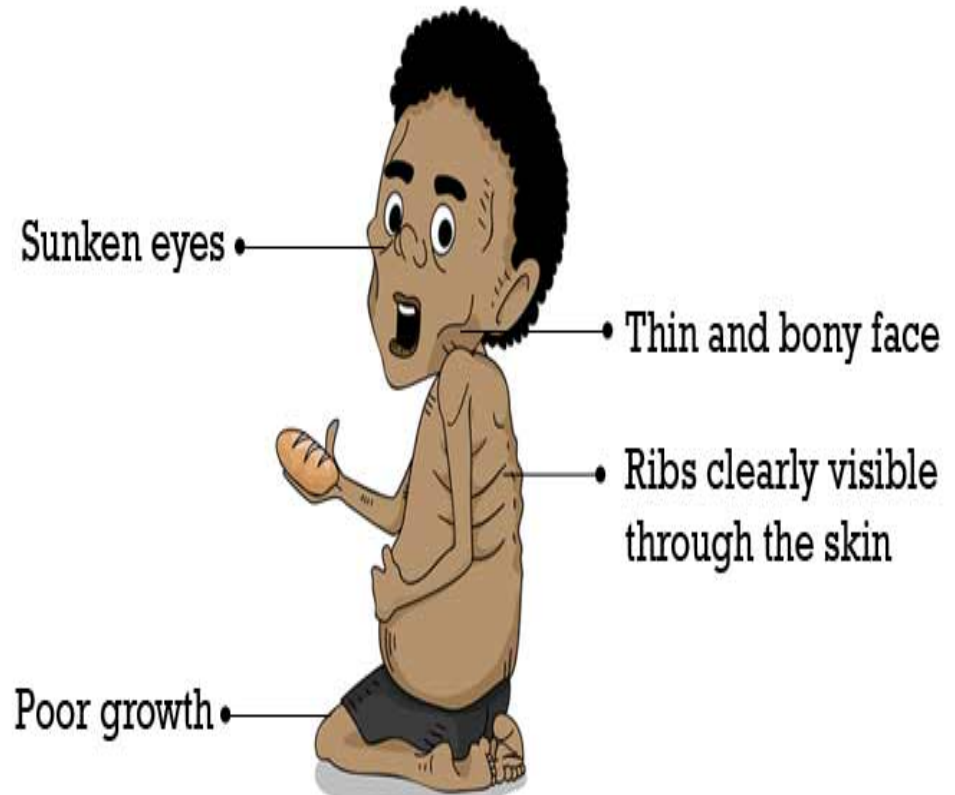
- ☐ Hair changes: coppery red colour
- ☐ Skin changes: In many cases, dermatosis (changes in skin) is seen.
- ☐ Moon face

Marasmus

- ☐ Marasmus is common in children below the age of 2 years.
 - ☐ Oedema is absent and there are bone and skin
 - ☐ Majorly CHO deficiency
-

Symptoms & Signs of Marasmus

- ❑ **Signs and Symptoms of Marasmus**
- ❑ Extreme muscle wasting - "skin and bones"
- ❑ Loose and hanging skin folds
- ❑ Old man's or monkey faces
- ❑ Absolute weakness

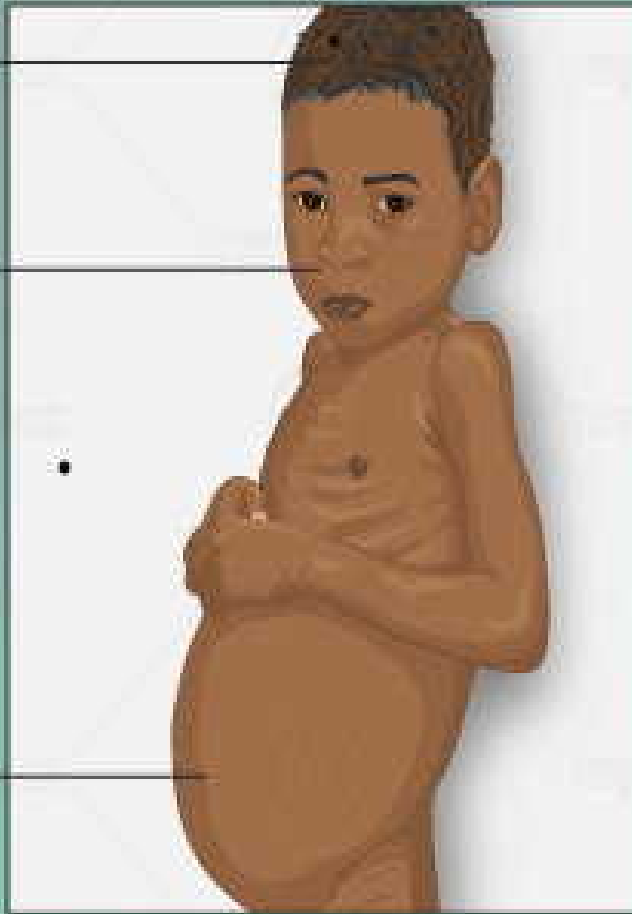


Two Types of Malnutrition



Thin and brittle hair

Moon face



Bulging abdomen (ascites)

Kwashiorkor

Thin and brittle hair

Simian facies



Loss of muscle mass

Marasmus

Marasmic Kwashiorkor

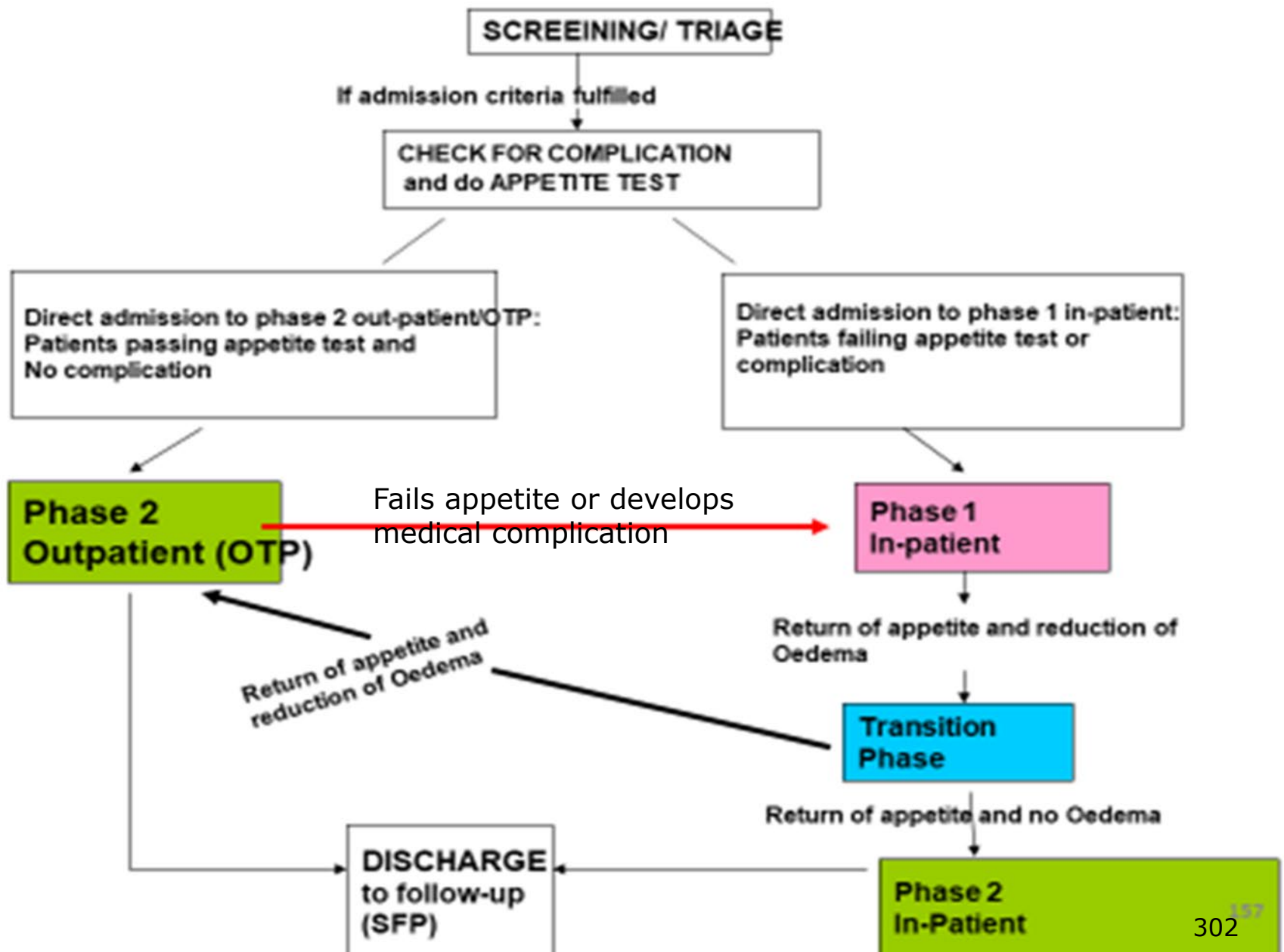
- ❑ children exhibit the features of both kwashiorkor and marasmus.
- ❑ extreme wasting of different degrees (**representing marasmus**) and also oedema (**a sign of kwashiorkor**).

Signs and symptoms of Marasmic Kwashiorkor

- ❑ Extreme muscle wasting - "skin and bones"
 - ❑ Loose and hanging skin folds
 - ❑ Old man's or monkey's face
 - ❑ Absolute weakness and **Oedema**
-

Initial assessment of the severely malnourished child

- ☐ **Take a history**
- ☐ **physical examination**
- ✓ Take vital signs
- ✓ Do growth assessment
- ✓ Signs of shock and dehydration (cold hands, slow capillary refill, weak and rapid pulse)
- ☐ **Investigation:** CBC, stool exam, chest X-ray, blood culture, and HIV test



Admission criteria

less than 6 months or less than 3 Kg

- ☐ Weight –for- Length (WFL) less than 70% or $< -3Z$ score OR
- ☐ Visible Severe Wasting if it is difficult to determine WFL
- ☐ Presence of pitting Oedema of both feet

Admission criteria cont....

Children 6 months to 5 years:

- ☐ Weight –for- Length (WFL) / WFH less than 70 % or $< -3Z$ score
OR
- ☐ Presence of pitting Oedema of both feet, OR
- ☐ MUAC < 11.5 cm with length > 65 cm

Out-patient Treatment Program (OTP):

- ❑ Children with SAM and **has no medical complications**, and who **pass the appetite test**
- ❑ classified as uncomplicated SAM and they are treated in OTP with RUTF and routine medicines.
- ❑ These are taken at home, and the child attends the outpatient care site every one week.

In-patient care (SC/TFU)

- Infants **below six months of age with SAM** OR
- **Children 6 months to 5 years with SAM** who have any one of the medical complications or failed appetite test OR
- Children with **SAM and referred from OTP for in-patient care** OR
- When **OTP is not available in your working area** OR
- where the care taker leave or **if the care taker's choice is inpatient care**, all SAM children need to be admitted for inpatient care even if they do not fulfill the In-patient admission criteria

Medical Complications

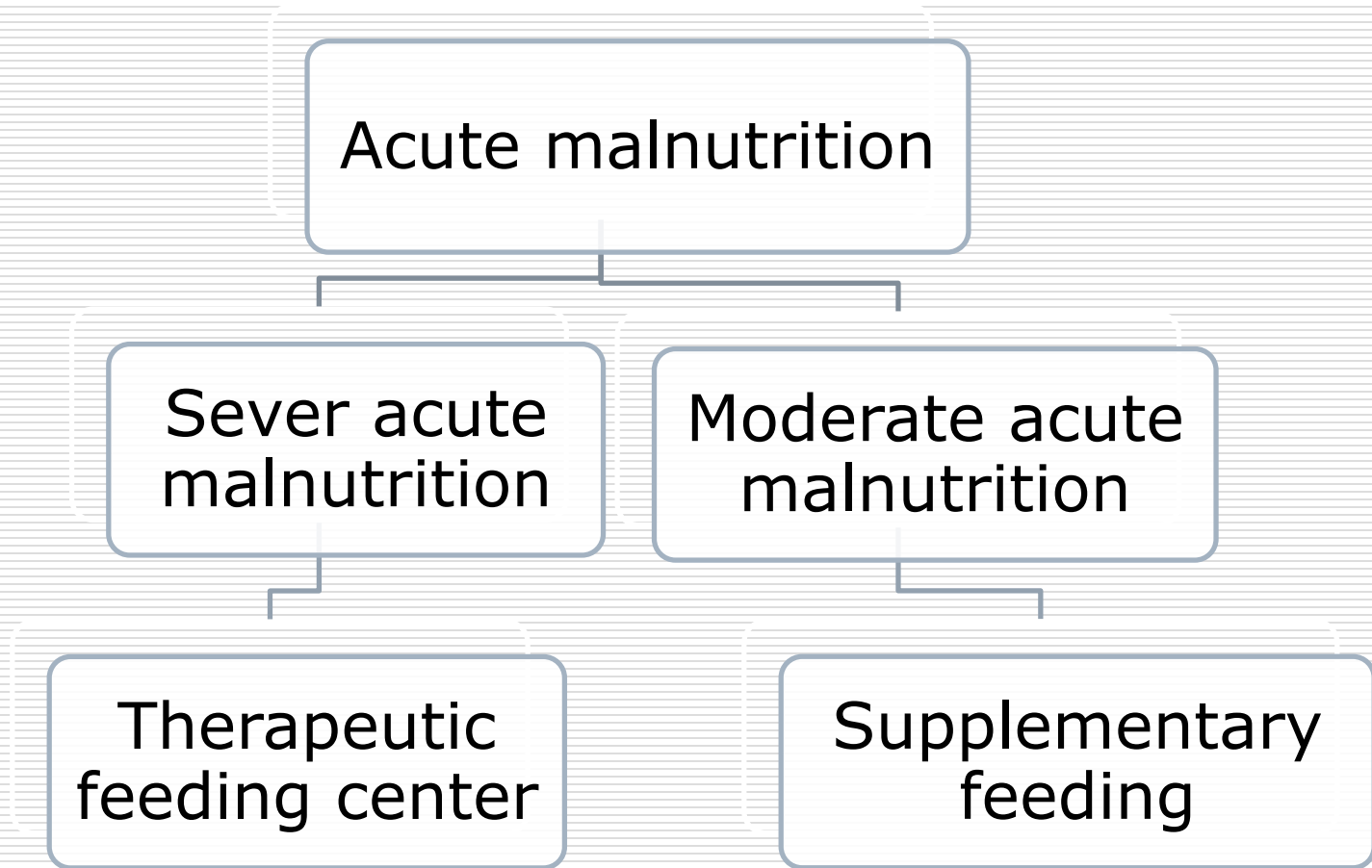
- ☐ Hypoglycemia
- ☐ Severe anemia
- ☐ Watery diarrhoea and/or vomiting and dehydration
- ☐ Infections
- ☐ Hypothermia: axillary temp $< 35.5\text{ }^{\circ}\text{C}$ (new evidence $35\text{ }^{\circ}\text{C}$)
- ☐ Fever $\geq 38.5\text{ }^{\circ}\text{C}$ (new evidence $39\text{ }^{\circ}\text{C}$)
- ☐ Pneumonia/severe pneumonia

management of sever acute Malnutrition

Treatment Approach

- Treatment of complications
- Routine medicines
- Dietary Treatment like Phase I, Transition phase, Phase II

WHO classification for selecting treatment modalities



Management has three phase

Phase 1

- ☐ admitted to an in-patient facility for Phase 1 treatment
- ☐ The formula used during this phase (F75)
- ☐ promotes recovery of normal metabolic function and nutrition-electrolytic balance.
- ☐ Rapid weight gain at this stage is dangerous, that is why F75 is formulated so that patients do not gain weight during this stage.

Transition Phase

- ❑ Children improving in PI, uses F100, lasts 2-4days until edema subsides, appetite has returned, no illness, no NGT or IV
- ❑ weight gain should be around 6 g/kg/day
- ❑ The quantity of F100 given is equal to the quantity of F75 given in Phase 1

Phase 2

- ❖ The last phase, all from TP, uses F100
- ❖ Children stay until edema has disappeared for 10days or WFH is >85% or target weight (more than 8 g/kg/day) is met for 2 consecutive days

The formula milks

- ❑ F-75 a starter/stabilizes/ formula for 2-7days, it has low protein 0.9gm and high carbohydrate 75kcal per 100ml
- ❑ F-100 a catch up formula to rebuild wasted tissue, it has high pr 2.9gm and CHO 100kcal, per 100ml

Constituent	Amount per 100 ml	
	F-75	F-100
Energy	75 kcal _{th} (315 kJ)	100 kcal _{th} (420 kJ)
Protein	0.9 g	2.9 g
Lactose	1.3 g	4.2 g
Potassium	3.6 mmol	5.9 mmol
Sodium	0.6 mmol	1.9 mmol
Magnesium	0.43 mmol	0.73 mmol
Zinc	2.0 mg	2.3 mg
Copper	0.25 mg	0.25 mg
Percentage of energy from:		
protein	5%	12%
fat	32%	53%
Osmolarity	333 mOsmol/l	419 mOsmol/l

Criteria to move from Phase I to transition phase or SC/TFU Exit criteria

- Return of appetite
- Edema reduced
- Clinically well
- No NGT or IV line

Criteria to go Back from TRP to PI if

- Sign of fluid over load
- Develops medical complications
- Poor appetite (feeds <75%)
- Rapid wt. gain >10gm/kg/day
- Develops edema or increase in edema
- If there is a need for NGT or IV

Criteria to Move from TRP to PII or OTP If

- ☐ Good appetite (eats/drinks $>90\%$ RUTF or F100 in TRP)
- ☐ Edema reduced if to OTP or edema subsides if to PII

in-patient

Criteria to move back from PII to PI or from OTP to SC/TFU

- ✓ Any serious medical complication
- ✓ Failure of appetite test
- ✓ Develops or increased edema
- ✓ Fulfilling the failure to respond criteria
- ✓ Weight loss for two consecutive visits
- ✓ Weight loss more than 5% body weight at any time
- ✓ Static weight for 3 consecutive visits
- ✓ Major illness or death of main caretaker and the substitute prefers
In- patient care

Failure to respond

Criteria for failure to respond	
Primary failure to respond (phase 1)	
Failure to regain appetite	Day 4
Failure to start to loose Edema	Day 4
Oedema still present	Day 10
Failure to enter phase 2 and gain $> 5\text{g/kg/d}$	Day 10
Secondary failure to respond	
Failure to gain more than 5g/kg/d for 3 successive days	During Phase 2

Criteria of failure to respond for outpatients

Criteria for failure to respond	Time after admission
Primary failure to respond	
Failure to gain any weight (non-oedematous children)	21 days
Failure to start to loose Edema	14 days
Oedema still present	21 days
Secondary failure to respond	
Failure of Appetite test	At any visit
Weight loss of 5% of body weight	At any visit
Weight loss for two successive visits	During OTP care
Failure to gain more than 2.5g/kg/d for 21days (after loss of oedema (kwashiorkor) or after day 14 (marasmus))	During OTP care

Discharge criteria

Age 6 months – 5 years Discharge Criteria

- ☐ weight-for-length or height $\geq 85\%$ and
- ☐ no oedema for 10 days (In-patient) or
- ☐ Target weight gain if the child is admitted with MUAC

Cont....

If admitted with bilateral pitting oedema, discharge cured when:

- No bilateral pitting oedema for 2 consecutive visits AND
- MUAC ≥ 12.5 cm or WFH/WFL ≥ -2 z-score AND
- Clinically well and alert

If admitted based on MUAC, discharge cured when:

- MUAC ≥ 12.5 cm AND
- No bilateral pitting oedema AND
- Clinically well and alert

If admitted based on WFH/WFL, discharge cured when:

- WFH/WFL ≥ -2 z-score AND
- No bilateral pitting oedema AND
- Clinically well and alert

Cont....

Age <6 months on breastfeeding or with a caretaker willing to breast feed

- Infant gaining weight on breast milk alone
- No medical problem
- There are no anthropometric criteria for discharge of the fully breast-fed infant who is gaining weight

Age <6 months who cannot be breast-fed

- When they reach 85% weight for length they can be switched to infant formula.

Routine medicines

- ☐ Vitamin A(one dose at admission and one dose at discharge)
- ☐ Folic acid-if sign of anemia
- ☐ Antibiotics-phase one plus 4 day on transition.
- ☐ Measles-
- ☐ Deworming-
- ☐ Anti malarial if appropriate

VITAMIN A

- ☐ On the day of admission (day 1), give vitamin A for all children except those with oedema or those who received vitamin A in the past 6 months
- ☐ Give vitamin A to every patient on the day of discharge (in-patient care) or at the 4th week of the treatment for those in out-patient care
- ☐ Dose: <6 months: 50,000IU PO at admission
- ☐ 6-11 months: 100,000IU
- ☐ ≥ 12 months: 200,000IU

Folic Acid

- On the day of admission, one single dose of folic acid (5mg for 6 months to 5 years and 2.5mg stat for infants) can be given to children with clinical signs of anaemia.

Antibiotics

- ❑ Antibiotics should be given to every patient with SAM, even if they do not have clinical signs of systemic infection
- ❑ If no complications, give **first line antibiotics**: oral Amoxicillin for 7 days (preferred) or cotrimoxazolen (**new evidence ampicillin**)
- ❑ If complications present, give **second line antibiotics**: Gentamicin, plus IV Ampicillin for 2 days followed by Amoxicillin depending on the condition of the child, for a total treatment duration of 7 days

Measles vaccine

- All children from 9 months without a vaccination card should be given measles vaccine both on admission and discharge after Phase 2

Deworming

- ❑ Albendazole or Mebendazole is given for children ≥ 2 year at the start of Phase 2 for patients that will remain as in patients.
- ❑ For both those transferred from in-patients to Phase 2 as out-patients and those admitted directly to OTP de-worming is given at the 2nd out-patient visit (after 7 days)
- Reference : National Guideline for the Management of Acute Malnutrition in Ethiopia May 2019

CHAPTER SIX

1. New born disorders (7hrs)

- Birth Asphaxia
- HMD(Respiratory Distress Syndrome)
- Meconium Aspiration Syndrome
- Jaundice
- Sepsis, Meningitis, Necrotizing Enter colitis, Neonatal tetanus, Congenital infection, TORCHS

- Infants of diabetic mother

Birth Asphyxia

Objectives

☐ **At the end of the lesson you will be able to:**

- ▶ Define Perinatal asphyxia
- ▶ Identify the main features of perinatal asphyxia
- ▶ Identify factors of perinatal asphyxia
- ▶ Identify management protocol of perinatal asphyxia
- ▶ Identify who needs resuscitation
- ▶ Explain steps of resuscitation

Perinatal Asphyxia

- World Health Organization (WHO) defines birth asphyxia as:
 - ▶ **Failure to initiate and sustain breathing at birth.**
- It is placental or pulmonary **gas exchange impairment** leading to **hypoxemia** and **hypercarbia**.
- **Asphyxia** is failure of the baby to **breathe spontaneously at birth.**

Cont...

- ❑ **Asphyxia can start before or after the baby is born.**
- ❑ **If the baby has asphyxia:**
 - ▶ The baby has trouble in breathing (**gasping** or **breathing very irregularly or no breathing**),
 - ▶ The baby's color is **pale** or **blue**. This is when the skill of newborn resuscitation can save the life of the newborn.

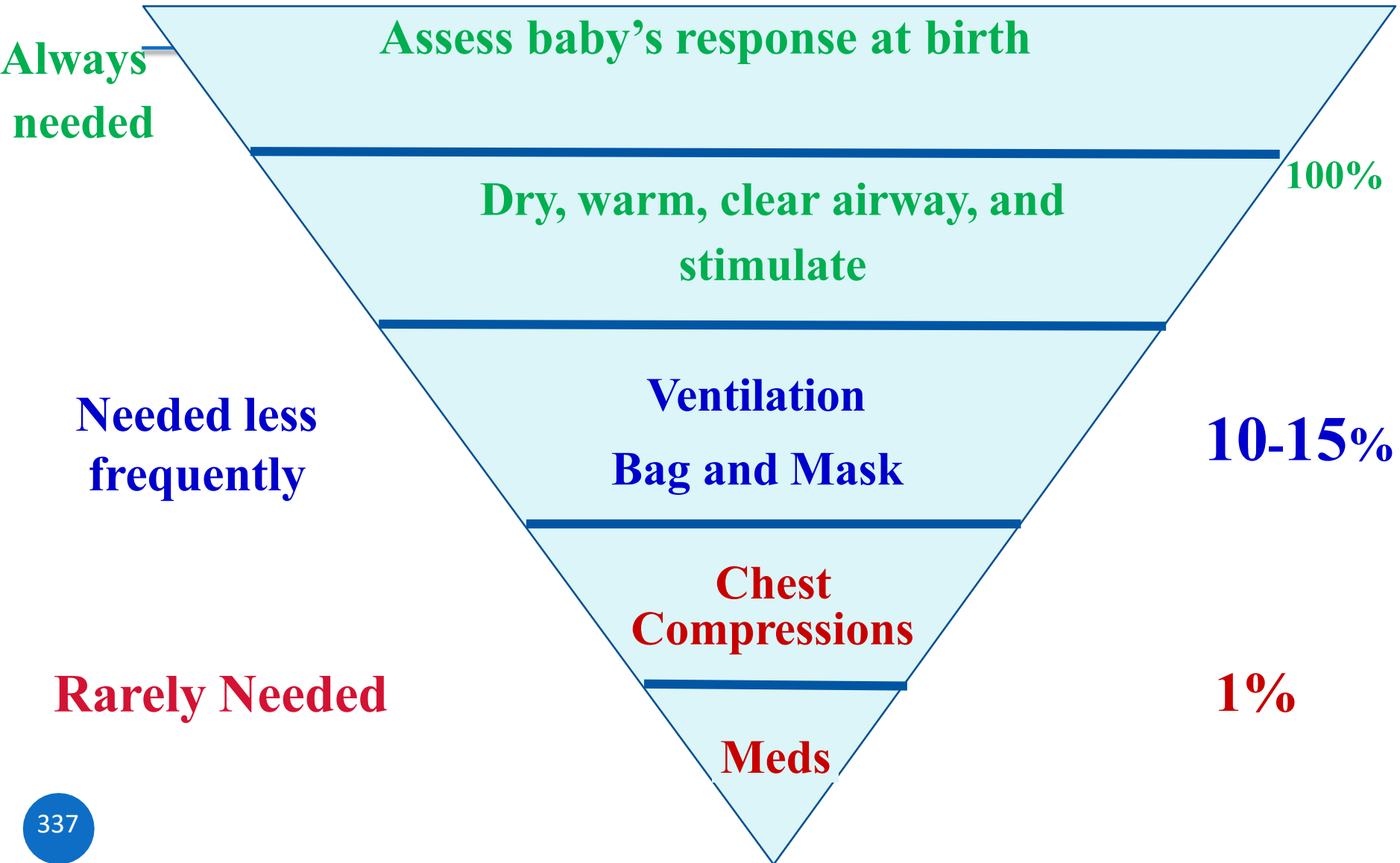
Management of asphyxia in the newborn

- ☐ Perinatal management of high risk pregnancies and early detection of fetal hypoxia
- ☐ Neonatal resuscitation

What is neonatal resuscitation?

- ☐ Neonatal resuscitation means to **revive** or **restore** life to a baby.
- ☐ **Who need's neonatal resuscitation???**

Basic Steps in Resuscitation



ABC's of Resuscitation

☐ **A**irway

- ▶ Position and clear

☐ **B**reathing

- ▶ Stimulate to breathe

☐ **C**irculation

- ▶ Assess heart rate

Risk Factors Associated with the Need for Resuscitation

☐ Maternal Risk Factors Before Birth:

- ▶ Ruptured membranes over 18 hours
- ▶ Pre-eclampsia and Eclampsia
- ▶ Maternal infection:- Malaria, HIV, etc.
- ▶ Premature labor
- ▶ Multiple births
- ▶ Post-term gestation

Cont...

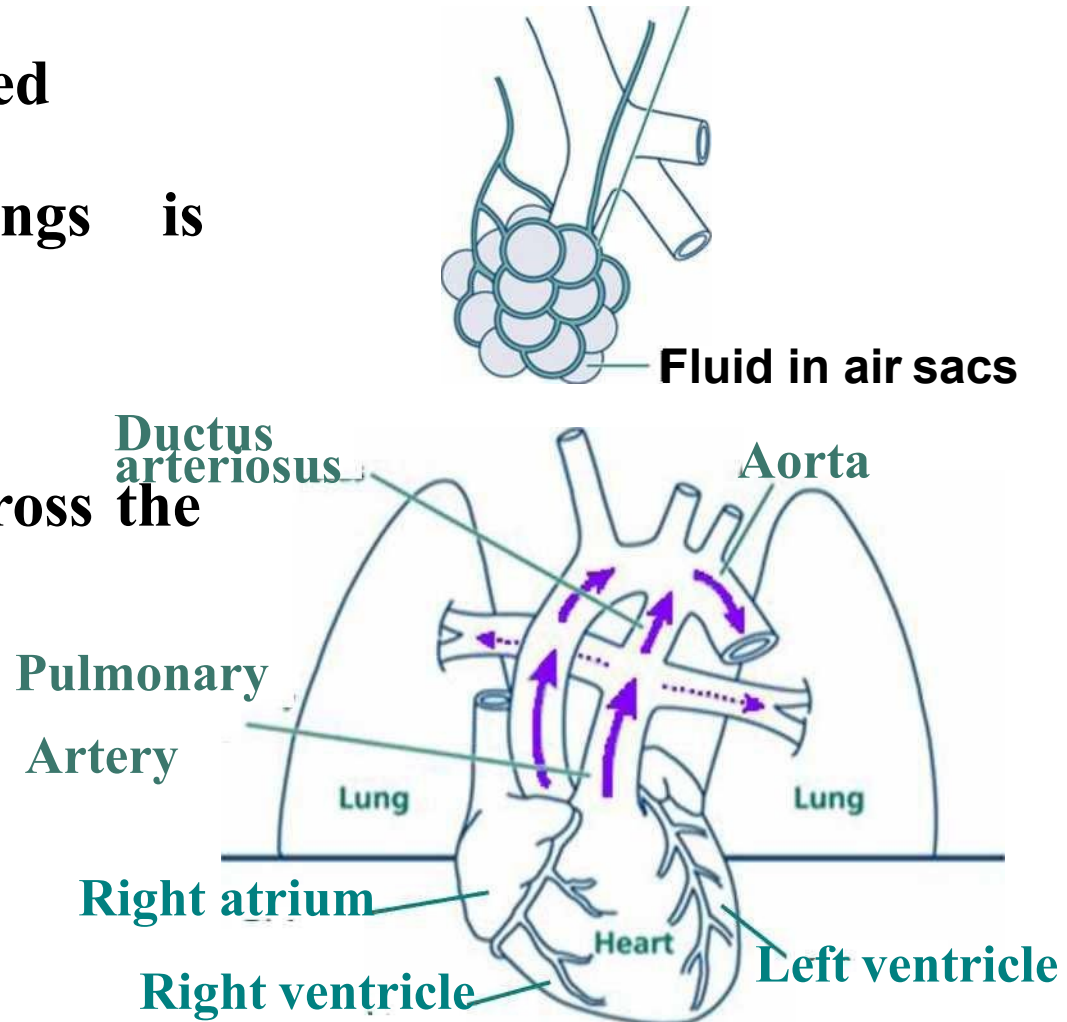
- ☐ Bleeding in second or third trimester
- ☐ Maternal diabetes
- ☐ Age <16 or >35 years
- ☐ Anemia
- ☐ No prenatal care
- ☐ Previous fetal or neonatal death
- ☐ Diminished fetal activity

Risk Factors During labor

- ▶ Excessive bleeding
- ▶ Breech presentation
- ▶ Meconium
- ▶ Abnormal fetal heart tones
- ▶ Prolapsed or nuchal cord
- ▶ Rapid, hard labor
- ▶ Foul smelling fluid
- ▶ Prolonged labor
- ▶ Shoulder dystocia

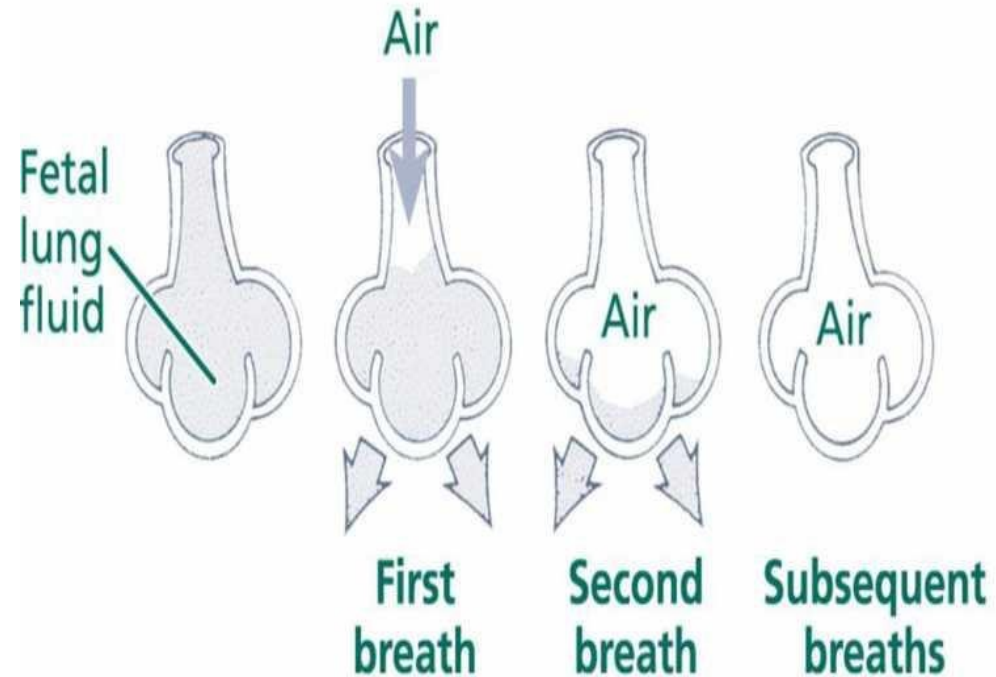
Blood Flow Before Delivery

- ☐ Blood vessels are constricted
- ☐ Blood flow in the lungs is decreased
- ☐ Blood flow is diverted across the ductus arteriosus



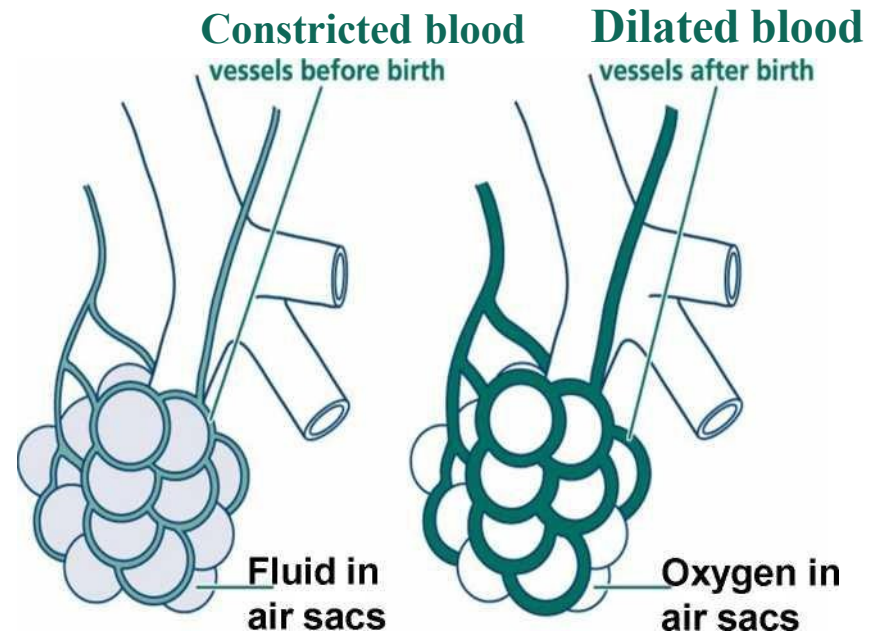
Blood Flow Changes at Birth

- ☐ Lungs expand with air
- ☐ Fetal lung fluid leaves air sacs
- ☐ Fluid replaced by air in air sacs



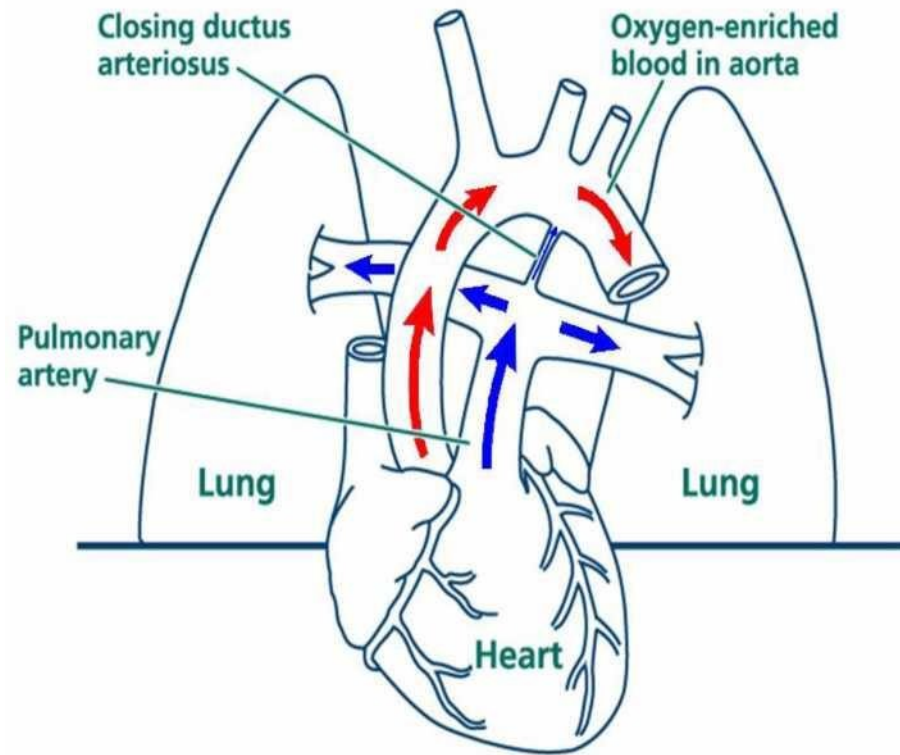
Blood Flow Changes at Birth...

- ☐ Umbilical cord clamped
- ☐ Blood flow in the lungs increases
- ☐ Blood vessels in the lung tissue relax



Blood Flow Changes at Birth...

- ☐ Blood oxygen levels rise
- ☐ Ductus arteriosus begins to close
- ☐ Blood flows through lungs to pick up oxygen



Normal Transition

- ☐ No meconium
- ☐ Breathing/crying
- ☐ Good muscle tone
- ☐ Term



Abnormal Transition

☐ Gasping, ineffective or no breathing

☐ Poor muscle tone

☐ Central cyanosis (blue)

**All need assisted
ventilation!**



Premature Infants at Higher Risk

- ☐ Immature lungs - increased respiratory distress
- ☐ Increased heat loss and poor temperature control
- ☐ Possible infection
- ☐ Prone to bleeding in the brain
- ☐ Feeding problems



Preparation for a delivery

☐ Basic equipment:

☐ Heat source

☐ Linens

☐ Suction

☐ Bag and mask

☐ A person trained in neonatal resuscitation

Cont...

- At every delivery, there should be at least one person whose primary responsibility is the baby and who is capable of performing resuscitation.

Initial Steps

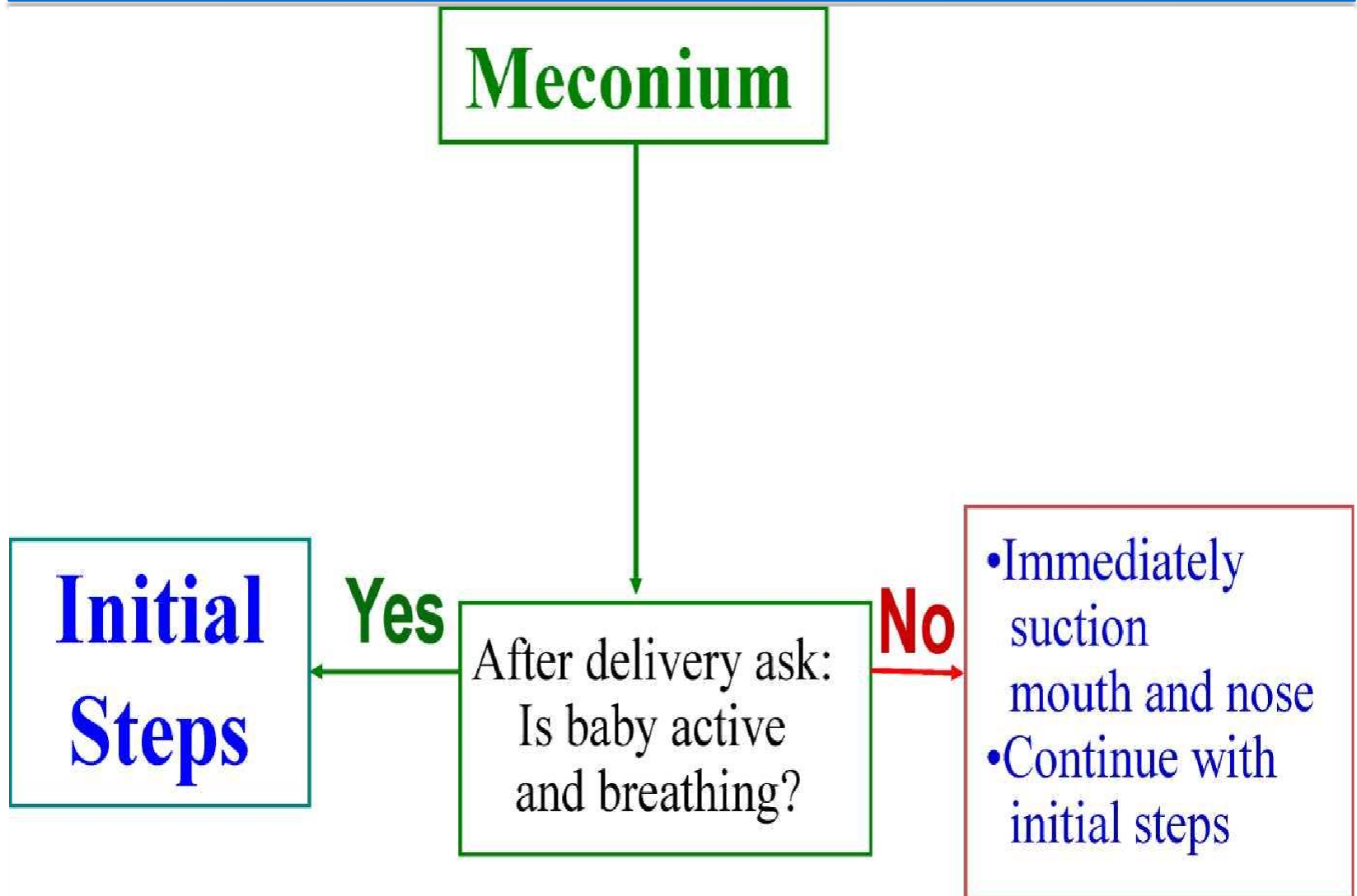
The First 30 Seconds

- ☐ Manage meconium if present
- ☐ Provide warmth
- ☐ Open the airway
- ☐ Suction mouth and nose
- ☐ Dry, stimulate, and reposition head
- ☐ Evaluate

Meconium



Management of Meconium



Provide Warmth

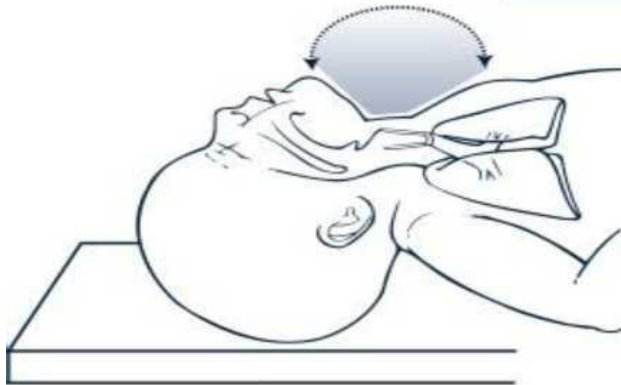
☐ Place baby in warm location:

- ▶ Warm surface, warm blankets
- ▶ Away from draft
- ▶ Heat lamp
- ▶ Overhead warmer, if available
- ▶ Skin to skin, **if routine care**

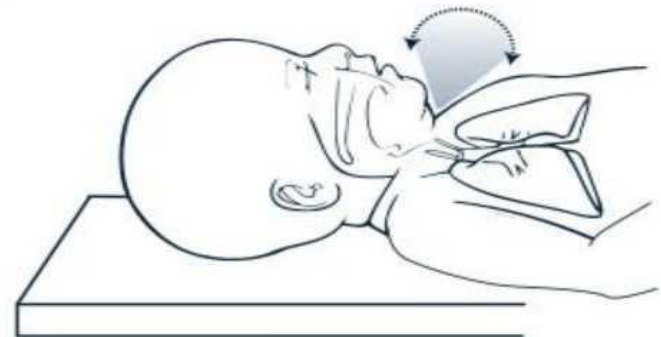
Positioning the Airway



Correct



Incorrect (Hyperextension)



Incorrect (Flexion)

Figure 2.2. Correct and incorrect head positions for resuscitation

Clear the mouth and nose

- ☐ Bulb syringe:
- ☐ Suction mouth first, then nose (“m” before “n”) **why???**
- ☐ No bulb syringe:
 - ▶ Clear secretions with clean, dry cloth
 - ▶ No deep suctioning with bulb syringe or catheter!
- ☐ Bradycardia (slowing of the heart rate) can result from deep suctioning

Cont...



Mouth first..



then nose

Drying the Infant

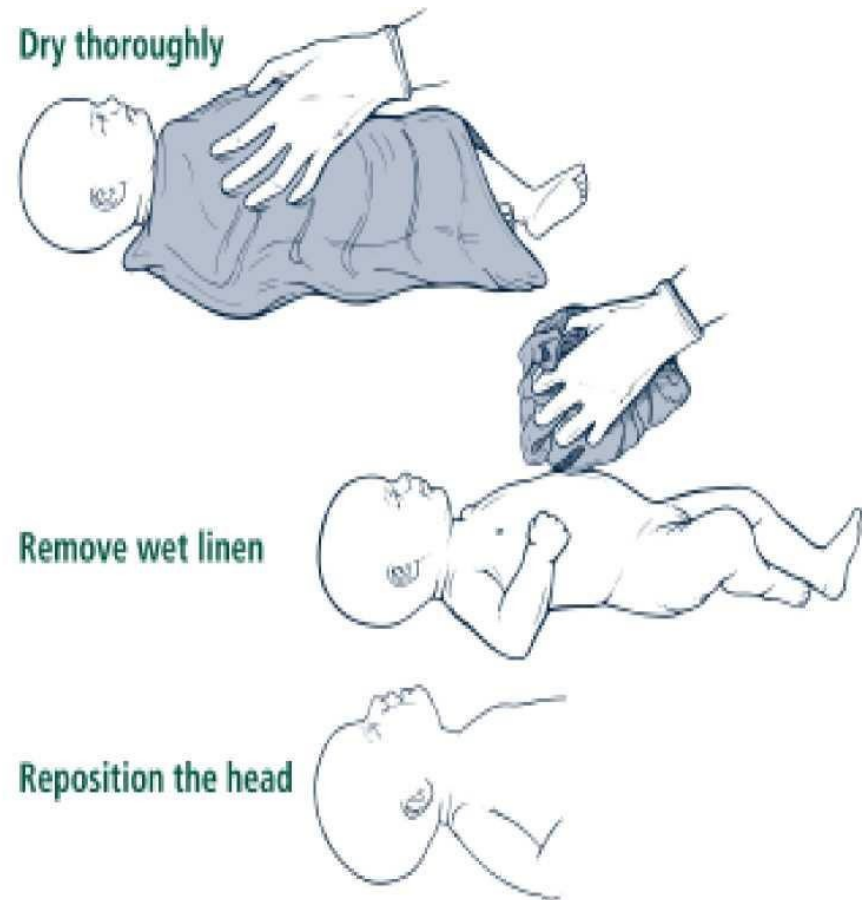


Figure: Drying and removing wet linen to prevent heat loss and repositioning the head to ensure an open air way.

Tactile Stimulation

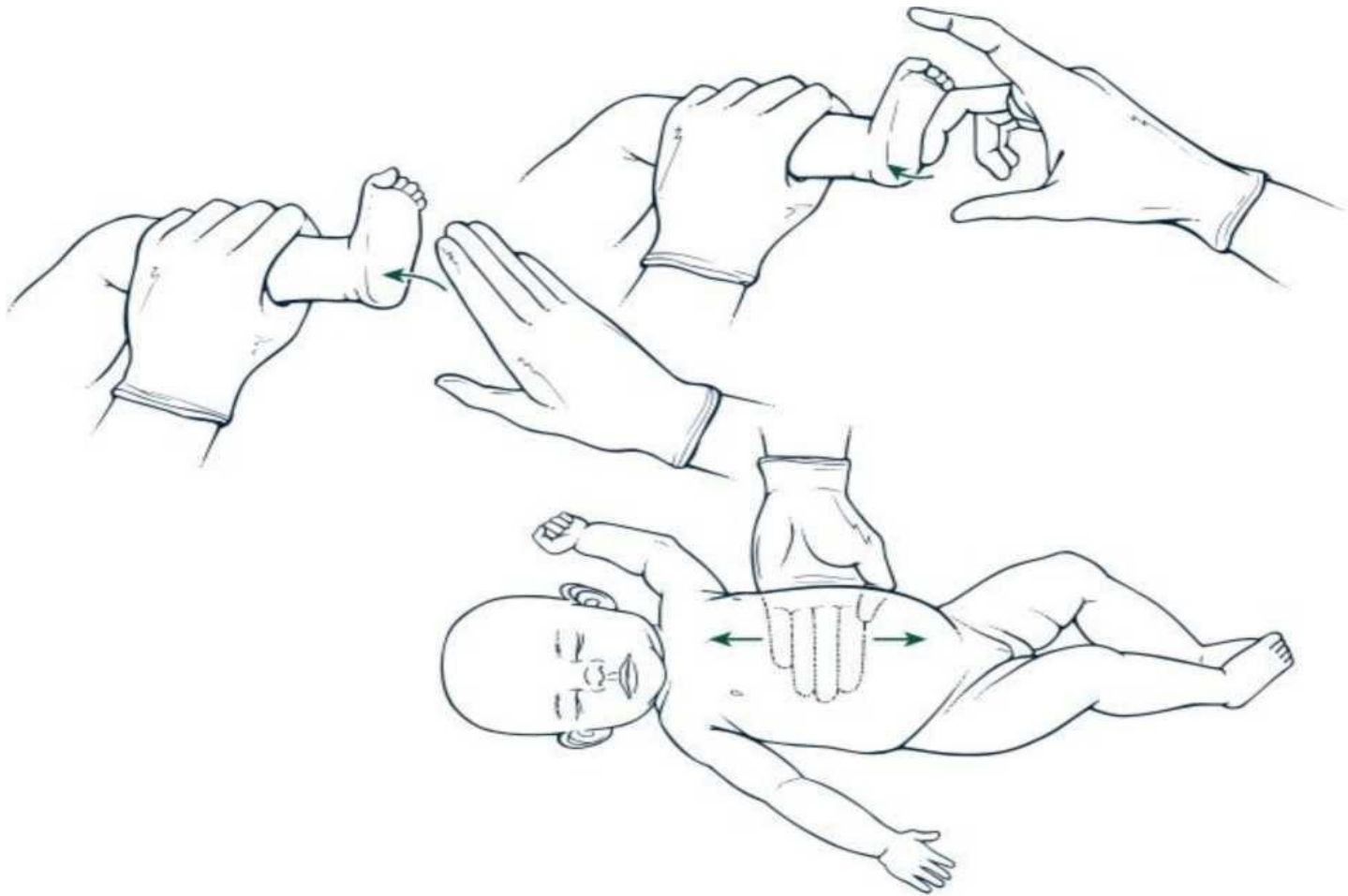


Figure 2.7. Acceptable methods of stimulating a baby to breathe

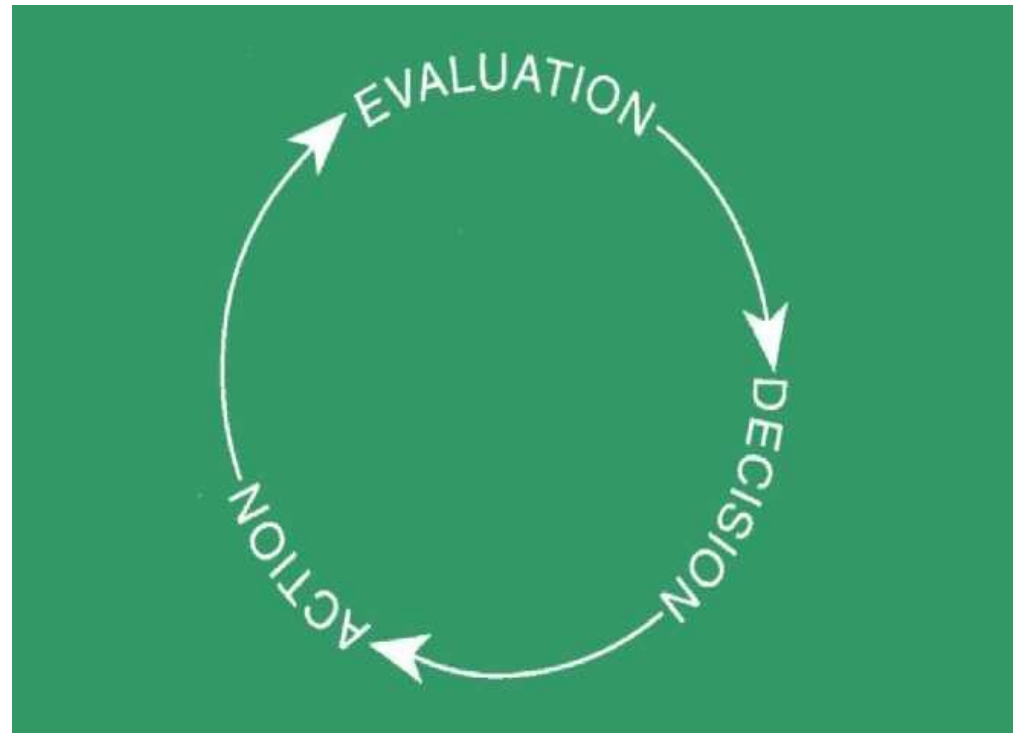
Do **NOT** Stimulate By:

- ☐ Slapping the back
- ☐ Squeezing the rib cage
- ☐ Forcing thighs into the abdomen
- ☐ Hot or cold compresses or baths
- ☐ Shaking

How to Prioritize Actions

□ Evaluation is based on:

- ▶ Respirations (breathing/crying)
- ▶ Heart rate
- ▶ Color



Evaluate Respirations

☐ YES

- ▶ Unlabored breathing/crying
- ▶ Listen with stethoscope

☐ NO

- ▶ If shallow breathing, gasping, or not breathing at all, give **positive-pressure ventilation**

Evaluate Heart Rate

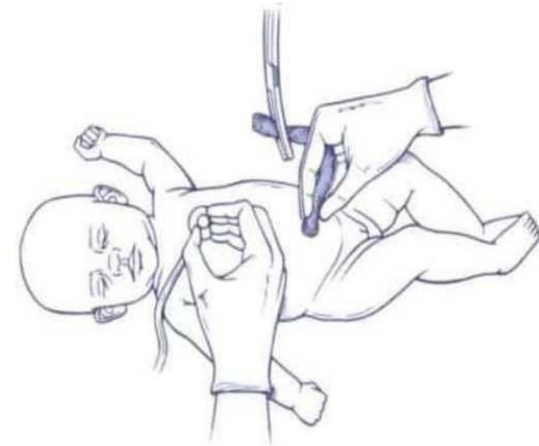
☐ Feel pulse at base of **umbilical cord** or listen with stethoscope at the apex

☐ Count for 6 seconds and multiply by 10

☐ Example:

▶ If you counted 13 pulsations in 6 seconds the baby's heart rate would be $13 \times 10 = 130$

☐ Heart rate < 100 requires **positive- pressure ventilation**



Evaluate Color

□ Peripheral cyanosis (a bluish color):

- ▶ Pink centrally: assess lips and gums
- ▶ Bluish color of hands and feet
- ▶ Requires **no further action**

□ Central cyanosis:

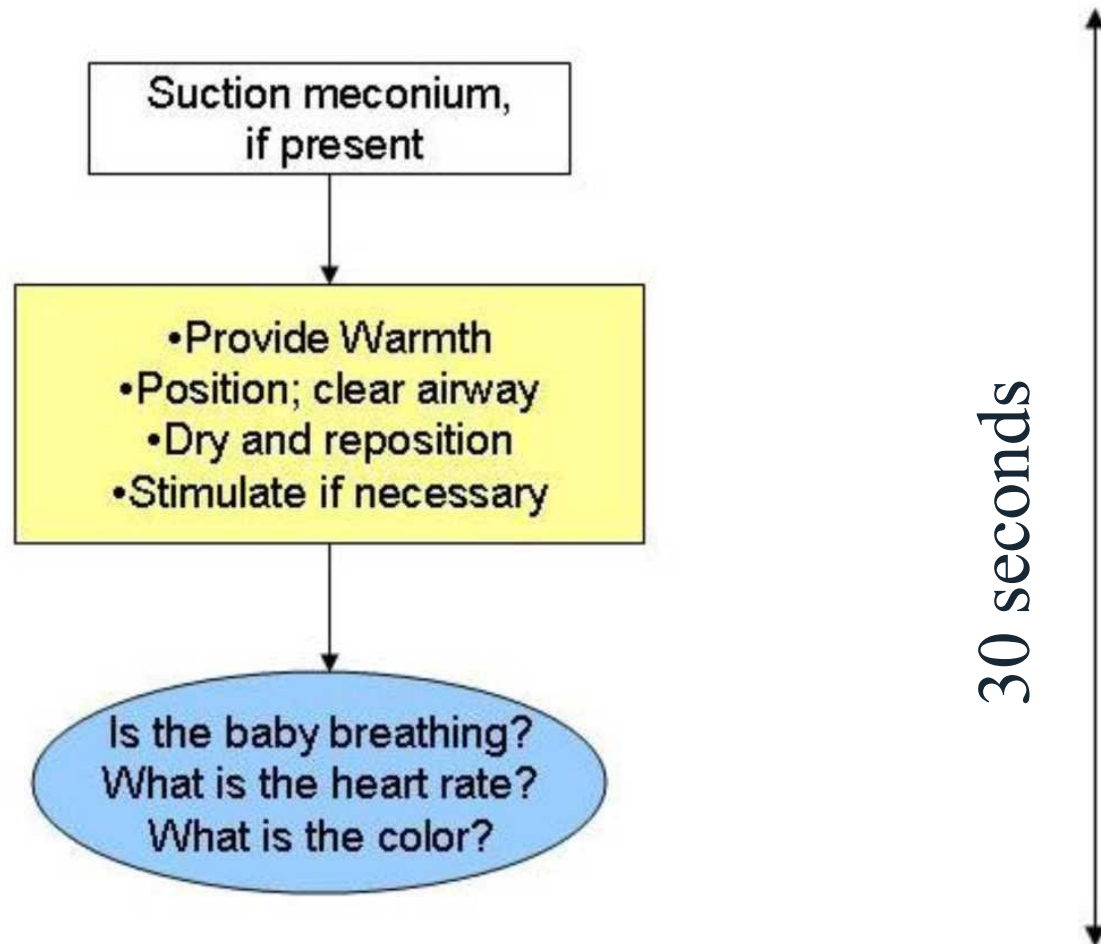
- ▶ Bluish color of the trunk and limbs
- ▶ Bluish color of lips and gums
- ▶ Use free-flow oxygen, if available, or positive-pressure ventilation

Free-flow Oxygen

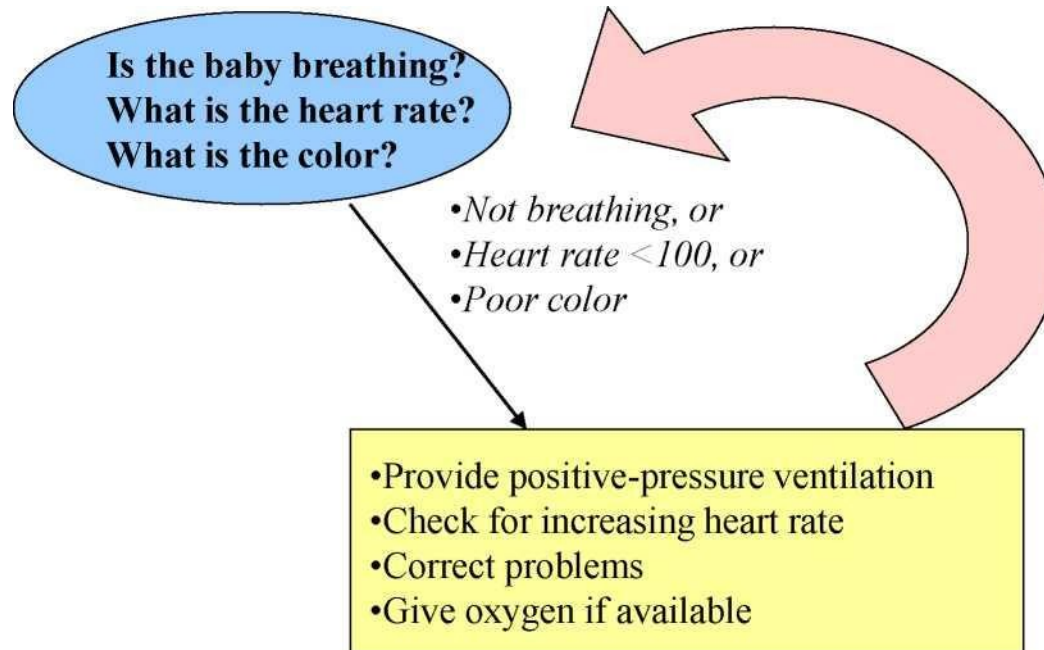
- ☐ If the infant is breathing but there is persistent central cyanosis, give free-flow oxygen.
- ☐ Flow rate at approximately 5-10 L/min
- ☐ If not available, use positive- pressure (bag-and-mask) ventilation



Newborn Resuscitation Flowchart Initial Steps



If the evaluation is NOT normal



2nd 30 seconds

Evaluate These Newborns



Positive-Pressure Ventilation

The Second 30 Seconds * •

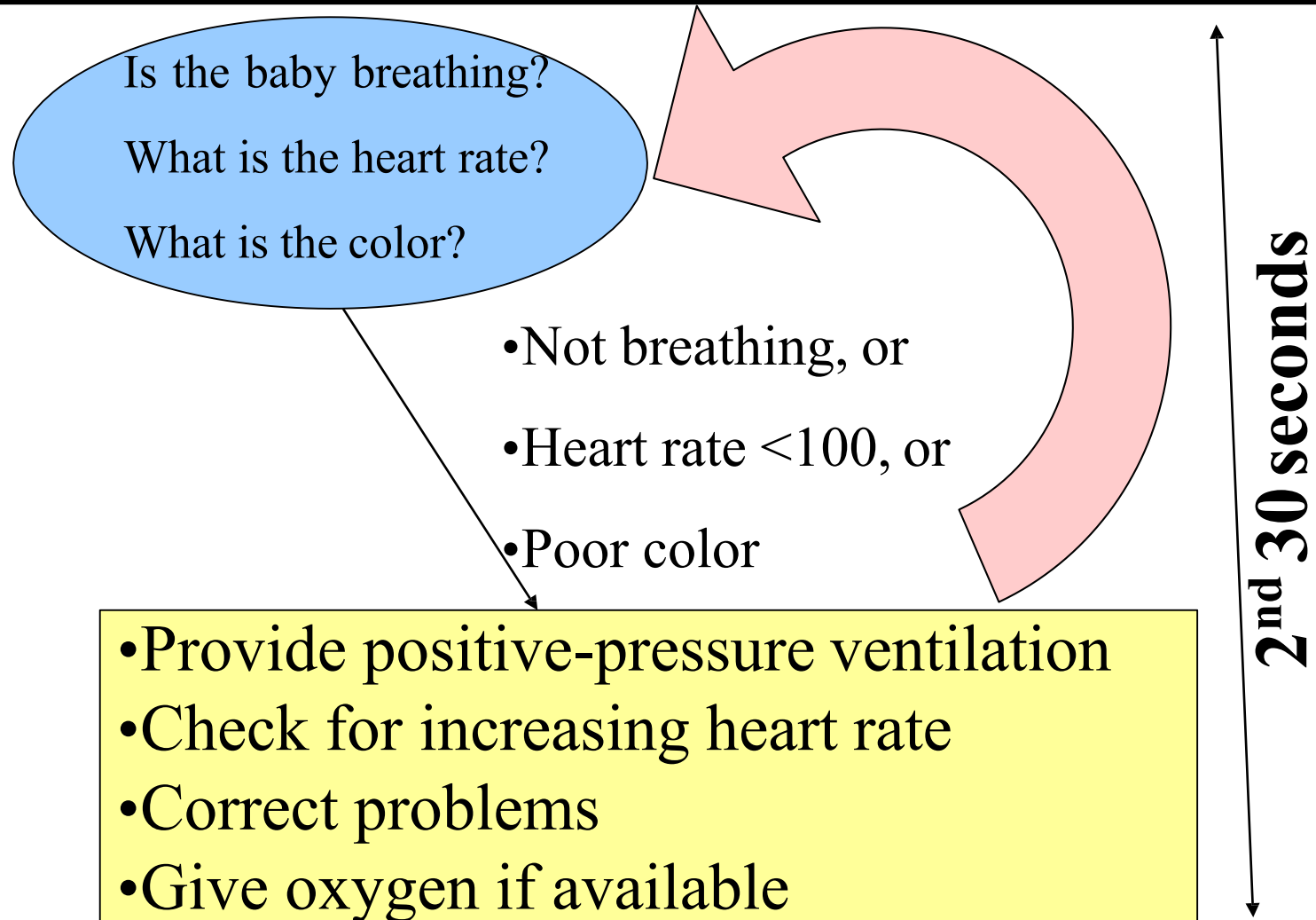
Content:

- ▶ When to ventilate
- ▶ Self-inflating bag and mask
- ▶ Important characteristics of face mask
- ▶ Testing bag and mask
- ▶ Preparing for positive-pressure ventilation
- ▶ What if the baby does not improve

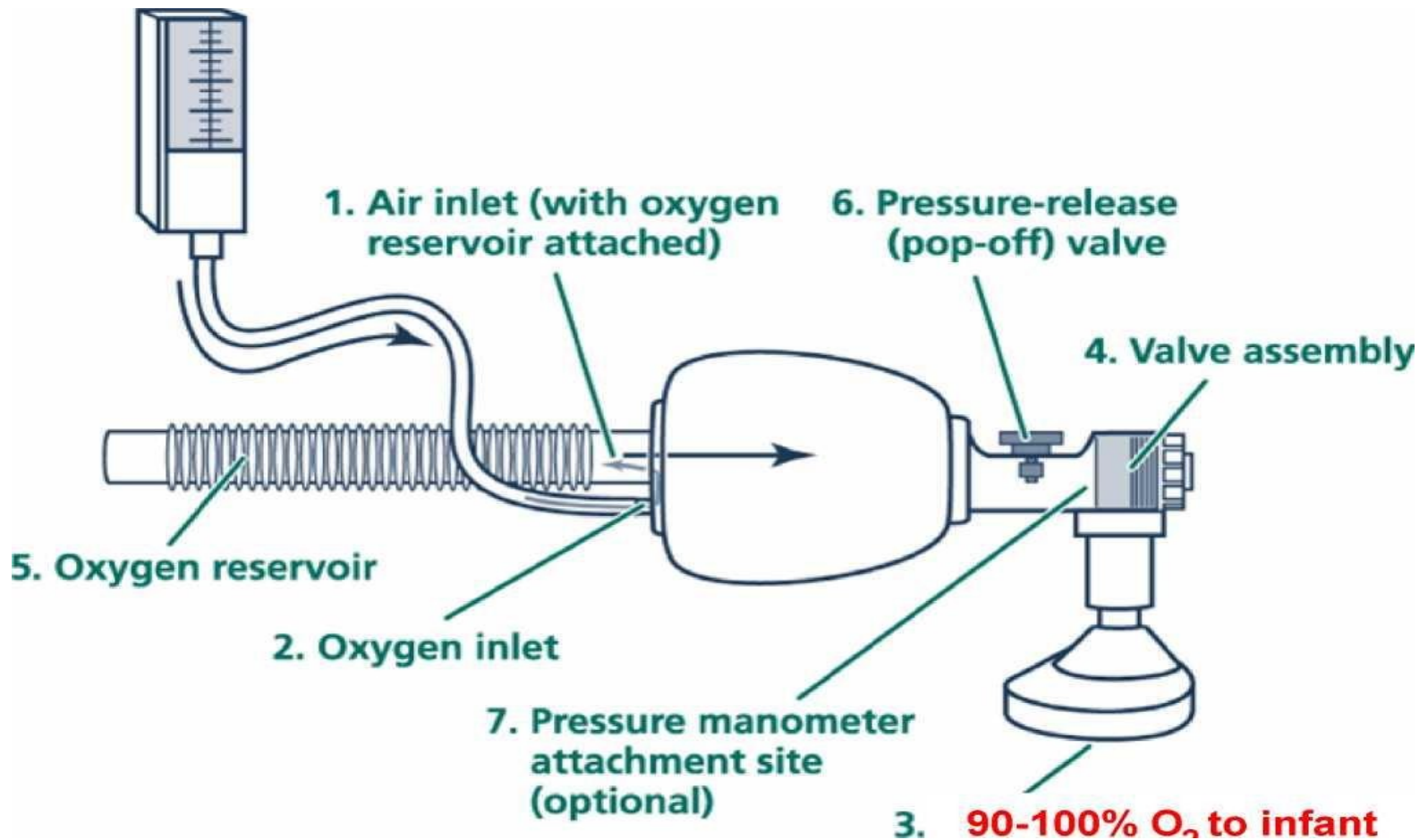
Ventilation

- ☐ Ventilation of the lungs is the single most important and effective step of resuscitation.

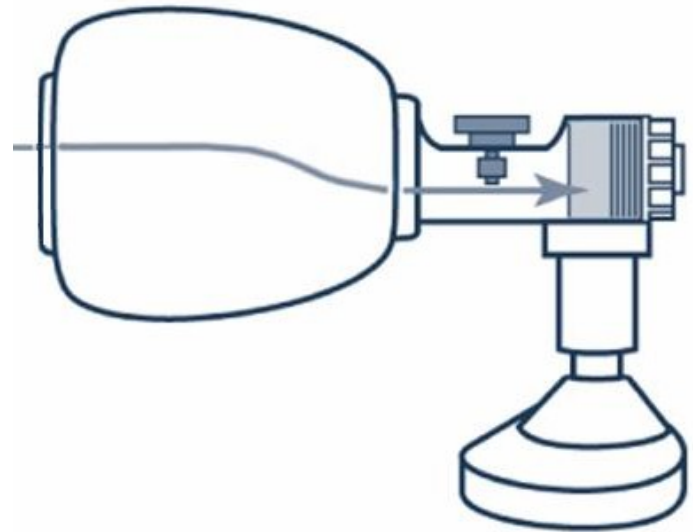
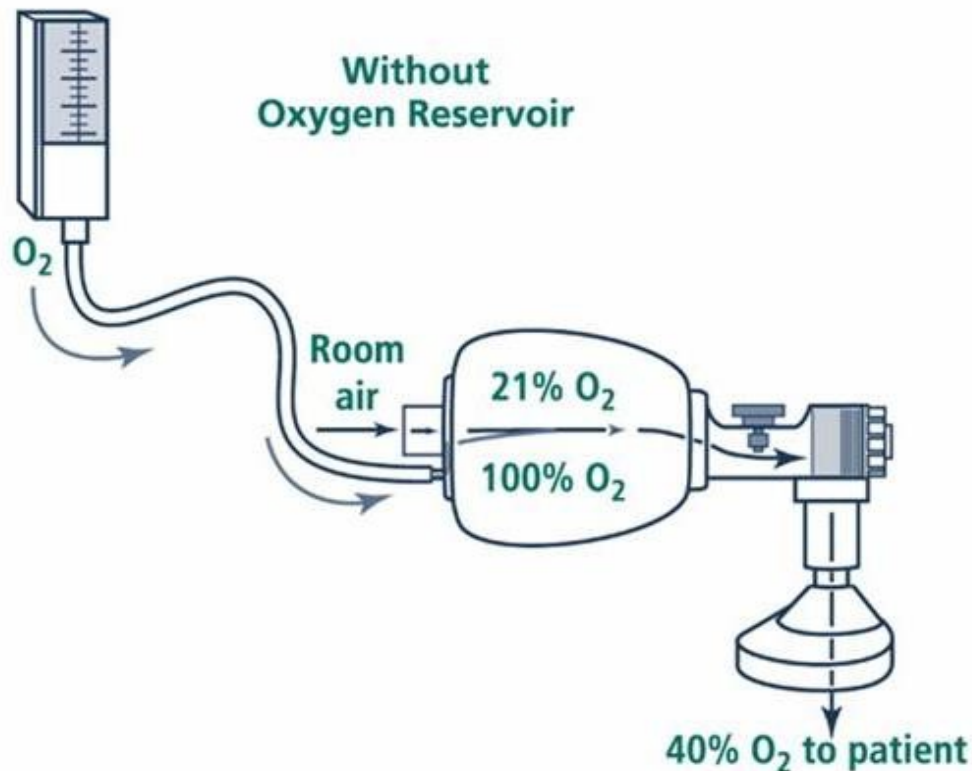
If the evaluation is **NOT** normal



Basic Parts of Bag and Mask



Room air may be used for resuscitation



Room air to infant

Operation of Bag and Mask

Pressure determined by 3 factors:

1. How hard the bag is squeezed
2. Any leaks between mask and face
3. Set point of pressure-release (pop-off) valve

Face Mask Placement

Correct: Covers mouth, nose, and chin

Incorrect:

Too large - covers eyes and extends over chin

Too small - does not cover nose and mouth well



Correct
Covers mouth, nose, and chin but not eyes



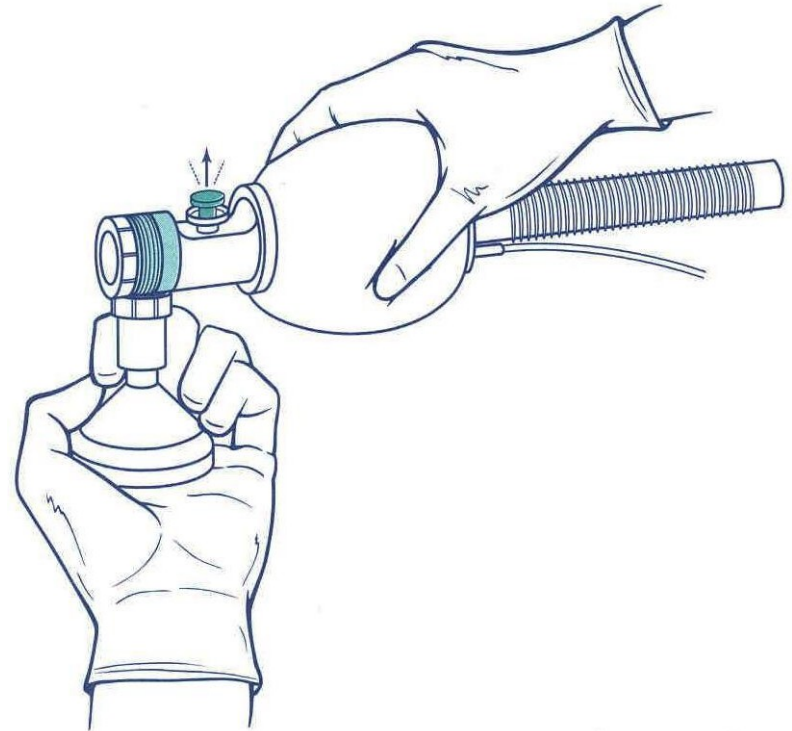
Incorrect
Too large: covers eyes and extends over chin



Incorrect
Too small: does not cover nose and mouth well

Testing Bag and Mask

- ☐ Pressure against your hand?
- ☐ Pressure-release valve opens?

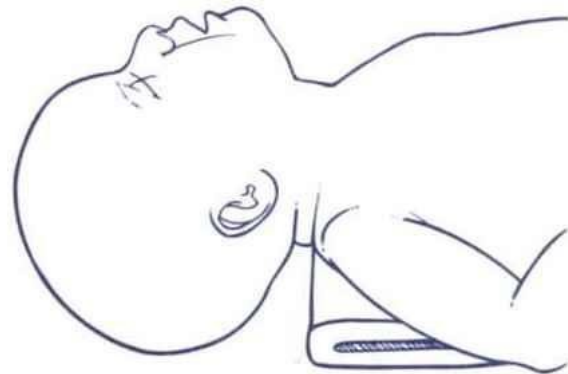


Face Mask Placement

- **Correct Positioning: Cover mouth, nose, and tip of chin, but not the eyes**



- **Correct position for assisted ventilation**



Preparing for Positive Pressure Ventilation

- ☐ Select correct-sized mask
- ☐ Clear airway
- ☐ Position head
- ☐ Position yourself at side or head of baby



Effective Use of Bag

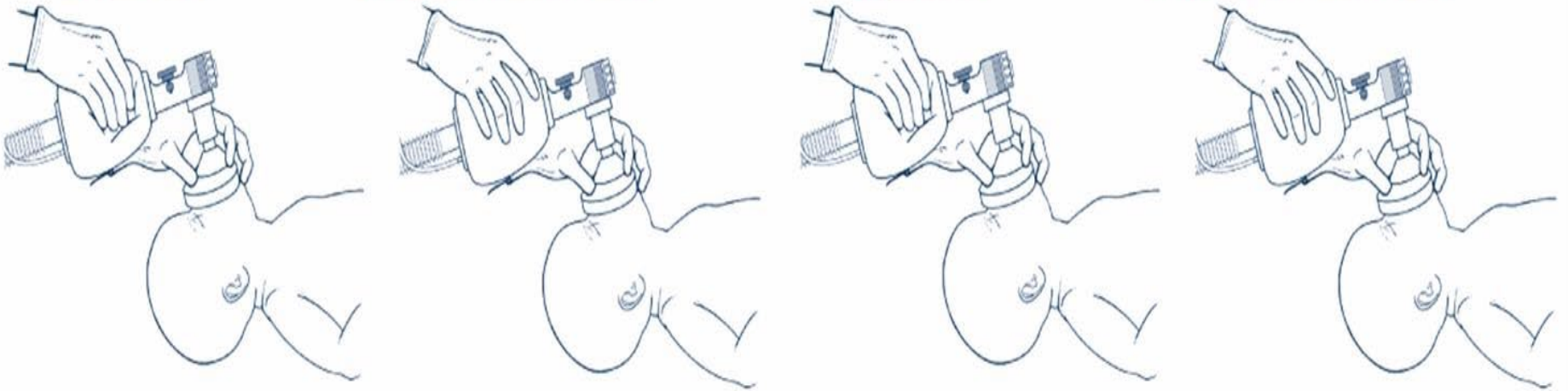
- ☐ The primary sign of good ventilation is an increase in heart rate
- ☐ Apply enough pressure to **create noticeable, gentle rise and fall of chest**
- ☐ Bilateral breath sounds
- ☐ Improvement of **color and heart rate**
- ☐ Initial breaths may require higher pressures
- ☐ If baby appears to be taking a very deep breath, too much pressure is being used

How often should you squeeze the bag?

40-60 breaths per minute

Count out loud: “Breathe—two—three”

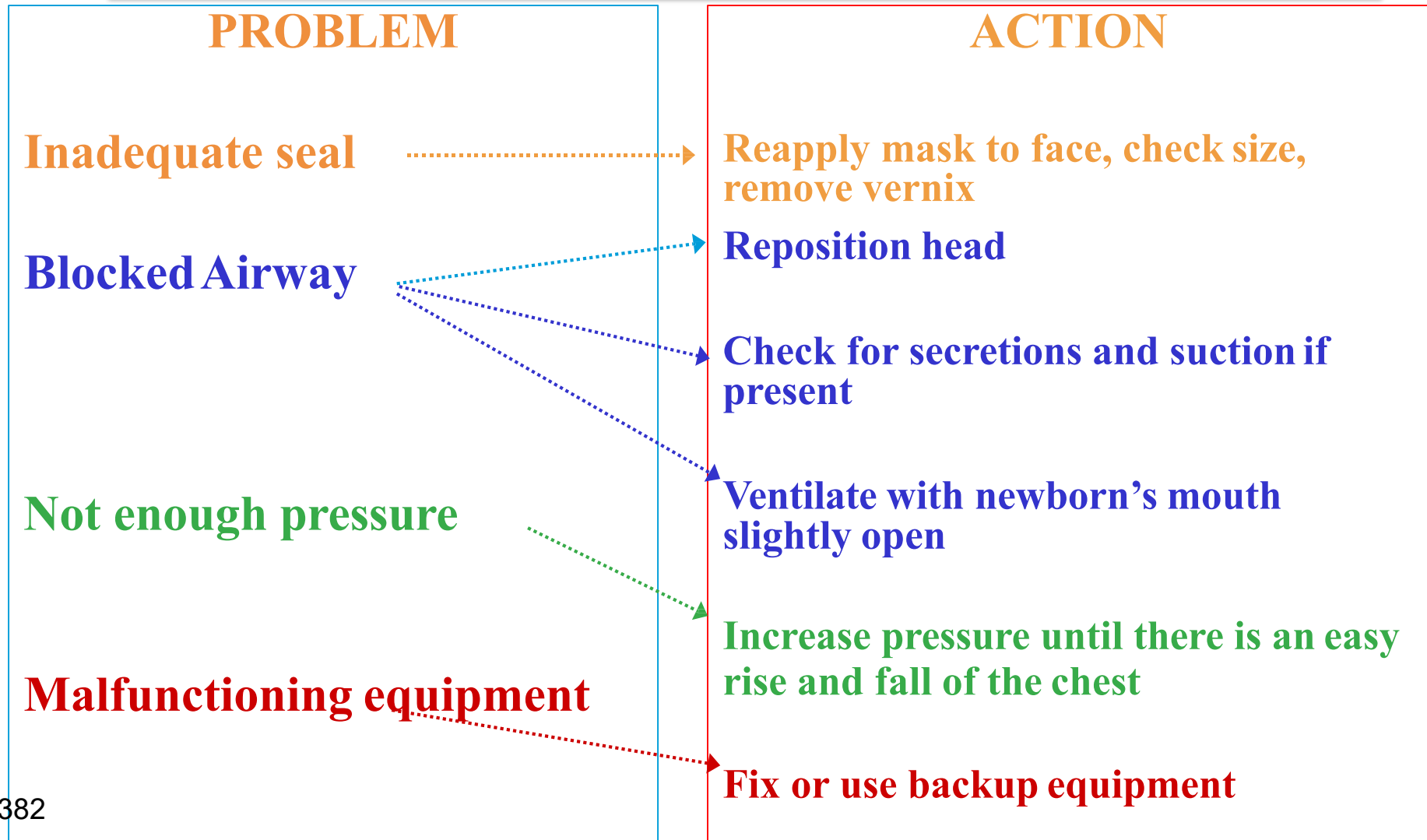
Breathe Two Three Breathe Two Three
(squeeze) (release) (squeeze) (release)



Signs of Improvement

1. **Increasing Heart Rate (>100)**
2. **Improving color**
3. **Spontaneous breathing or crying**
4. **Improving muscle tone**

Inadequate Chest Expansion



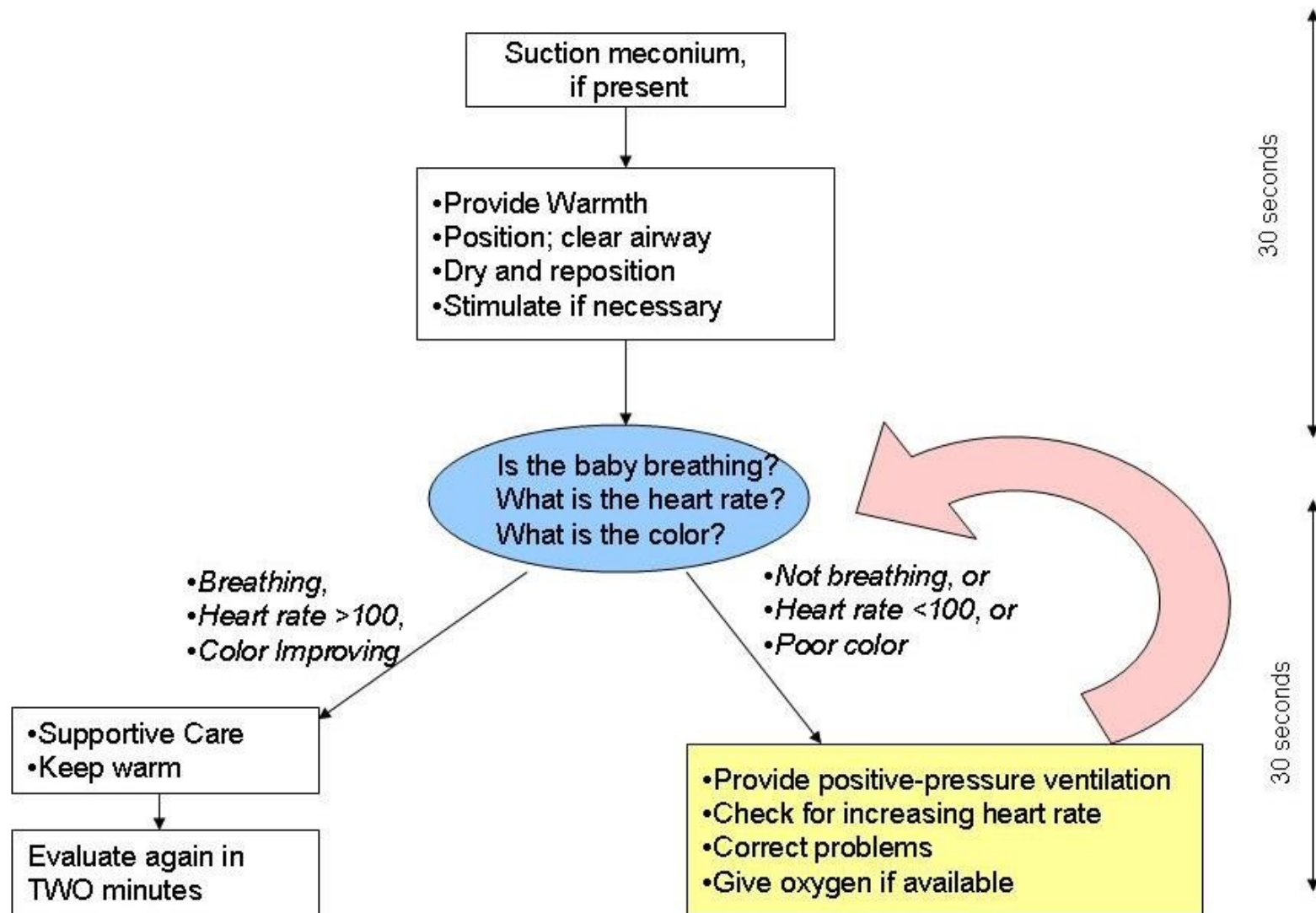
Is adequate oxygen being administered?

If available, is 100% oxygen being given?

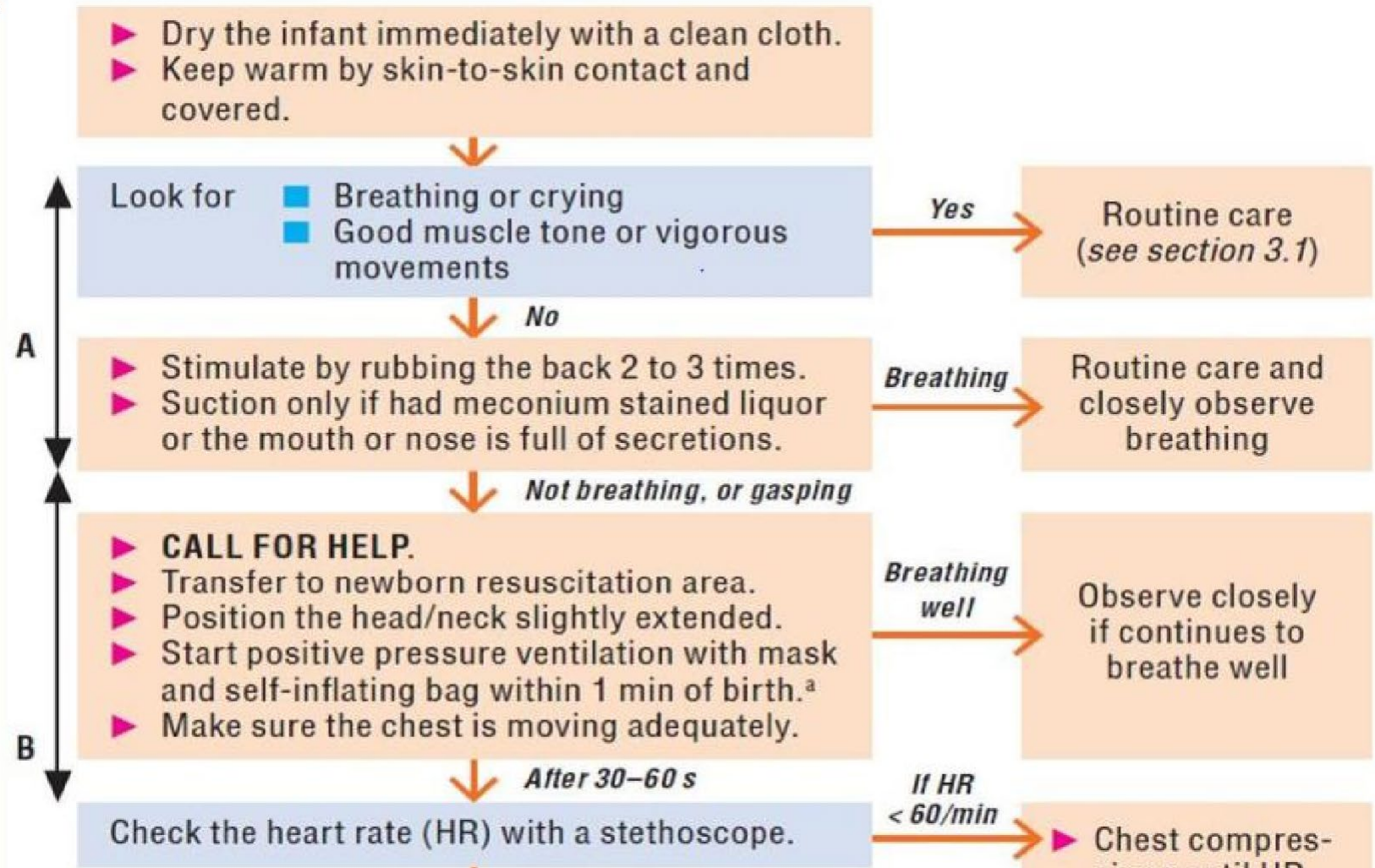
- ▶ Oxygen tubing attached to the bag and the oxygen source?
- ▶ Gas turned on?
- ▶ Oxygen reservoir attached?
- ▶ Oxygen in the tank?

If supplemental oxygen is not available, give positive-pressure ventilation using room air!

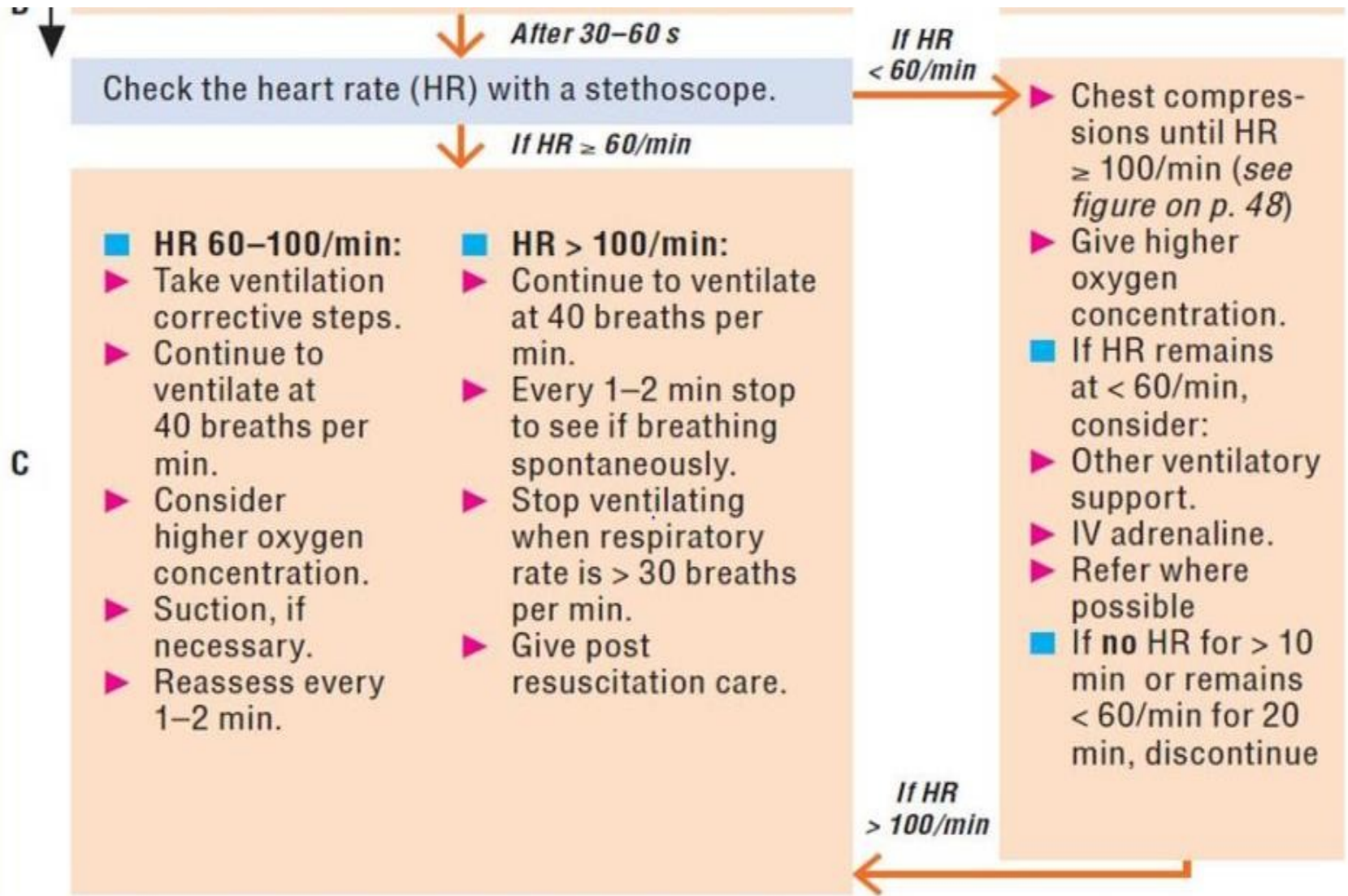
Newborn Resuscitation Flowchart



Neonatal resuscitation Flow chart



Cont...



Key to Success



- Establishing effective ventilation is the key to nearly all successful neonatal resuscitations!

Postnatal management of asphyxia

- ☐ **Keep NPO** until the neonate is passing meconium, has clear gastric content, and normoactive bowel sound.
- ☐ **Then Trophic feeding can be started**

Cont....

- A. Fluid Management-** two third of the maintenance fluid (avoid both overload and inadequate circulating volume)
- B. Oxygenation** it should be maintained in the normal range (Saturation between 90-95%)
- C. Do not warm asphyxiated babies.** Cooling therapy is the standard treatment for hypoxic ischemic encephalopathy. However, it is not feasible in Ethiopian setup.

Cont...

D. Correction of Metabolic States– Blood glucose has to be kept in the normal range.

- Hypoglycemia is often seen in asphyxiated newborns. It increases energy deficit. It has to be treated with **2ml/kg of 10% dextrose** (4ml/kg (in the presence of seizure) followed by maintenance.)
- Hypocalcemia (can cause seizure and decreased cardiac contractility) administer **1- 2ml/kg of 10% calcium gluconate** QID with strict cardiac monitoring.

E. Seizure Treatment (refer to the guideline on seizure treatment)

❖ **Complication of asphyxia**

Hypoxic-Ischemic encephalopathy (HIE)

It is an important cause of permanent damage to CNS cells that may result in :

Neonatal death 15-20 %,

And 25-50% of survivors are left with permanent neurodevelopmental abnormalities (cerebral palsy, mental retardation).

1. Infant Respiratory Distress Syndrome(HMD)

Defn:

-
- ❖ also called neonatal respiratory distress syndrome or
 - ❖ respiratory distress syndrome of newborn,
 - ❖ previously called hyaline membrane disease, is a syndrome or a common lung disorders caused in premature infants.
 - ❖ It is a breathing problem or decreasing of blood perfusion to the lungs/problem is on the system/lung.

RESPIRATORY DISTRESS SYNDROME



Hyaline Membrane Disease(HMD)



Causes

Primarily insufficiency surfactant production 2nd to prematurity of lungs.

Genetic problem with the production of surfactant associated proteins.

Predisposing Factors

-Obstruction of the air passage with thick secretions or maternal blood.

-Infant born from:

- ✓ **diabetic mother,**
- ✓ **infant born by C\S,**
- ✓ **perinatal asphyxia,**
- ✓ **low birth weight babies.**

Clinical Manifestation

Cyanosis

Expiratory grunting(make a low inarticulate sound to express effort)

Nasal flaring(noise)

Tachypnea

Tachycardia

Retractions

Treatment

Thermoregulation

- ❖ Optimum thermal environment to minimize oxygen consumption and oxygen requirements.
- ❖ Body temperature should be between 36.5 and 37°C.

Fluid

- ❖ Maintenance fluid requirements usually begin at 60-80 ml/kg/days as a 10 % dextrose solution and increases gradually to 150 ml/kg/day by the 5th day of life.

Antibiotics

❖ **The potentially fulminant course of congenital**

pneumonia and the difficulty in distinguishing it from HMD have led to the recommendation that all infants with significant respiratory distress receive antibiotic after appropriate cultures.

❖ **A penicillin combined with an aminoglycoside is the antibiotic regimen of choice.**

Surfactant therapy

❖ **Multi-dose endotracheal instillation of exogenous surfactant for the treatment of HMD has dramatically improved**

survival and reduced the incidence of pulmonary air leaks.

Ventilatory support

❖ **Warm humidified oxygen should be provided to keep arterial levels between 55 and 70 mmHg**

❖ **Nasal CPAP with continuous monitoring (see Neonatal Procedure)**

❖ **Infants with severe HMD or those who develop complications resulting in persistent apnea require assisted mechanical ventilation.**

Prevention

❖ **Most important is prevention of prematurity**

Administration of dexamethasone to women 48 hours before delivery of fetuses less than 34 weeks of gestation significantly reduces the incidence and the mortality and morbidity of HMD.

.

3. Meconium aspiration syndrome



Meconium Aspiration Syndrome

- ❖ Meconium staining of the amniotic fluid is usually considered to be indicator of fetal distress.
- ❖ The passage of meconium in utero accompanies 8 to 20% of all deliveries and is seen predominantly in **small for gestational age and post mature infants.**
- ❖ **Five percent** of such infants develop meconium aspiration pneumonia.

Clinical features

- ❖ The infant frequently exhibits the classic signs of post maturity with evidence of weight loss together with nails, skin and umbilical cord that are heavily stained with a yellowish pigment.
- ❖ The infants are often depressed at birth secondary to the hypoxic insult precipitating the passage of meconium.
- ❖ Respiratory distress with cyanosis, grunting, flaring retractions and marked tachypnea soon ensues.
- ❖ Characteristically the **chest acquires an over inflation appearance, and rales** may be audible on auscultation.

Pathophysiology

❖ Due to prolonged labour or other causes (foetal distress in the uterus) → Anal sphincter relaxes → Meconium passed in to the amniotic fluid → Meconium it may be strain in the skin → During the gestation or at birth first breath, meconium is aspirated in to the lungs → Mechanical plug causes partial air way obstructed (infant struggles to take in more air during inspiration - air hunger) → child causes respiratory distress (due to hyperinflation, hypoxemia and acidosis results in increased pulmonary vascular resistance)

➤ **Roentgenogram:** Coarse, irregular pulmonary densities, increased anterior posterior diameter and flattening of the diaphragm

➤ **Prevention**

- The risk of meconium aspiration syndrome can be decreased by
- Paying careful attention to fetal distress and initiating prompt delivery in the presence of fetal distress.
- Careful, thorough suctioning of the oropharynx after the head is delivered.
- Suctioning via endotracheal tube is indicated for depressed infants before initiation of positive pressure ventilation.

Treatment

- ❖ Depressed infant should undergo endotracheal intubation and suction should be applied directly to the endotracheal tube to remove meconium from the airway.
- ❖ Treatment includes supportive care and standard management for respiratory distress.
- ❖ Supportive management may include the use of antibiotics, because bacterial sepsis may have precipitated the passage and subsequent aspiration of meconium, and bacterial pneumonia may be indistinguishable roentgenographically from meconium aspiration syndrome. Further meconium enhances bacterial growth in vitro.
- ❖ Vigorous infant with lightly stained amniotic fluid may not require treatment.

Neonatal Jaundice

❖ Objectives

At the end of this lesson the student will be able to:

- Define jaundice
- Differentiate between physiologic and pathologic jaundice
- Identify the causes of jaundice
- Investigate and Manage jaundice

Neonatal Jaundice

- ☐ Is a yellowish discoloration of the skin and or sclera due to bilirubin deposition.
- ☐ In newborns jaundice appears when total **bilirubin (TB)** is **more than 7 mg /dl** and almost 97 % healthy full term babies have biochemical hyperbilirubinemia.
- ☐ It could be physiologic or pathologic

Comparison b/n physiological from Pathological Jaundice

<i>No</i>	<i>Features</i>	<i>Physiologic Jaundice</i>	<i>Pathological Jaundice</i>
1	Clinical onset of jaundice (after birth)	>24 hrs	<24 hrs
2	Jaundice still clinically visible (day after birth)	Term < 8 days Preterm < 14 days	Term $\geq$ 8 days Preterm > 14 days
3	Peak TSB	Term < 12 mg/dl Preterm < 15 mg/dl	Term > 12 mg/dl Preterm > 15 mg/dl
4	Rise in TSB	< 5mg/dl/24 hrs	> 5mg/dl/24 hrs
5	Conjugated serum bilirubin level	<2mg/dl	>2mg/dl or 15 % of TB

Causes of Jaundice

☐ Isoimmunization

- ▶ RH incompatibility,
- ▶ ABO incompatibility,
- ▶ Other blood group incompatibility

☐ Infection

- ▶ Bacterial, viral, protozoal

☐ Sequestered blood

- ▶ Subgallial hemorrhage, cephal hematoma, ecchymosis, hemangioma

☐ Erythrocyte biochemical defect

- ▶ G6PD deficiency, Hexokinase deficiency

Causes of Jaundice...cont

☐ Structural abnormalities of erythrocytes:

- ▶ Hereditary spherocytosis, eliptocytosis

☐ Disorder of hepatic uptake:

- ▶ Gilbert syndrome

☐ Disorder of conjugation:

- ▶ Crigler – Najar syndrome (absence of UGT activity)

☐ Disorder of enterohepatic circulation

- ▶ Breast feeding Jaundice

☐ Hypothyroidism:

Clinical manifestations

- ☐ Yellowish discoloration of sclera, skin, mucus membranes
- ☐ A newborn may present with signs of bilirubin encephalopathy.

Risk factors for bilirubin encephalopathy

- ☐ Prematurity
- ☐ Metabolic acidosis,
- ☐ Hypoglycemia,
- ☐ Sepsis,
- ☐ Temperature instability,
- ☐ Significant lethargy
- ☐ Low serum albumin

Investigations

- ☐ Total bilirubin
- ☐ Direct and indirect bilirubin
- ☐ Maternal and neonatal blood group and RH
- ☐ Direct/indirect Coombs test
- ☐ Hemoglobin (Hgb) or hematocrit (HCT)
- ☐ Peripheral RBC morphology

Investigations

- ☐ Reticulocyte production index(RPI)
- ☐ Serum albumin level
- ☐ albumin to bilirubin ratio
- ☐ Liver function test (LFT)
- ☐ Septic work up.
- ☐ Abdominal ultrasound with indication.

Principles of treatment

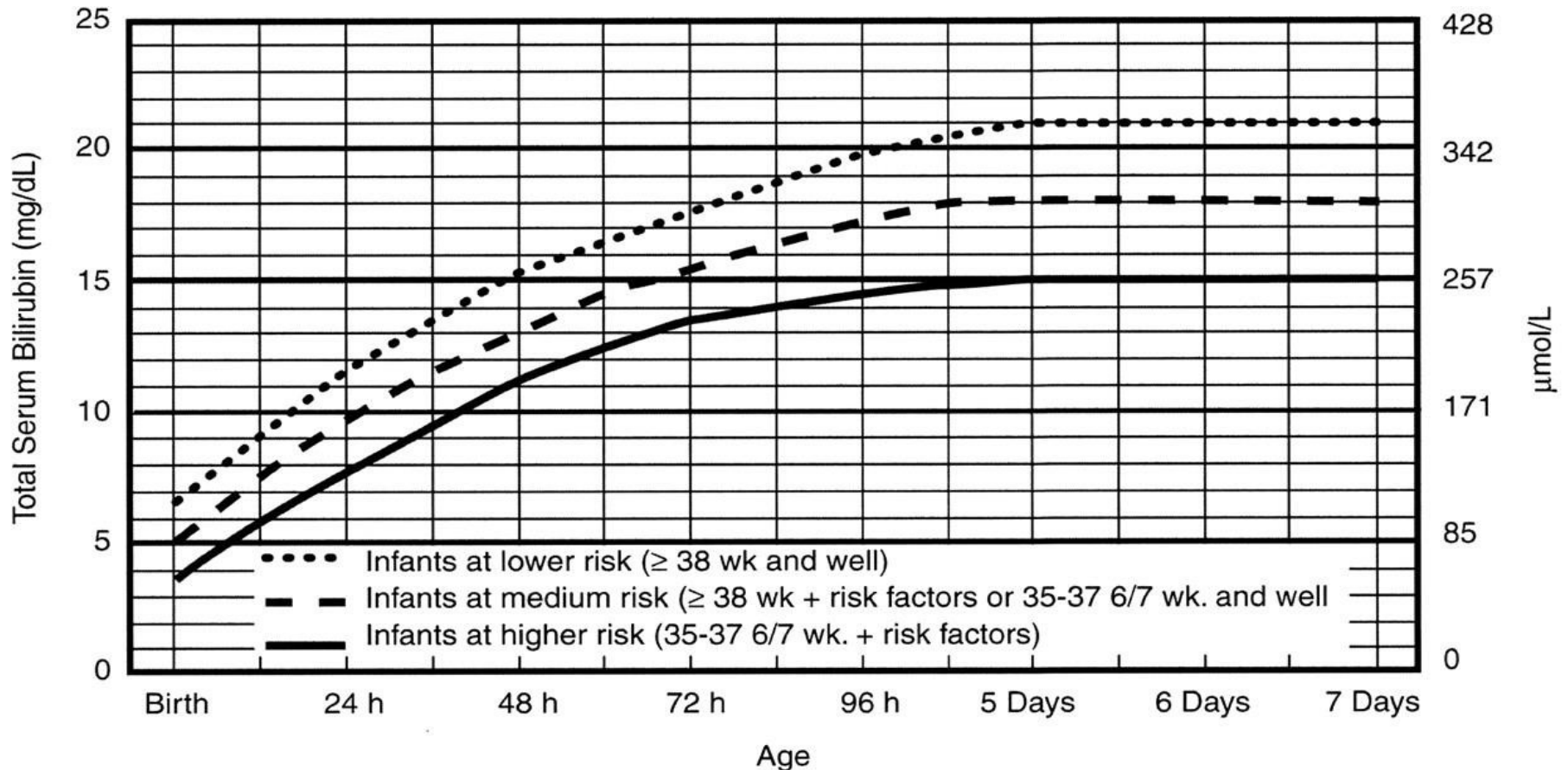
☐ Phototherapy

☐ Exchange transfusion

☐ Other medical managements

☐ *Note: Use Butanic curve for choosing the management options.*

Butanic curve



- Use total bilirubin. Do not subtract direct reacting or conjugated bilirubin.
- Risk factors = isoimmune hemolytic disease, G6PD deficiency, asphyxia, significant lethargy, temperature instability, sepsis, acidosis, or albumin $< 3.0\text{g/dL}$ (if measured)
- For well infants 35-37 6/7 wk can adjust TSB levels for intervention around the medium risk line. It is an option to intervene at lower TSB levels for infants closer to 35 wks and at higher TSB levels for those closer to 37 6/7 wk.
- It is an option to provide conventional phototherapy in hospital or at home at TSB levels 2-3 mg/dL (35-50mmol/L) below those shown but home phototherapy should not be used in any infant with risk factors.

Phototherapy

- ☐ Use Bhutanic curve for determination of the management of hyperbilirubinemia
- ☐ Bilirubin absorbs light maximally in the blue range (420-470 nm).
 - ▶ Use either
 - Blue, or special narrow-spectrum (super) blue lights
- ☐ Cover the baby's eye and put diaper with maximum body surface area being exposed.
- ☐ Measure weight daily.
- ☐ Increase fluid intake by 25 %
- ☐ Give a bolus of fluid with Normal saline 20ml/kg if bilirubin remains high.

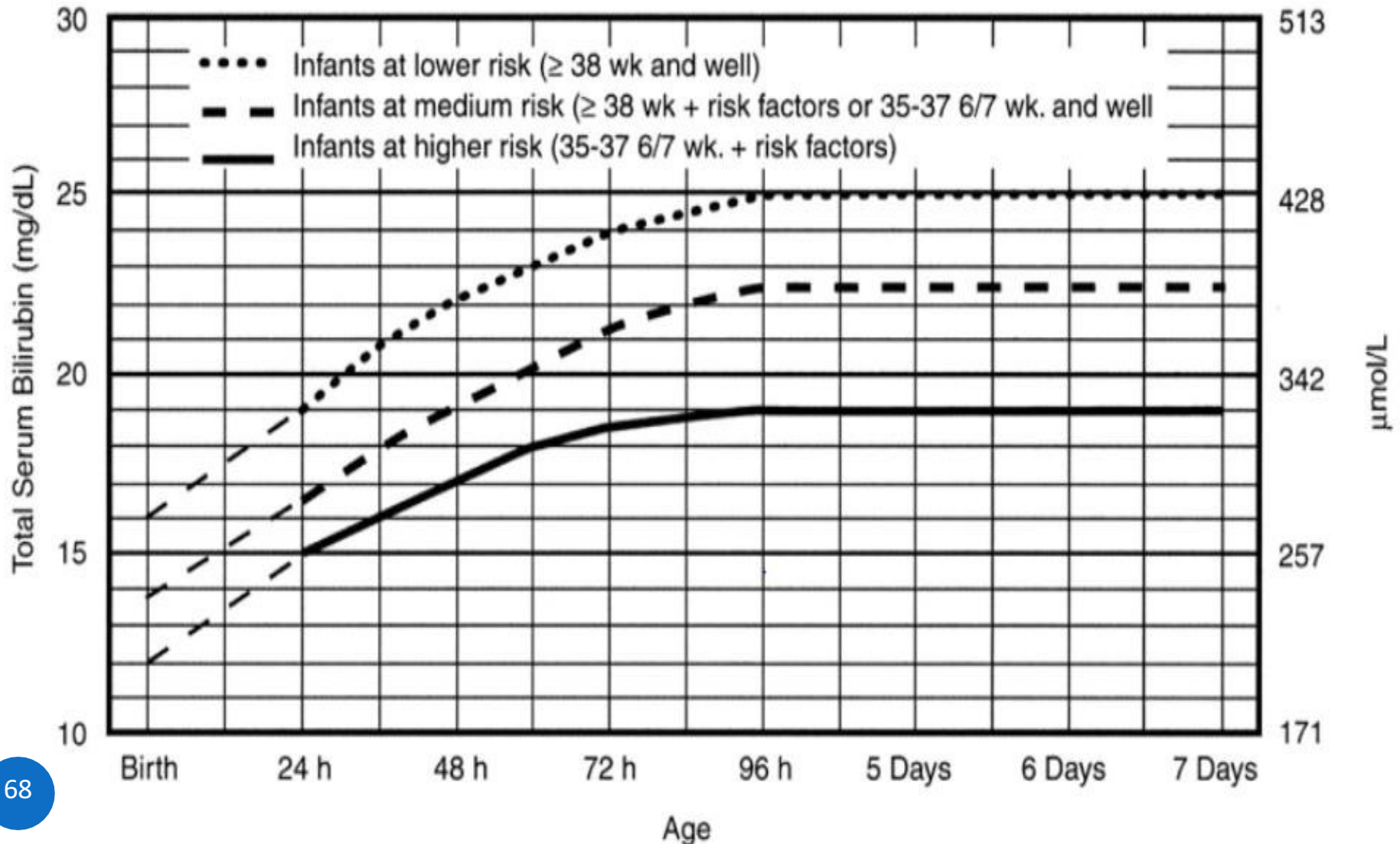
Side effects of phototherapy

- ☐ Insensible water loss
- ☐ Watery and frequent stool
- ☐ Retinal damage
- ☐ Erythema and increased blood flow
- ☐ Bronze baby syndrome (with increased CB)
- ☐ Low calcium level (in preterm)
- ☐ Interferes with maternal infant bonding

Double Volume Exchange transfusion

- ☐ The amount of blood volume to be exchanged is equivalent to 2x the blood volume of the baby (85ml/kg)
- ☐ Use fresh blood less than 7 days old
- ☐ Do procedure after umbilical catheterization using aseptic technique
- ☐ Heparinize the catheter
- ☐ The amount of blood to be removed at a time is 5ml to 20 ml

Bhutani curve: Exchange transfusion in infants of 35 or more weeks' gestation



Type of blood to be transfused

☐ Rh hemolytic disease

- ▶ give blood group compatible to the baby and RH of the mother.

☐ ABO hemolytic disease

- ▶ give blood group of the mother and Rh compatible to the newborn

☐ **O negative blood is the most preferred type of blood for exchange transfusion.**

Monitoring

- ☐ Strictly monitor the vital signs during the procedure.
- ☐ Determine post transfusion HCT 4-6 hours after the procedure.
- ☐ Determine bilirubin 4 hourly after the procedure.
- ☐ Monitor RBS every 30-60 minutes during the procedure and 2-4 hourly for the first 24 hours after procedure.

Monitoring

- ☐ Calcium gluconate slowly via a peripheral vein under strict cardiac monitoring after every 100ml of blood is exchanged.
- ☐ Cloxacillin 50mg/Kg bid for 2- 3 days and gentamicin 5mg/kg BID for 2-3 days.
- ☐ Keep baby NPO for 3 hours before and after procedure.

Other treatment modalities

- ☐ Phenobarbital 5 mg/kg to stimulate liver enzyme in Crigler – Najjar syndrome.
- ☐ High dose of IV immunoglobulin.
- ☐ In case of Breast milk jaundice discontinuation of breast milk for 48 hrs

Complications of hyperbilirubinamia

□ 1. Acute bilirubin encephalopathy has three phases

▶ Phase -I

- 1st – 2 Days of Age
- Poor motor reflex, high pitched cry, Decreased tone, lethargy, poor feeding

▶ Phase-II

- Middle of 1st week
- Hypertonia, seizure and depressed sensorium, fever, opisthotonos posturing, paralysis of upward gazing.

▶ Phase -III

- After 1 week of age
- Hypertonia decreases, Hearing and visual abnormality, poor feeding, Athetosis and seizure may also occur.

Complications

- 2. Chronic bilirubin encephalopathy (Kernicterus)
 - ▶ Choreoathetoid cerebral palsy
 - ▶ Upward gaze palsy
 - ▶ Sensorineural hearing loss

Summary

- ☐ Jaundice is yellowish discoloration of mucus membranes and skin
- ☐ Isoimmune Hemolytic disease is the leading cause of pathologic jaundice
- ☐ The feared complication of hyper bilirubinemia is kernicterus.
- ☐ Management is mainly by phototherapy and sometimes double volume exchange transfusion.

Neonatal Sepsis

- Neonatal sepsis is defined as a clinical syndrome of **bacteremia** with systemic signs and symptoms of **infection** in the first 4 weeks of life.
- Neonatal Sepsis can be divided as early and late onset depending on the time of occurrence

Early-onset neonatal sepsis

- ☐ Occurs within the first 7 days of life.
- ☐ Usually due to vertical transmission by ascending contaminated amniotic fluid (chorioamnionitis) or during vaginal delivery from bacteria colonizing or infecting the mother's lower genital tract.

Early-onset neonatal sepsis....

- ☐ Is caused by organisms prevalent in the maternal genital tract, in the labor room or operation theatre.
- ☐ Majority are caused by **Gp.B streptococcus, E-coli, klebsiela** and **enterobacter**.
- ☐ Majority manifest with respiratory distress due to an early intrauterine pneumonia.

Risk factors

- ☐ Birth asphyxia
- ☐ Unclean vaginal examination
- ☐ Foul smelling liquor
- ☐ Prolonged labor (>24 hrs)
- ☐ Preterm and low birth weight neonate
- ☐ Prolonged rupture of membrane (>18 hrs)
- ☐ Intrapartum Maternal pyrexia ($> 38^{\circ}\text{C}$)
- ☐ Five minute Apgar score ≤ 6

Late onset neonatal sepsis

- ❑ It occurs after first 7 days of life
- ❑ Most are caused by gram negative bacteria's **Klebsiela, enterobacter, E-coli, pseudomonas** and **salmonella**
- ❑ Gram positives like staph aureus also contribute as causes of late onset sepsis.

Late onset neonatal sepsis...

□ Late-onset infections are acquired as nosocomial after delivery in the normal newborn nursery, neonatal intensive care unit (NICU) or the community.

□ Usual sources of infection include:-

- ▶ Incubators
- ▶ Resuscitation equipment's
- ▶ Feeding bottles
- ▶ Catheters
- ▶ Infusion sets and sites

Bacterial species	Frequency of isolation	
	Early-onset	Late-onset
Group B Streptococcus	+++	+++
Escherichia coli	+++	++
Klebsiella spp.	+	+
Enterobacter spp.	+	+
Listeria monocytogenes	+	+
Other enteric gram-negatives	+	+
Non-enteric gram-negatives*	+	+
Viridans streptococci	+	+
Staphylococcus aureus	+	+++
Citrobacter spp.	0	+
Salmonella spp.	0	+
Coagulase-negative staphylococci	0	+
Enterococcus spp.	0	+

+++ : commonly associated; ++ : frequently associated; + : occasionally associated; 0 : rarely associated.

* Includes non typable Hemophilus influenzae and Neisseria meningitidis .

Source :©2013 UpToDate ® Common bacterial agents causing neonatal sepsis in term infants

Clinical manifestation

☐ Systemic manifestation of danger signs

☐ Not feeding well

☐ Sever chest indrawing

☐ Convulsion

☐ Fever $>38^{\circ}\text{C}$

☐ Drowsy or unconscious

☐ Decreased body temperature
 $<35.5^{\circ}\text{C}$

☐ Movement when stimulated or
no movement at all

☐ Central cyanosis

☐ Fast breathing (≥ 60 breath
per minute)

☐ Sever jaundice

☐ Sever abdominal distension

☐ Crying

Clinical manifestation cont...

- ☐ Localizing signs of infection.
- ☐ Signs of pneumonia
- ☐ Many or severe skin pustules
- ☐ Bulging fontanelle
- ☐ Painful joints
- ☐ Joint swelling
- ☐ Reduced movement...

Laboratory evaluation

- ☐ Consider blood culture and sensitivity whenever possible and modify that treatment accordingly.
- ☐ CBC (Complete Blood Count with differential).
- ☐ Concern for sepsis if:
 - ▶ Total WBC is abnormal ($<5,000$ or $>20,000$)
 - ▶ Differential with granulocytes $>70\%$.
- ☐ ESR or CRP. Concern for sepsis if positive.

Laboratory evaluation Cont...

- ☐ Consider **urinalysis** and gram stain if symptoms of urinary tract infection or more general concerns for sepsis in infant **>1week old**
- ☐ Consider lumbar puncture if concern for meningitis (lethargy, irritability, convulsions, bulging fontanel, meningismus).
- ☐ Consider chest x-ray if respiratory distress or oxygen desaturation.

Sepsis with meningitis

□ A diagnosis of meningitis should be made based on clinical evidence:-

▶ Abnormal neurological exam: seizures, abnormal tone and full fontanel) and risk of infection for babies less than 72 hrs of age

▶ For babies age greater than 72 hrs of age diagnosed with sepsis CSF analysis should be done to rule out

meningitis despite absence of overt signs of meningitis.

CSF analysis suggestive of meningitis:

- ☐ Identification of organism on gram stain or culture.
- ☐ WBC count greater than or equal to 20 cells/mm³
- ☐ Low glucose (less than two third of serum value) and
- ☐ Protein greater than 150 mg/dl

Differential Diagnosis

- ☐ **Viral infections** – Enteroviruses, herpes simplex virus, cytomegalovirus, influenza viruses, respiratory syncytial virus, etc.
- ☐ **Spirochetal infections** – Syphilis
- ☐ **Parasitic infections** – Congenital malaria, toxoplasmosis
- ☐ **Fungal infection** – Candidiasis
- ☐ **Other bacterial infections** include urinary tract infection (particularly in the setting of a congenital genitourinary tract malformation), osteomyelitis or septic arthritis, pneumonia, and tuberculosis

Management

□ **General supportive measures**, including respiratory and hemodynamic management, are combined with antibiotic treatment.

□ **Antibiotic therapy:**

- ▶ Empirical antibiotics should be given to children with suspected neonatal sepsis.
- ▶ Admit to hospital and when possible, do septic work up before starting antibiotics.

For early onset (less than 7 days)

□ Antibiotic – **Ampicillin and Gentamycin**

□ Duration: If positive cultures – **minimum 10 days.**

- ▶ If **negative** cultures, and **clinically well**, with normal CRP or ESR– stop after 48 hours
- ▶ If **negative** cultures, but **not clinically well**, abnormal CXR or elevated CRP – treat as **confirmed sepsis.**
- ▶ If no improvement after 48 hours, or worsens, after repeating blood cultures (if possible) and considering further investigations, consider changing to: **Ceftriaxone and**

Gentamicin

Cont...

- ☐ **Ampicillin** 150 mg/kg/dose IV every 12 hours **plus** **Gentamicin** 3-4 mg/kg IV once a day and once in 48 hrs in very preterm babies for 10 days.
- ☐ If meningitis ruled out: Ampicillin 50 mg/kg/dose IV every 12 hours. If no improvement after 48 hrs:
- ☐ Ceftriaxone 50 mg/kg IV every 12 hours **plus** Gentamicin 3-4 mg/kg IV once a day and once in 48 hrs in very preterm babies for 10 days

For late onset (7-28 days)

- ❑ Antibiotic – Ampicillin and Gentamicin
- ❑ If the baby is critically sick or staphylococcal infection is likely (pustular skin rash, osteomyelitis...) start with triple antibiotics (**Cloxacillin + ampicillin + gentamicin**)
- ❑ If no improvement after 48 hours, or the infant's condition worsens. Consider changing antibiotics to:
 - ▶ **Cloxacillin + Ceftriaxone + Gentamicin** or
 - ▶ **Vancomycin + Gentamicin**

Treatment of neonatal sepsis with meningitis

- ☐ Antibiotics the same as for sepsis but with higher dose and prolonged duration, for **21 days** and
- ☐ Gentamycin should be stopped on the **14th day**

Cont...

☐ If the age of the baby is ≤ 14 days:-

☐ **Ampicillin** 150 mg/kg/dose IV every 12 hours **plus**

Gentamicin 3-4 mg/kg IV once a day and once in 48 hrs
in very preterm babies for 10 days.

☐ If the age of the baby is >14 days:-

☐ **Ampicillin** 50 mg/kg/dose IV every 6 hours **plus**

Gentamicin 3-4 mg/kg IV once a daily for 10 days

☐ **Meningitis:** 100 mg/kg/dose IV every 6hr.

Cont...

☐ If not improved

☐ Cloxacillin 50mg/kg/dose every 6 hours **plus**

Ceftriaxone 50 mg/kg/dose every 12 hours **plus**

Gentamicin 4 mg/kg IV once a day for 10 days.

☐ Ceftriaxone is contraindicated in setting of jaundice or within 48 hours of IV calcium administration

Neonatal Tetanus

Neonatal Tetanus

- It is a generalized tetanus caused by bacterium **Clostridium tetani** which are universally present in the soil.
- The disease is caused by the action of a **potent neurotoxin** produced during the growth of the bacteria in dead tissues,
 - ▶ e.g. In dirty wounds or in the umbilicus following non sterile delivery.
- Without good supportive care, case fatality rates can exceed 90%.

Clinical presentation

- ☐ Irritability
- ☐ Excessive crying
- ☐ Poor feeding
- ☐ Rigidity
- ☐ Facial grimacing
- ☐ Severe spasms with touch
- ☐ Opisthotonos

Management and treatment

☐ Treatment of neonatal tetanus includes:

- ▶ Administration of tetanus antitoxin and
- ▶ Muscle relaxants and parenteral feeding

Management and treatment....

☐ Control of muscle spasms:

☐ The patient should be admitted to a **quiet, darkened room** where all possible **auditory, visual, tactile, or other stimuli** are minimized.

☐ Diazepam controls spasms better and safer than other options listed below.

▶ The initial dose of 0.1–0.2mg/kg every 3–6 hr given intravenously is subsequently titrated to control the tetanic spasms.

Management and treatment....

- ☐ Other drugs which can be used in combination with diazepam include:
- ☐ **Chlorpromazine** (1-5mg/kg/dose q8hr)
- ☐ **Phenobarbitone** (loading dose 20mg/kg, then 2.5mg/kg/dose q12hr, increased to max 5mg/kg/dose q12hr).

Management and treatment....

☐ Antitoxin therapy:

☐ Human tetanus immunoglobulin should be given intramuscularly in a single dose (3,000 to 6,000 IU).

▶ If human serum immunoglobulin is unavailable, **tetanus antitoxin** should be given, assuming sensitivity reactions to horse serum are negative.

☐ The antitoxin is given intravenously and intramuscularly (half of the dose via each route).

Management and treatment....

☐ Antimicrobial therapy.

- ▶ Metronidazole (30 mg/kg/day, given at six hour intervals; maximum 4 g/day) or
- ▶ Parenteral penicillin G (100,000 U/kg/day) is an alternative.

☐ Treatment for 10 to 14 days is recommended.

Management and treatment....

☐ Wound treatment.

- ▶ After the patient has been sedated and received antitoxin, the
 - Wound should be thoroughly cleansed and debrided.

☐ Supportive treatment.

- ▶ Oxygen should be available.
- ▶ During early stages, oral feeding should be avoided because of the danger of aspiration.
- ▶ A continuous intra- venous infusion can provide fluid (such as water and plasma), electrolytes, glucose, and amino acids.

Management and treatment....

☐ Tracheotomy.

- ▶ The combination of heavy sedation, difficulty in swallowing, laryngospasm, and accumulation of secretions may lead to obstruction of the airway.
- ▶ A tracheotomy can be lifesaving if performed when appropriately indicated.

Neonatal Hypoglycemia

BACKGROUND and PATHOPHYSIOLOGY

- Glucose is the major energy source for fetus and neonate.
- The newborn brain depends upon glucose almost exclusively.
- Up to **90%** of total glucose used is consumed by the brain.
- Alternate fuels (e.g., **ketones**, **lactate**) are produced in very low quantities.

Cont...

- ❑ The usual rate of glucose utilization is **4-8 mg/kg/min**.
- ❑ Glucose regulatory mechanisms are **sluggish** at birth.
- ❑ The infant is susceptible to hypoglycemia when glucose demands are increased or when **exogenous** or **endogenous** glucose supply is limited.
- ❑ Severe or prolonged hypoglycemia may result in long term neurologic damage.

Hypoglycemia

- Hypoglycemia in the first few days after birth is defined as **blood glucose <40 mg/dL**.
- In preterm infants, repeated blood glucose levels below **50mg/dL** may be associated with neurodevelopmental delay.

Who should be evaluated?

☐ Newborns at risk for hypoglycemia that needs evaluation include:

- ▶ Preterm infants
- ▶ Small for gestational age(SGA)
- ▶ Large for gestational age(LGA)
- ▶ Infants of diabetic mothers(IDM)
- ▶ Sick infants who require intensive care (e.g. sepsis, asphyxia, respiratory distress)

Who should be evaluated?....

- ☐ Post exchange blood transfusion
- ☐ Infants on intravenous fluids or parenteral fluids
- ☐ Infants whose mothers were treated with beta adrenergic or oral hypoglycemic agents
- ☐ Intrapartum dextrose infusions
- ☐ Infants with polycythemia
- ☐ Hypothermic newborns

ETIOLOGY

1. Decreased substrate availability:

- ▶ Intra-uterine growth retardation
- ▶ Glycogen storage disease
- ▶ Inborn errors (e.g., fructose intolerance)
- ▶ Prematurity
- ▶ Prolonged fasting without IV glucose

2. Hyperinsulinemia:

- ▶ Infant of diabetic mother
- ▶ Islet cell hyperplasia

Cont...

1. Other endocrine abnormalities: 3. Miscellaneous conditions:

- ▶ Pan-hypopituitarism
- ▶ Hypothyroidism
- ▶ Adrenal insufficiency
- ▶ Polycythemia
- ▶ Congenital heart disease
- ▶ CNS abnormalities

2. Increased glucose utilization:

- ▶ Cold stress
- ▶ Increased work of breathing
- ▶ Sepsis
- ▶ Perinatal asphyxia

SIGNS AND SYMPTOMS

➤ Jitteriness

➤ Apnea

➤ Irritability

➤ Grunting and

➤ Lethargy

➤ Sweating (uncommon).

➤ Seizures

❑ Hypoglycemic infants may not always be symptomatic.

▶ Therefore, routine glucose monitoring for at-risk infants is mandatory.

❑ Lack of symptoms does not guarantee absence of long term sequelae.

DIAGNOSTIC WORKUP:

- ☐ Supportive perinatal history (risk factors).
- ☐ Signs and symptoms of hypoglycemia.
- ☐ Whole blood glucose less than 40mg/dl.
 - ▶ Specimens for measurement of glucose should be obtained from:-
 - Heelstick
 - Venipuncture

MANAGEMENT OF HYPOGLYCEMIA:

☐ Treatment of asymptomatic hypoglycemia

☐ Feeding

- ▶ Offer breast feeding immediately.
- ▶ Check blood glucose 30 minutes after feeding to insure normal glucose level before the next feeding.
- ▶ If repeated blood glucose is $> 40\text{mg/dl}$ continue to offer feedings at 2-3hours interval.

Treatment of asymptomatic hypoglycemia...

□ Indications of IV infusions in asymptomatic hypoglycemia:-

- ▶ Blood glucose $< 25\text{mg/dl}$.
- ▶ Blood glucose remains $< 40\text{mg/dl}$ after one attempt of feeding
- ▶ If infant becomes symptomatic
- ▶ If oral feeding is contraindicated

Treatment of symptomatic hypoglycemia

☐ Immediate treatment

- ▶ Secure IV line
- ▶ Give 2ml/kg of 10% glucose IV bolus over one minute if signs other than seizure
- ▶ Give 4 ml/kg of 10% glucose as a bolus over one minute if seizure is present

Continuous treatment

- ☐ Put on 10% glucose infusion at glucose infusion rate (GIR) of 6mg/kg/minutes ($\sim$ 90ml/kg/day).
- ☐ Recheck blood glucose after 30 minutes.
- ☐ If it remains above 40mg/dl check blood glucose 1 hourly and then every 6 hourly.
- ☐ If blood glucose remains <40 mg/dl, increase the GIR by 2mg/kg/minutes every 30 minutes till repeat values are above 40mg/dl.

Continuous treatment

- ☐ Once the blood glucose values stabilize above 40mg/dl for 24 hours, taper the GIR by 2 ml/kg/min every 6 hourly with proportional increment of oral feeds.
- ☐ If the neonate requires $\text{GIR} > 12\text{mg/kg/minutes}$, persistent hypoglycemia should be considered.

Neonatal hyperglycemia

- ☐ It is usually defined as whole blood glucose level >125 mg/dl.

Diagnosis of hyperglycemia will be made based on:

- ☐ Perinatal risk factors (like asphyxia, sepsis, prematurity)
- ☐ Clinical manifestations (weight loss, signs of dehydration, polyuria).
- ☐ Laboratory (blood glucose >125 mg/dl., ketone & glucose on urine analysis)

Management of neonatal hyperglycemia

- ☐ Treat the neonate if in shock or dehydrated accordingly.
- ☐ Decrease glucose infusion by 2mg/kg/min (30ml/kg/day) every 6 hrs and stop gradually if on IV infusion.
- ☐ If the newborn was not on infusion put him/her on 5% glucose with rate of 4mg/kg/min.(~ 60ml/kg/day).
- ☐ Look for and treat underlying causes.
- ☐ **Insulin therapy**

Congenital syphilis

- **Syphilis is a chronic, generalized infectious disease caused by a slender spirochete, *treponema pallidum*.**
- **Congenital syphilis results from transplacental transmission of spirochetes.**
- **Pregnant women with 10 or 20 syphilis and spirochetemia are more likely to transmit infection to the fetus than are women with latent infection.**

Cont...

- **Transmission may occur throughout pregnancy.**
- **Syphilis during pregnancy has a transmission rate approaching 100%.**
- **Fetal and perinatal deaths occur in 40% of affected infants.**

Clinical manifestations

- **Early congenital syphilis** Early congenital syphilis appears during the first 2 years of life.
- **Approximately two-thirds of the infants are asymptomatic at birth, if they are untreated, symptoms develop within weeks or months.**
- ❖ **Manifestations include:**
 - ✓ **hepatosplenomegaly,**
 - ✓ **lymphadenopathy,**
 - ✓ **hemolytic anemia**
 - ✓ **thrombocytopenia and mucocutaneous rash.**
 - ✓ **Rhinitis (snuffles) and condylomatous lesions are highly characteristic features of mucous membrane involvement.**
 - ✓ **CNS abnormalities & failure to thrive**
 - ✓ **Chorioretinitis, nephritis and nephrotic syndrome may also be seen.**
 - ✓ **Roentgenographic abnormalities include:**
 - Multiple sites osteochondritis at the wrist, elbow, ankle and knees and** ☐ **Periostitis of the long bones.**

Late congenital syphilis

- ✓ **The manifestations result primarily from chronic inflammation of bone, teeth, and the CNS.**
- ✓ **Skeletal changes include frontal bossing, thickening of the sterno clavicular portion of the clavicle and an anterior bowing of the mid portion of the tibia (Saber shins).**
- ✓ **Hutchinson's triad includes: Hutchinson's teeth (peg shaped upper central incisors) □ 8th nerve deafness and interstitial keratitis**
- ✓ **The clutton joint represents synovitis involving the lower extremities (usually the knee).**
- ✓ **Saddle nose is a result of rhinitis.**

Diagnosis Serology

✓ VDRL (Venereal Disease Research Laboratory)

✓ RPR (Rapid Plasma Reagin)

➤ These tests are rapid and non-expensive but they are not specific.

➤ Maternal antibody crosses the placenta and accordingly a false positive VDRL can occur in an uninfected infant delivered to a VDRL positive mother.

➤ This passively transferred antibody is not demonstrable by 3 months of age.

Treponemal tests include:

TPI (Treponema Pallidum Immobilization test)

FTA-ABs (Fluorescent Treponemal Antibody Absorption test)

Dark field microscopy or direct fluorescent antibody test of scrapings from primary lesions can reveal T. pallidum and often permits a **definitive diagnosis. Because doing these investigation is not always possible, it is recommended:**

it is recommended:

A presumptive diagnosis of congenital syphilis should be made and be treated in an infant whose mother had untreated or inadequately treated syphilis or whose treatment status is unknown.

Inadequate treatment means she was treated ≤ 30 days before delivery or treated with non penicillin regimen.

All infants with a presumptive diagnosis of congenital syphilis should be treated with regimen effective for neurosyphilis because this cannot be reliably excluded.

Treatment

✓ Treatment involves:

✓ **Crystalline penicillin** (100,000-150,000 IU/kg/d iv) or

✓ *Procaine penicillin* (50,000 IU/kg/d im) for 10-14 days.

❖ Treatment is recommended for any infant if maternal treatment is unknown, inadequate or given during the last 4 wks of pregnancy or if the treatment was by any drug other than penicillin

